CLINICAL NEUROLOGY CURRICULUM • MASTER HANDBOOK

Clinical Neuro-Localisation Challenges

A Comprehensive 10-Stage Spiral Curriculum in Anatomical Diagnosis: From First-Principles Gross Triage to Advanced Multi-Tier Discriminators (150 Worked Cases).

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Core Pedagogical Framework: Based on the 5-Question Method & 13 Golden Rules of Neuro-Localisation.
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10 Stages • 150 Worked Cases

Chapter 0: Foundations of Clinical Neuro-Localisation

Instructions to Students, The 5-Step Diagnostic Protocol & The 13 Golden Rules of Localization

1. The 2-Minute Clinical Orientation & Philosophy

Neurological localisation is pattern recognition based on anatomy. The greatest pitfall in clinical neurology is rushing to name an etiology (such as *"stroke"*, *"multiple sclerosis"*, or *"tumour"*) before establishing where the anatomical lesion resides.

Always state the anatomical locus and pathways involved first, for example:
“This is an acute right caudal pontine lesion involving the right CN VII fascicle, right CN VI fascicle, and the descending left corticospinal tract.”
Only after this spatial locus is fixed should you inquire whether the underlying pathological mechanism is vascular, inflammatory, demyelinating, or neoplastic.

2. The 5-Question Diagnostic Method

For every clinical patient, systematically ask these five sequential questions:

- **Question 1: What neurological functions are abnormal?** Map all affected systems: motor power, sensation, cranial nerves, visual fields, language/speech, cerebellar coordination, and autonomic sphincters.
- **Question 2: Is the lesion Central or Peripheral?** Look for Upper Motor Neuron (UMN) signs (spasticity, hyperreflexia, extensor plantar) versus Lower Motor Neuron (LMN) signs (flaccidity, wasting, fasciculations, areflexia).
- **Question 3: Is there an anatomical "Level"?** Cortical signs → Cortex; Crossed face-body signs → Brainstem; Sharp sensory level → Spinal cord; Dermatomal distribution → Spinal nerve root; Named nerve territory → Peripheral nerve.
- **Question 4: Which side is the lesion on?** Cerebral hemisphere → Contralateral body; Cerebellar hemisphere → Ipsilateral limbs; Brainstem → Ipsilateral cranial nerve + contralateral body; Hemicord → Ipsilateral weakness/proprioception + contralateral pain/temperature loss.
- **Question 5: Which neighbouring structures must lie together?** *The essence of localisation.* Determine the unique spatial junction where all affected tracts and cranial nerve nuclei converge.

3. The 13 Golden Rules of Neuro-Localisation

"Don't memorise 100 syndromes. Learn about 15 anatomical rules. Then every case becomes a combination of those rules."

1. **Cortical signs** (aphasia, neglect, seizure, agnosia) → **Cerebral Cortex**
2. **Same-side face + body sensory loss → Above Brainstem / Thalamocortical**
3. **Crossed face (one side) + body (opposite) → Brainstem lesion**
4. **Cranial nerve palsy + opposite hemiparesis → Brainstem crossed syndrome**
5. **Sensory level across trunk → Spinal Cord**
6. **LMN signs at level + UMN signs below → Segmental Spinal Cord**
7. **Symmetric saddle anesthesia + early bladder → Conus Medullaris**
8. **Asymmetric painful flaccid leg weakness → Cauda Equina (Roots)**
9. **Dermatomal sensory + myotomal motor → Spinal Nerve Root**
10. **Isolated named peripheral nerve territory → Peripheral Nerve**
11. **Same-side limb dysmetria & ataxia → Cerebellar Hemisphere**
12. **Impaired proprioception + positive Romberg → Dorsal Column / Large Fiber**
13. **Dense pure motor/sensory without cortical signs → Deep Hemisphere (Internal Capsule / Thalamus)**

4. Peter Gates' Brainstem Rule of 4s

Master brainstem crossed stroke syndromes in seconds using the 4 Medial ("M") vs 4 Lateral ("S") structural rule:

Rule Component	Midline (Medial) Structures — "4 M's"	Side (Lateral) Structures — "4 S's"
1. Cranial Nerves	Divisible into 12 (except 1 & 2): **CN III, IV, VI, XII** (Motor exit midline)	Non-divisible into 12: **CN V, VII, VIII, IX, X, XI** (Exit laterally)
2. Motor / Cerebellar	**Motor pathway:** Corticospinal tract *Deficit:* Contralateral hemiparesis	**Spinocerebellar pathway:** ICP / MCP *Deficit:* Ipsilateral limb ataxia
3. Sensory Lemniscus	**Medial lemniscus:** Dorsal column relay *Deficit:* Contralateral vibration/proprioception loss	**Spinothalamic tract:** Anterolateral system *Deficit:* Contralateral pain/temperature loss
4. Autonomic / Cranial	**Medial Longitudinal Fasciculus (MLF):** *Deficit:* INO (impaired adduction on lateral gaze)	**Sympathetic descending:** Horner syndrome **Spinal V:** Ipsilateral facial pain/temp loss

Level Geography: 4 above pons (CN I & II in cerebrum; III & IV in midbrain), 4 in pons (CN V, VI, VII, VIII), 4 in medulla (CN IX, X, XI, XII).

5. Upper Motor Neuron (UMN) vs Lower Motor Neuron (LMN) Matrix

Clinical Sign	Upper Motor Neuron (UMN) Lesion	Lower Motor Neuron (LMN) Lesion
Weakness Distribution	Pyramidal pattern (Extensors > Flexors in arms; Flexors > Extensors in legs)	Focal segmental (isolated peripheral nerve, root, or anterior horn cell pattern)
Muscle Tone	Spasticity (velocity-dependent clasp-knife rigidity)	Flaccidity / Hypotonia
Deep Tendon Reflexes	Hyperreflexia, clonus, spread of reflex zones	Hyporeflexia or complete areflexia
Plantar Response	Extensor (Babinski sign positive), Oppenheim / Chaddock signs	Flexor or absent
Muscle Bulk & Twitches	Disuse atrophy only (slow, mild); Fasciculations absent	Neurogenic wasting (early, severe); Fasciculations frequent

Spinal Shock Pitfall: Acute transverse myelopathy or sudden cord transaction may initially present with flaccidity, areflexia, and absent Babinski for days to weeks before spasticity emerges.

6. The 3-Question Aphasia Decision Flowchart

Assess language systematically: **(1) Spontaneous Fluency → (2) Auditory Comprehension → (3) Repetition**.

Aphasia Type	Fluency	Comprehension	Repetition	Classic Anatomical Localization
Broca	Nonfluent	Preserved	**Impaired**	Left inferior frontal gyrus (pars triangularis/opercularis, Brodmann 44/45)
Transcortical Motor	Nonfluent	Preserved	**Preserved**	Left dorsolateral prefrontal / SMA watershed area anterior to Broca
Wernicke	Fluent	**Impaired**	**Impaired**	Left posterior superior temporal gyrus (Brodmann 22)
Transcortical Sensory	Fluent	**Impaired**	**Preserved**	Left temporo-parieto-occipital junction (sparing Wernicke area)
Conduction	Fluent	Preserved	**Markedly Impaired**	Left arcuate fasciculus / supramarginal gyrus (Brodmann 40)
Global	Nonfluent	**Impaired**	**Impaired**	Entire left perisylvian language zone (combined MCA superior + inferior)
Mixed Transcortical	Nonfluent	**Impaired**	**Preserved**	Isolation of perisylvian language arc (bilateral watershed infarcts)
Anomic	Fluent	Preserved	Preserved	Dominant angular gyrus or anterior/inferior temporal lobe

7. Spinal Cord Syndromes & Cross-Sectional Anatomy

Cord Syndrome	Motor Deficits	Dorsal Columns (Vibration/JPS)	Spinothalamic (Pain/Temp)	Autonomic / Sphincters
Brown-Séquard	Ipsilateral UMN weakness below	Ipsilateral loss below	Contralateral loss 1-2 levels below	Usually spared
Anterior Cord	Bilateral paraplegia / quadriplegia	**Intact (Preserved)**	Bilateral loss below level	Acute urinary retention
Central Cord (Syrinx)	Hand intrinsic wasting (LMN) → leg spasticity	Preserved until late	Dissociated "cape-like" loss over arms/trunk	Late bladder dysfunction
Subacute Combined	Bilateral spastic paraparesis	Severe sensory ataxia, Romberg (+)	Usually spared or distal neuropathy	Late sphincter urgency

8. Peripheral Localisation: Root vs Plexus vs Peripheral Nerve

Diagnostic Challenge	Peripheral Nerve Entrapment	Radiculopathy (Spinal Nerve Root)	Plexopathy / Trunk Lesion
Foot Drop	**Common Peroneal:** Weak dorsiflexion & eversion. *Normal inversion (tibialis posterior) & normal hip abduction*. Normal ankle jerk.	**L5 Radiculopathy:** Weak dorsiflexion, eversion, *AND weak inversion + weak hip abduction (gluteus medius)*. Back / radicular pain.	**Lumbosacral Plexus / Sciatic:** Weak dorsiflexion, plantarflexion, inversion, eversion + weak hamstrings + *lost ankle jerk*.
Hand Wasting	**Ulnar Nerve:** Claw hand, interossei weak, spared thenar (APB). Medial 1.5 digit numbness. **Median Nerve (CTS):** Thenar wasting (APB weak), spared interossei. Palmar digits 1-3.5 numbness.	**C8 Radiculopathy:** Weakness crosses median + ulnar intrinsics (APB + interossei weak). Medial hand & forearm numbness. Triceps normal.	**Thoracic Outlet (Lower Trunk):** Severe thenar wasting > hypothenar (T1 motor) + medial forearm numbness (C8 sensory). Normal paraspinal EMG.
Shoulder Weakness	**Axillary Nerve:** Isolated deltoid + teres minor weakness; sensory badge over lateral shoulder. **Long Thoracic:** Winged scapula (serratus anterior).	**C5/C6 Radiculopathy:** Weak deltoid, biceps, brachioradialis + reduced biceps reflex + sensory loss to thumb.	**Erb's Palsy (Upper Trunk C5-C6):** "Waiter's tip" arm (adducted, internally rotated, pronated). Biceps + supinator lost.

9. Cranio-Ocular Motility & Pupil Localisation Rules

- **Pupil-Involving vs Pupil-Sparing CN III Palsy:**
  • *Pupil-Involving (Dilated, Fixed Pupil):* Surgical compression (PComm aneurysm, uncal herniation). Parasympathetic pupillomotor fibers ride on the superficial outer mantle of CN III.
  • *Pupil-Sparing (Reactive Pupil):* Microvascular ischemic infarction (diabetes, hypertension). Deep somatic motor core infarcted; peripheral pial blood supply preserves pupillary fibers.
- **Internuclear Ophthalmoplegia (INO):** Lesion of ipsilateral MLF. Contralateral horizontal gaze causes failure of ipsilateral eye adduction + nystagmus of contralateral abducting eye. *Convergence is intact* (midbrain circuit).
- **One-and-a-Half Syndrome:** Ipsilateral PPRF / CN VI nucleus + ipsilateral MLF. Complete conjugate gaze palsy to the lesion side + adduction failure to opposite side.
- **WEBINO:** Wall-Eyed Bilateral INO due to bilateral MLF lesions → bilateral adduction failure with primary position exotropia.
- **Conjugate Gaze: Frontal Eye Field (FEF) vs Pontine (PPRF):**
  • *Frontal Eye Field (FEF):* Eyes look *toward the hemispheric lesion* (away from hemiparesis). Overcome with oculocephalic maneuver (doll's eyes).
  • *Pontine PPRF / CN VI Nucleus:* Eyes look *away from the pontine lesion* (toward the hemiparesis). Cannot be overcome with doll's eye maneuvers.

CURRICULUM CHAPTER 1 • STAGE 1

Stage 1 — Gross Localisation: Central vs Peripheral (Brain, Cord, or Nerve?)

Chapter Learning Objective

Establish fundamental triage: Central (UMN) vs Peripheral (LMN), Brain vs Spinal Cord, Cord vs Roots without eponyms.

Pedagogical Localisation Anchor

Finding anchors: Cortical signs -> Cortex; Crossed signs -> Brainstem; Bilateral legs + sensory level -> Cord; Root/nerve LMN -> Peripheral.

Included Clinical Vignettes (12 Cases):
Case 6: Proportional Face, Arm, and Leg Weakness Without Cortical Signs • Case 7: Pure Hemi-Body Sensory Loss Sparing Motor Strength • Case 41: Dense Leg Weakness with Minimal Arm Involvement and Urinary Incontinence • Case 48: Bilateral Lower Extremity Weakness with a Sharp Sensory Level on Trunk • Case 65: Wrist and Finger Drop After Sleeping with Arm Compressed Over a Chair • Case 66: Foot Drop with Inability to Evert the Foot but Preserved Inversion • *...and 6 more cases*

6 Proportional Face, Arm, and Leg Weakness Without Cortical Signs

Adult patient • Stage 1: Central vs Peripheral • #1

Clinical Vignette: A 68-year-old man with long-standing hypertension wakes with weakness of the entire right side but has no difficulty speaking or understanding.

- Right UMN facial weakness
- Right arm power 3/5
- Right leg power 3/5
- Brisk right-sided reflexes
- Right extensor plantar response
- No aphasia
- No neglect
- No visual field defect
- Normal sensation

Q1: CENTRAL OR PERIPHERAL?
Central (Deep subcortical / thalamocortical / internal capsule — dense deficits with absent cortical signs).

Q2: NEURAXIS LEVEL
Left internal capsule

Q3: STRUCTURES & TRACTS INVOLVED
A dense proportional face-arm-leg weakness without cortical signs strongly suggests compact corticospinal and corticobulbar fibres in the internal capsule.

Q4: DEFINITIVE DIAGNOSIS & SYNDROME
pure motor lacunar syndrome

Teaching Pearl: A dense proportional face-arm-leg weakness without cortical signs strongly suggests compact corticospinal and corticobulbar fibres in the internal capsule.

Anatomical Mechanism & Discriminators: A dense proportional face-arm-leg weakness without cortical signs strongly suggests compact corticospinal and corticobulbar fibres in the internal capsule. — *Anchor: Left internal capsule*

7 Pure Hemi-Body Sensory Loss Sparing Motor Strength

60-year-old woman suddenly • Stage 1: Central vs Peripheral • #2

Clinical Vignette: A 60-year-old woman suddenly develops numbness affecting the entire left half of her body, including her face.

- Reduced pinprick and temperature over left face
- Similar sensory loss in left arm, trunk and leg
- Normal power
- Normal reflexes
- No aphasia
- No neglect

Q1: CENTRAL OR PERIPHERAL?
Central (Deep subcortical / thalamocortical / internal capsule — dense deficits with absent cortical signs).

Q2: NEURAXIS LEVEL
Right thalamus

Q3: STRUCTURES & TRACTS INVOLVED
Face and body sensory loss together without weakness or cortical signs suggests involvement of the thalamic VPM and VPL sensory relay nuclei.

Q4: DEFINITIVE DIAGNOSIS & SYNDROME
Right thalamus

Teaching Pearl: Face and body sensory loss together without weakness or cortical signs suggests involvement of the thalamic VPM and VPL sensory relay nuclei.

Anatomical Mechanism & Discriminators: Face and body sensory loss together without weakness or cortical signs suggests involvement of the thalamic VPM and VPL sensory relay nuclei. — *Anchor: Right thalamus*

41 Dense Leg Weakness with Minimal Arm Involvement and Urinary Incontinence

68-year-old man suddenly • Stage 1: Central vs Peripheral • #3

Clinical Vignette: A 68-year-old man suddenly develops weakness of the left leg. His family also notices that he has become unusually quiet and has urinary urgency.

- Left leg power 2/5
- Left arm power 4+/5
- Left lower-limb hyperreflexia
- Left extensor plantar response
- Grasp reflex present
- Reduced spontaneous speech and initiative, without language impairment
- Urinary incontinence

Q1: CENTRAL OR PERIPHERAL?

Central (Cerebral cortex — presence of cortical signs such as aphasia, neglect, apraxia, or visual field deficits).

Q2: NEURAXIS LEVEL

Right medial frontal lobe — ACA territory

Q3: STRUCTURES & TRACTS INVOLVED

The leg area of the motor cortex lies on the medial cerebral hemisphere.

Q4: DEFINITIVE DIAGNOSIS & SYNDROME

Anterior cerebral artery stroke

Teaching Pearl: The leg area of the motor cortex lies on the medial cerebral hemisphere. ACA lesions therefore cause leg-predominant weakness. Medial frontal involvement can also produce abulia, frontal release signs and bladder dysfunction.

Anatomical Mechanism & Discriminators: The leg area of the motor cortex lies on the medial cerebral hemisphere. ACA lesions therefore cause leg-predominant weakness. Medial frontal involvement can also produce abulia, frontal release signs and bladder dysfunction. — *Anchor: Right medial frontal lobe — ACA territory*

48 Bilateral Lower Extremity Weakness with a Sharp Sensory Level on Trunk

49-year-old man • Stage 1: Central vs Peripheral • #4

Clinical Vignette: A 49-year-old man develops progressive heaviness of both legs over several weeks, followed by urinary urgency.

- Spastic paraparesis
- Brisk knee and ankle reflexes
- Bilateral extensor plantar responses
- Upper limbs normal
- Reduced pinprick sensation below the nipple line
- Vibration impaired in legs
- Bladder urgency

Q1: CENTRAL OR PERIPHERAL?

Central (Spinal cord myelopathy — UMN signs below lesion + sensory level ± LMN signs at level).

Q2: NEURAXIS LEVEL

Thoracic spinal cord around T4

Q3: STRUCTURES & TRACTS INVOLVED

Bilateral UMN leg weakness plus a truncal sensory level strongly localises to the spinal cord.

Q4: DEFINITIVE DIAGNOSIS & SYNDROME

Thoracic myelopathy

Teaching Pearl: Bilateral UMN leg weakness plus a truncal sensory level strongly localises to the spinal cord. The nipple line roughly corresponds to the T4 dermatome, although the anatomical cord lesion may lie somewhat above the clinically identified sensory level.

Anatomical Mechanism & Discriminators: Bilateral UMN leg weakness plus a truncal sensory level strongly localises to the spinal cord. The nipple line roughly corresponds to the T4 dermatome, although the anatomical cord lesion may lie somewhat above the clinically identified sensory level. — *Anchor: Thoracic spinal cord around T4*

65 Wrist and Finger Drop After Sleeping with Arm Compressed Over a Chair

Adult patient • Stage 1: Central vs Peripheral • #5

Clinical Vignette: A 35-year-old man falls asleep with his right arm draped over the back of a chair after drinking heavily. He wakes unable to straighten his wrist or fingers.

- Right wrist drop
- Weak extension of fingers and thumb
- Weak thumb abduction
- Triceps strength preserved
- Brachioradialis preserved
- Small patch of sensory loss over the first dorsal web space
- Intrinsic hand muscles normal

Q1: CENTRAL OR PERIPHERAL?

Peripheral (Lower motor neuron / peripheral nerve / plexus / root distribution).

Q2: NEURAXIS LEVEL

Right radial nerve at the spiral groove of the humerus

Q3: STRUCTURES & TRACTS INVOLVED

Compression at the spiral groove spares the triceps and brachioradialis, whose branches leave the nerve more proximally.

Q4: DEFINITIVE DIAGNOSIS & SYNDROME

Saturday night palsy — radial nerve compression

Teaching Pearl: Compression at the spiral groove spares the triceps and brachioradialis, whose branches leave the nerve more proximally. The result is wrist and finger extension weakness with only a small area of dorsal hand sensory loss.

Anatomical Mechanism & Discriminators: Compression at the spiral groove spares the triceps and brachioradialis, whose branches leave the nerve more proximally. The result is wrist and finger extension weakness with only a small area of dorsal hand sensory loss. — *Anchor: Right radial nerve at the spiral groove of the humerus*

66 Foot Drop with Inability to Evert the Foot but Preserved Inversion

48-year-old woman • Stage 1: Central vs Peripheral • #6

Clinical Vignette: A 48-year-old woman develops a right foot drop after several hours of squatting while gardening. She has no back pain.

- Weak dorsiflexion of the right foot
- Weak eversion
- Inversion preserved
- Plantarflexion preserved
- High-stepping gait
- Sensory loss over the lateral leg and dorsum of the foot
- Ankle jerk normal
- No back tenderness or radicular pain

Q1: CENTRAL OR PERIPHERAL?

Peripheral (Lower motor neuron / peripheral nerve / plexus / root distribution).

Q2: NEURAXIS LEVEL

Right common peroneal (fibular) nerve at the fibular head

Q3: STRUCTURES & TRACTS INVOLVED

Preserved inversion (tibialis posterior — tibial nerve, L5) and a normal ankle jerk separate a peroneal nerve lesion at the fibular head from an L5 radiculopathy.

Q4: DEFINITIVE DIAGNOSIS & SYNDROME

Common peroneal mononeuropathy

Teaching Pearl: Preserved inversion (tibialis posterior — tibial nerve, L5) and a normal ankle jerk separate a peroneal nerve lesion at the fibular head from an L5 radiculopathy. Compression at the fibular head after squatting, leg crossing or weight loss is the usual mechanism.

Anatomical Mechanism & Discriminators: Preserved inversion (tibialis posterior — tibial nerve, L5) and a normal ankle jerk separate a peroneal nerve lesion at the fibular head from an L5 radiculopathy. Compression at the fibular head after squatting, leg crossing or weight loss is the usual mechanism. — *Anchor: Right common peroneal (fibular) nerve at the fibular head*

67 Foot Drop Accompanied by Weak Foot Inversion and Pain Radiating Down Leg

44-year-old man • Stage 1: Central vs Peripheral • #7

Clinical Vignette: A 44-year-old man develops acute low-back pain radiating down the lateral leg to the top of the foot after lifting a heavy box. He notices the foot slaps when he walks.

- Weak dorsiflexion and eversion
- Weak inversion
- Weak hip abduction on the right
- Sensory loss over the lateral leg and dorsum of the foot
- Ankle jerk preserved
- Straight-leg-raise test positive
- Lower-limb reflexes otherwise normal

Q1: CENTRAL OR PERIPHERAL?

Peripheral (Lower motor neuron / peripheral nerve / plexus / root distribution).

Q2: NEURAXIS LEVEL

Right L5 nerve root

Q3: STRUCTURES & TRACTS INVOLVED

Unlike a peroneal palsy, an L5 root lesion also weakens inversion and hip abduction, because those muscles receive L5 fibres through other nerves.

Q4: DEFINITIVE DIAGNOSIS & SYNDROME

L5 radiculopathy — lumbosacral disc herniation

Teaching Pearl: Unlike a peroneal palsy, an L5 root lesion also weakens inversion and hip abduction, because those muscles receive L5 fibres through other nerves. Radicular pain, a positive straight-leg raise and the L5 dermatomal sensory loss confirm root rather than peripheral nerve involvement.

Anatomical Mechanism & Discriminators: Unlike a peroneal palsy, an L5 root lesion also weakens inversion and hip abduction, because those muscles receive L5 fibres through other nerves. Radicular pain, a positive straight-leg raise and the L5 dermatomal sensory loss confirm root rather than peripheral nerve involvement. — *Anchor: Right L5 nerve root*

20 Severe Low Back Pain with Perineal Numbness and Incontinence

Adult patient • Stage 1: Central vs Peripheral • #8

Clinical Vignette: A 52-year-old man with severe low-back pain develops progressive weakness of both legs, numbness around the perineum and inability to pass urine.

- Bilateral distal leg weakness, markedly asymmetric
- Reduced knee reflexes
- Absent ankle jerks
- Flexor plantar responses
- Saddle anesthesia
- Reduced anal tone
- Urinary retention

Q1: CENTRAL OR PERIPHERAL?

Peripheral (Nerve root / Cauda equina level — Lower Motor Neuron signs predominate).

Q2: NEURAXIS LEVEL

Cauda equina

Q3: STRUCTURES & TRACTS INVOLVED

The cauda equina is composed of lumbosacral nerve roots, so deficits are predominantly LMN, often asymmetric, with saddle anesthesia and sphincter dysfunction.

Q4: DEFINITIVE DIAGNOSIS & SYNDROME

Cauda equina

Teaching Pearl: The cauda equina is composed of lumbosacral nerve roots, so deficits are predominantly LMN, often asymmetric, with saddle anesthesia and sphincter dysfunction.

Anatomical Mechanism & Discriminators: The cauda equina is composed of lumbosacral nerve roots, so deficits are predominantly LMN, often asymmetric, with saddle anesthesia and sphincter dysfunction. — *Anchor: Cauda equina*

49 Early Bladder Dysfunction and Saddle Sensory Loss with Preserved Knee Jerks

58-year-old man • Stage 1: Central vs Peripheral • #9

Clinical Vignette: A 58-year-old man develops severe low-back pain followed by numbness around the anus and rapidly progressive urinary retention.

- Saddle anesthesia
- Markedly reduced anal tone
- Urinary retention
- Bilateral leg weakness, relatively symmetric
- Ankle jerks reduced
- Knee reflexes may be preserved
- Little radicular pain compared with neurological deficit

Q1: CENTRAL OR PERIPHERAL?
Central / Distal Medullary Cord (Symmetric saddle sensory loss and early autonomic/sphincter dysfunction).

Q2: NEURAXIS LEVEL
Conus medullaris

Q3: STRUCTURES & TRACTS INVOLVED
Conus lesions tend to produce: early bladder and bowel dysfunction prominent saddle anesthesia relatively symmetric deficits Cauda equina lesions are usually more asymmetric and commonly produce marked radicular pain.

Q4: DEFINITIVE DIAGNOSIS & SYNDROME
Conus medullaris syndrome

Teaching Pearl: Conus lesions tend to produce: early bladder and bowel dysfunction prominent saddle anesthesia relatively symmetric deficits Cauda equina lesions are usually more asymmetric and commonly produce marked radicular pain.

Anatomical Mechanism & Discriminators: Conus lesions tend to produce: early bladder and bowel dysfunction prominent saddle anesthesia relatively symmetric deficits Cauda equina lesions are usually more asymmetric and commonly produce marked radicular pain. *— Anchor: Conus medullaris*

90 Foot Drop with Spared Foot Inversion and Preserved Hip Abduction

26-year-old man falls asleep with his legs tightly crossed on a hard chair and • Stage 1: Central vs Peripheral • #10

Clinical Vignette: A 26-year-old man falls asleep with his legs tightly crossed on a hard chair and wakes with an inability to lift his right foot.

- Weakness of right ankle dorsiflexion (power 1/5)
- Weakness of right foot eversion (power 2/5)
- Preserved strength of foot inversion (tibialis posterior power 5/5)
- Preserved strength of hip abduction (gluteus medius power 5/5)
- Sensory loss over the anterolateral lower leg and dorsum of the right foot
- Normal patellar and Achilles reflexes

Q1: CENTRAL OR PERIPHERAL?
Peripheral (Lower motor neuron / peripheral nerve / plexus / root distribution).

Q2: NEURAXIS LEVEL
Right common peroneal (fibular) nerve at the fibular neck

Q3: STRUCTURES & TRACTS INVOLVED
The common peroneal nerve divides at the fibular head into superficial (foot eversion) and deep (dorsiflexion) branches.

Q4: DEFINITIVE DIAGNOSIS & SYNDROME
Common peroneal nerve palsy

Teaching Pearl: The common peroneal nerve divides at the fibular head into superficial (foot eversion) and deep (dorsiflexion) branches. Spared foot inversion (tibialis posterior / tibial nerve / L5) and hip abduction (gluteus medius / superior gluteal nerve / L5) rule out an L5 radiculopathy or lumbosacral plexopathy.

Anatomical Mechanism & Discriminators: The common peroneal nerve divides at the fibular head into superficial (foot eversion) and deep (dorsiflexion) branches. Spared foot inversion (tibialis posterior / tibial nerve / L5) and hip abduction (gluteus medius / superior gluteal nerve / L5) rule out an L5 radiculopathy or lumbosacral plexopathy. *— Anchor: Right common peroneal (fibular) nerve at the fibular neck*

93 Nocturnal Hand Paresthesias and Weakness of Thumb Abduction

46-year-old woman • Stage 1: Central vs Peripheral • #11

Clinical Vignette: A 46-year-old woman wakes with burning pain and tingling in her right thumb, index and middle fingers. She shakes the hand for relief.

- Wasting of the right thenar eminence
- Weak thumb abduction and opposition
- Numbness over the thumb, index, middle and radial half of the ring finger
- Tinel sign over the wrist
- Phalen manoeuvre reproduces the symptoms
- Hypothenar muscles normal

Q1: CENTRAL OR PERIPHERAL?
Peripheral (Lower motor neuron / peripheral nerve / plexus / root distribution).

Q2: NEURAXIS LEVEL
Right median nerve at the carpal tunnel

Q3: STRUCTURES & TRACTS INVOLVED
Median nerve compression at the wrist spares the forearm flexors (which branch above the wrist) while affecting the thenar muscles and the radial 3.

Q4: DEFINITIVE DIAGNOSIS & SYNDROME
Carpal tunnel syndrome

Teaching Pearl: Median nerve compression at the wrist spares the forearm flexors (which branch above the wrist) while affecting the thenar muscles and the radial 3.5 digits. Sparing of the hypothenar eminence separates this from an ulnar nerve lesion.

Anatomical Mechanism & Discriminators: Median nerve compression at the wrist spares the forearm flexors (which branch above the wrist) while affecting the thenar muscles and the radial 3.5 digits. Sparing of the hypothenar eminence separates this from an ulnar nerve lesion. — *Anchor: Right median nerve at the carpal tunnel*

94 Claw Hand Deformity with Hypothenar Wasting and Ulnar Border Numbness

38-year-old man • Stage 1: Central vs Peripheral • #12

Clinical Vignette: A 38-year-old man develops wasting of the right hand and numbness of the little and ring fingers after repeatedly leaning on his elbow.

- Wasting of the hypothenar and interosseous muscles
- Weak finger abduction and adduction
- Clawing of the ring and little fingers
- Froment sign positive
- Numbness over the little finger and ulnar half of the ring finger
- Tinel sign behind the medial epicondyle

Q1: CENTRAL OR PERIPHERAL?
Peripheral (Lower motor neuron / peripheral nerve / plexus / root distribution).

Q2: NEURAXIS LEVEL
Right ulnar nerve at the elbow (cubital tunnel)

Q3: STRUCTURES & TRACTS INVOLVED
The ulnar nerve supplies all intrinsic hand muscles except the radial two lumbricals and the thenar group.

Q4: DEFINITIVE DIAGNOSIS & SYNDROME
Ulnar neuropathy at the elbow

Teaching Pearl: The ulnar nerve supplies all intrinsic hand muscles except the radial two lumbricals and the thenar group. Clawing (metacarpophalangeal hyperextension with interphalangeal flexion) of the ring and little fingers plus ulnar-border sensory loss localises to the ulnar nerve.

Anatomical Mechanism & Discriminators: The ulnar nerve supplies all intrinsic hand muscles except the radial two lumbricals and the thenar group. Clawing (metacarpophalangeal hyperextension with interphalangeal flexion) of the ring and little fingers plus ulnar-border sensory loss localises to the ulnar nerve. — *Anchor: Right ulnar nerve at the elbow (cubital tunnel)*

CURRICULUM CHAPTER 2 • STAGE 2

Stage 2 — The 3 Major Long Tracts: Motor & Sensory Longitudinal Systems

Chapter Learning Objective

Master the 3 great highways: Corticospinal (UMN weakness), Dorsal Columns (proprioception/vibration/Romberg), and Spinothalamic (pain/temp).

Pedagogical Localisation Anchor

Cord localisation = which longitudinal tracts + which spinal segments are involved?

Included Clinical Vignettes (10 Cases):
Case 18: Penetrating Spinal Trauma with Asymmetric Motor and Sensory Deficits • Case 19: Sudden Paraplegia and Dissociated Sensory Loss Following Aortic Surgery • Case 118: Sudden Loss of Proprioception and Vibration in Both Legs with Sparing of Pain Sensation • Case 16: Bilateral Dissociated Sensory Loss Across Shoulders and Upper Chest • Case 60: Spastic Leg Weakness Combined with Absent Ankle Jerks and Impaired Vibration • Case 89: Sensory Ataxia, Spastic Paraparesis, Babinski Signs, and Absent Ankle Jerks • *...and 4 more cases*

18 Penetrating Spinal Trauma with Asymmetric Motor and Sensory Deficits

29-year-old man • Stage 2: Major Long Tracts • #1

Clinical Vignette: A 29-year-old man sustains a penetrating injury to the right side of his lower thoracic spine.

- Right leg weak and spastic
- Right knee and ankle reflexes brisk
- Right extensor plantar response
- Loss of vibration and joint position in right leg
- Loss of pain and temperature in left leg beginning a few segments below the lesion

Q1: CENTRAL OR PERIPHERAL?
Central (Spinal cord myelopathy — UMN signs below lesion + sensory level ± LMN signs at level).

Q2: NEURAXIS LEVEL
Right spinal cord hemisection

Q3: STRUCTURES & TRACTS INVOLVED
A hemicord lesion causes ipsilateral corticospinal and dorsal-column deficits with contralateral spinothalamic loss.

Q4: DEFINITIVE DIAGNOSIS & SYNDROME
Brown-Séquard syndrome

Teaching Pearl: A hemicord lesion causes ipsilateral corticospinal and dorsal-column deficits with contralateral spinothalamic loss.

Anatomical Mechanism & Discriminators: A hemicord lesion causes ipsilateral corticospinal and dorsal-column deficits with contralateral spinothalamic loss. — *Anchor: Right spinal cord hemisection*

19 Sudden Paraplegia and Dissociated Sensory Loss Following Aortic Surgery

67-year-old man • Stage 2: Major Long Tracts • #2

Clinical Vignette: A 67-year-old man develops sudden paraplegia and urinary retention shortly after major aortic surgery.

- Bilateral leg paralysis
- Initially flaccid legs, later hyperreflexia
- Bilateral extensor plantar responses
- Loss of pain and temperature below T10
- Vibration and joint-position sense preserved
- Bladder dysfunction

Q1: CENTRAL OR PERIPHERAL?
Central (Spinal cord myelopathy — UMN signs below lesion + sensory level ± LMN signs at level).

Q2: NEURAXIS LEVEL
Anterior two-thirds of spinal cord

Q3: STRUCTURES & TRACTS INVOLVED
The corticospinal and spinothalamic pathways are damaged while the posterior columns remain relatively spared.

Q4: DEFINITIVE DIAGNOSIS & SYNDROME
anterior spinal artery syndrome

Teaching Pearl: The corticospinal and spinothalamic pathways are damaged while the posterior columns remain relatively spared.

Anatomical Mechanism & Discriminators: The corticospinal and spinothalamic pathways are damaged while the posterior columns remain relatively spared. — *Anchor: Anterior two-thirds of spinal cord*

118 Sudden Loss of Proprioception and Vibration in Both Legs with Sparing of Pain Sensation

65-year-old man • Stage 2: Major Long Tracts • #3

Clinical Vignette: A 65-year-old man develops sudden severe unsteadiness after an episode of profound hypotension.

- Marked loss of vibration below T8
- Loss of joint-position sense in both legs
- Positive Romberg sign
- Severe sensory ataxia
- Pinprick and temperature relatively preserved
- Power essentially normal
- Plantar responses flexor
- No saddle anesthesia

Q1: CENTRAL OR PERIPHERAL?
Central (Spinal cord myelopathy — UMN signs below lesion + sensory level ± LMN signs at level).

Q2: NEURAXIS LEVEL
Posterior spinal cord / dorsal columns

Q3: STRUCTURES & TRACTS INVOLVED
The opposite pattern to anterior spinal artery syndrome is seen.

Q4: DEFINITIVE DIAGNOSIS & SYNDROME
Posterior spinal artery syndrome

Teaching Pearl: The opposite pattern to anterior spinal artery syndrome is seen. The posterior columns are preferentially damaged, producing loss of: vibration proprioception discriminative touch Pain and temperature pathways remain relatively intact.

Anatomical Mechanism & Discriminators: The opposite pattern to anterior spinal artery syndrome is seen. The posterior columns are preferentially damaged, producing loss of: vibration proprioception discriminative touch Pain and temperature pathways remain relatively intact. — *Anchor: Posterior spinal cord / dorsal columns*

16 Bilateral Dissociated Sensory Loss Across Shoulders and Upper Chest

38-year-old man • Stage 2: Major Long Tracts • #4

Clinical Vignette: A 38-year-old man has gradually progressive hand weakness and frequently burns his hands without realizing it.

- Loss of pain and temperature over both shoulders and upper arms
- Similar loss across upper chest
- Light touch preserved
- Vibration preserved
- Wasting of small muscles of both hands
- Reduced upper-limb reflexes
- Brisk knee reflexes

Q1: CENTRAL OR PERIPHERAL?
Central (Spinal cord myelopathy — UMN signs below lesion + sensory level ± LMN signs at level).

Q2: NEURAXIS LEVEL
Central cervical cord

Q3: STRUCTURES & TRACTS INVOLVED
A central cord cavity first damages crossing spinothalamic fibres, causing dissociated “cape-like” sensory loss.

Q4: DEFINITIVE DIAGNOSIS & SYNDROME
syringomyelia

Teaching Pearl: A central cord cavity first damages crossing spinothalamic fibres, causing dissociated “cape-like” sensory loss. Expansion may involve anterior horn cells and corticospinal tracts.

Anatomical Mechanism & Discriminators: A central cord cavity first damages crossing spinothalamic fibres, causing dissociated “cape-like” sensory loss. Expansion may involve anterior horn cells and corticospinal tracts. — *Anchor: Central cervical cord*

60 Spastic Leg Weakness Combined with Absent Ankle Jerks and Impaired Vibration

60-year-old woman on a strict vegan diet • Stage 2: Major Long Tracts • #5

Clinical Vignette: A 60-year-old woman on a strict vegan diet develops months of progressive leg stiffness, unsteadiness and tingling in her hands and feet.

- Spastic paraparesis
- Bilateral extensor plantar responses
- Absent ankle jerks; knee jerks variable
- Markedly impaired vibration and joint-position sense in the legs
- Positive Romberg sign
- Mild glove-and-stocking pinprick loss
- Macrocytic anaemia on blood film

Q1: CENTRAL OR PERIPHERAL?

Peripheral (Lower motor neuron / peripheral nerve / plexus / root distribution).

Q2: NEURAXIS LEVEL

Dorsal and lateral columns of the spinal cord, plus peripheral nerves

Q3: STRUCTURES & TRACTS INVOLVED

The diagnostic clue is the combination of upper motor neuron signs (spasticity, extensor plantars from corticospinal tract involvement) with absent reflexes (peripheral neuropathy) and dorsal column loss.

Q4: DEFINITIVE DIAGNOSIS & SYNDROME

Subacute combined degeneration — vitamin B12 deficiency

Teaching Pearl: The diagnostic clue is the combination of upper motor neuron signs (spasticity, extensor plantars from corticospinal tract involvement) with absent reflexes (peripheral neuropathy) and dorsal column loss. This pattern should always prompt a search for B12 deficiency.

Anatomical Mechanism & Discriminators: The diagnostic clue is the combination of upper motor neuron signs (spasticity, extensor plantars from corticospinal tract involvement) with absent reflexes (peripheral neuropathy) and dorsal column loss. This pattern should always prompt a search for B12 deficiency. — *Anchor: Dorsal and lateral columns of the spinal cord, plus peripheral nerves*

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Sensory Ataxia, Spastic Paraparesis, Babinski Signs, and Absent Ankle Jerks

64-year-old strict vegan • Stage 2: Major Long Tracts • #6

Clinical Vignette: A 64-year-old strict vegan presents with six months of progressive unsteadiness walking in the dark, tingling in both feet, and leg stiffness.

- Severe loss of vibration and joint-position sense extending up to the iliac crests
- Positive Romberg sign
- Spastic paraparesis with brisk knee jerks
- Absent ankle jerks bilaterally
- Bilateral extensor plantar responses (Babinski sign present)
- "Stocking-glove" pinprick sensory attenuation

Q1: CENTRAL OR PERIPHERAL?

Peripheral (Lower motor neuron / peripheral nerve / plexus / root distribution).

Q2: NEURAXIS LEVEL

Dorsal columns and lateral corticospinal tracts of the spinal cord with concurrent peripheral large-fiber neuropathy

Q3: STRUCTURES & TRACTS INVOLVED

Vitamin B12 deficiency selectively demyelinates the posterior columns (sensory ataxia, loss of vibration/proprioception) and lateral corticospinal tracts (spasticity, UMN signs), often coexisting with peripheral neuropathy (attenuated distal ankle jerks).

Q4: DEFINITIVE DIAGNOSIS & SYNDROME

Subacute combined degeneration (SCD) of the spinal cord (Vitamin B12 deficiency)

Teaching Pearl: Vitamin B12 deficiency selectively demyelinates the posterior columns (sensory ataxia, loss of vibration/proprioception) and lateral corticospinal tracts (spasticity, UMN signs), often coexisting with peripheral neuropathy (attenuated distal ankle jerks). 90 * Foot drop with preserved foot inversion

Anatomical Mechanism & Discriminators: Vitamin B12 deficiency selectively demyelinates the posterior columns (sensory ataxia, loss of vibration/proprioception) and lateral corticospinal tracts (spasticity, UMN signs), often coexisting with peripheral neuropathy (attenuated distal ankle jerks). 90 * Foot drop with preserved foot inversion — *Anchor: Dorsal columns and lateral corticospinal tracts of the spinal cord with concurrent peripheral large-fiber neuropathy*

59 Lightning Pains, Sensory Ataxia, and Unreactive Pupils to Light

Adult patient • Stage 2: Major Long Tracts • #7

Clinical Vignette: A 55-year-old man describes years of sudden stabbing "electric" pains in his legs. He has become unsteady in the dark and now has difficulty emptying his bladder.

- Severe loss of vibration and joint-position sense in the legs
- Positive Romberg sign
- Stamping, sensory-ataxic gait
- Absent knee and ankle jerks
- Pupils small, irregular, constricting to accommodation but not to light
- Urinary retention with overflow incontinence
- Power preserved

Q1: CENTRAL OR PERIPHERAL?
Central (Cerebral cortex — presence of cortical signs such as aphasia, neglect, apraxia, or visual field deficits).

Q2: NEURAXIS LEVEL
Dorsal columns and dorsal roots, with pretectal pupillary pathway involvement

Q3: STRUCTURES & TRACTS INVOLVED
Degeneration of the dorsal roots and columns produces lancinating pains, sensory ataxia, areflexia and bladder dysfunction.

Q4: DEFINITIVE DIAGNOSIS & SYNDROME
Tabes dorsalis — tertiary neurosyphilis

Teaching Pearl: Degeneration of the dorsal roots and columns produces lancinating pains, sensory ataxia, areflexia and bladder dysfunction. Argyll Robertson pupils — light-near dissociation — complete the classic picture.

Anatomical Mechanism & Discriminators: Degeneration of the dorsal roots and columns produces lancinating pains, sensory ataxia, areflexia and bladder dysfunction. Argyll Robertson pupils — light-near dissociation — complete the classic picture. — *Anchor: Dorsal columns and dorsal roots, with pretectal pupillary pathway involvement*

17 Unsteadiness in the Dark with Positive Romberg Sign

58-year-old man • Stage 2: Major Long Tracts • #8

Clinical Vignette: A 58-year-old man complains that he becomes very unsteady while walking in the dark and needs to watch his feet.

- Normal limb power
- Markedly reduced vibration at toes and ankles
- Impaired joint-position sense
- Pinprick relatively preserved
- Positive Romberg sign
- Stamping gait that worsens with eye closure

Q1: CENTRAL OR PERIPHERAL?
Central (Cerebral cortex — presence of cortical signs such as aphasia, neglect, apraxia, or visual field deficits).

Q2: NEURAXIS LEVEL
Posterior columns / large-fibre proprioceptive pathway

Q3: STRUCTURES & TRACTS INVOLVED
Loss of proprioception and vibration with preserved pain sensation indicates dorsal column dysfunction.

Q4: DEFINITIVE DIAGNOSIS & SYNDROME
Posterior columns / large-fibre proprioceptive pathway

Teaching Pearl: Loss of proprioception and vibration with preserved pain sensation indicates dorsal column dysfunction. Visual input compensates until the eyes are closed.

Anatomical Mechanism & Discriminators: Loss of proprioception and vibration with preserved pain sensation indicates dorsal column dysfunction. Visual input compensates until the eyes are closed. — *Anchor: Posterior columns / large-fibre proprioceptive pathway*

40 Bilateral Hand Weakness and Loss of Pain/Temperature with Normal Position Sense

45-year-old man • Stage 2: Major Long Tracts • #9

Clinical Vignette: A 45-year-old man develops sudden weakness of both hands after a hyperextension injury to the neck. Leg weakness is much less severe.

- Upper limbs weaker than lower limbs
- Hand weakness particularly marked
- Variable loss of pain and temperature below the lesion
- Sacral sensation relatively preserved
- Vibration and joint-position sense relatively preserved
- Legs mildly spastic after the acute phase
- Bladder dysfunction may occur

Q1: CENTRAL OR PERIPHERAL?

Central (Spinal cord myelopathy — UMN signs below lesion + sensory level ± LMN signs at level).

Q2: NEURAXIS LEVEL

Central cervical spinal cord

Q3: STRUCTURES & TRACTS INVOLVED

Central cervical cord injury classically produces disproportionately greater upper-limb than lower-limb weakness.

Q4: DEFINITIVE DIAGNOSIS & SYNDROME

Central cord syndrome

Teaching Pearl: Central cervical cord injury classically produces disproportionately greater upper-limb than lower-limb weakness. The central portions of the cord and crossing spinothalamic fibres are preferentially affected, while peripheral sacral fibres may be relatively spared. A typical setting is cervical hyperextension in a patient with pre-existing cervical canal narrowing. Here are 10 more localisation challenges, deliberately avoiding aphasia and diplopia. These focus on motor, sensory, visual, cerebellar, cranial nerve and spinal cord patterns.

Anatomical Mechanism & Discriminators: Central cervical cord injury classically produces disproportionately greater upper-limb than lower-limb weakness. The central portions of the cord and crossing spinothalamic fibres are preferentially affected, while peripheral sacral fibres may be relatively spared. A typical setting is cervical hyperextension in a patient with pre-existing cervical canal narrowing. Here are 10 more localisation challenges, deliberately avoiding aphasia and diplopia. These focus on motor, sensory, visual, cerebellar, cranial nerve and spinal cord patterns. — *Anchor: Central cervical spinal cord*

142 Ipsilateral Spastic Hemiparesis and Horner Syndrome with Contralateral Spinothalamic Loss

39-year-old man • Stage 2: Major Long Tracts • #10

Clinical Vignette: A 39-year-old man sustains a penetrating injury to the right lower cervical cord.

- Segmental weakness and wasting of right hand
- Right leg spastic weakness
- Right Babinski response
- Loss of vibration and position sense on right below lesion
- Loss of pain and temperature on left below lesion
- Right miosis and partial ptosis
- Facial sensation normal

Q1: CENTRAL OR PERIPHERAL?

Central (Spinal cord myelopathy — UMN signs below lesion + sensory level ± LMN signs at level).

Q2: NEURAXIS LEVEL

Right cervical hemicord with descending sympathetic fibres

Q3: STRUCTURES & TRACTS INVOLVED

This combines: ipsilateral corticospinal loss ipsilateral dorsal-column loss contralateral spinothalamic loss ipsilateral Horner syndrome Horner syndrome helps move the hemicord localisation up into the cervical region.

Q4: DEFINITIVE DIAGNOSIS & SYNDROME

Cervical Brown-Séquard syndrome with ipsilateral Horner syndrome

Teaching Pearl: This combines: ipsilateral corticospinal loss ipsilateral dorsal-column loss contralateral spinothalamic loss ipsilateral Horner syndrome Horner syndrome helps move the hemicord localisation up into the cervical region.

Anatomical Mechanism & Discriminators: This combines: ipsilateral corticospinal loss ipsilateral dorsal-column loss contralateral spinothalamic loss ipsilateral Horner syndrome Horner syndrome helps move the hemicord localisation up into the cervical region. — *Anchor: Right cervical hemicord with descending sympathetic fibres*

CURRICULUM CHAPTER 3 • STAGE 3

Stage 3 — Spinal Cord Localisation Vertically: Level & Segmental Syndromes

Chapter Learning Objective

Determine exact cranio-caudal level using LMN at level + UMN below + sensory level.

Pedagogical Localisation Anchor

Cervical -> arms +/- legs; Thoracic -> legs only, arms spared; Lumbar/Epiconus -> leg LMN/UMN mix.

Included Clinical Vignettes (11 Cases):
Case 15: Wasted Hands and Reduced Arm Reflexes with Spastic Legs • Case 116: Wasting and Weakness of Biceps with Hyperreflexia and Spasticity in Lower Limbs • Case 117: Acute Quadriparesis and Diaphragmatic Weakness with Normal Cranial Nerves • Case 138: Absent Biceps Reflex with Inverted Brachioradialis Reflex and Hyperactive Lower Limb Jerks • Case 115: Immediate Flaccid Areflexic Paraplegia and Atonic Bladder Following High-Impact Trauma • Case 121: Rapidly Evolving Paraplegia, Sharp Sensory Level at Mid-Thorax, and Bladder Retention • *...and 5 more cases*

15 Wasted Hands and Reduced Arm Reflexes with Spastic Legs

62-year-old man • Stage 3: Vertical Cord Level • #1

Clinical Vignette: A 62-year-old man has several months of progressive hand weakness, difficulty buttoning clothes and increasing stiffness while walking.

- Wasting of intrinsic hand muscles bilaterally
- Fasciculations in forearms
- Weak grip and finger abduction
- Reduced biceps and triceps reflexes
- Spastic lower limbs
- Brisk knee and ankle reflexes
- Bilateral extensor plantar responses

Q1: CENTRAL OR PERIPHERAL?

Central (Spinal cord myelopathy — UMN signs below lesion + sensory level ± LMN signs at level).

Q2: NEURAXIS LEVEL

Cervical spinal cord with segmental anterior horn/root involvement

Q3: STRUCTURES & TRACTS INVOLVED

LMN signs in the arms indicate involvement at the cervical level, while corticospinal tract damage produces UMN signs in the legs below the lesion.

Q4: DEFINITIVE DIAGNOSIS & SYNDROME

cervical myelopathy

Teaching Pearl: LMN signs in the arms indicate involvement at the cervical level, while corticospinal tract damage produces UMN signs in the legs below the lesion.

Anatomical Mechanism & Discriminators: LMN signs in the arms indicate involvement at the cervical level, while corticospinal tract damage produces UMN signs in the legs below the lesion. — *Anchor: Cervical spinal cord with segmental anterior horn/root involvement*

116 Wasting and Weakness of Biceps with Hyperreflexia and Spasticity in Lower Limbs

61-year-old man • Stage 3: Vertical Cord Level • #2

Clinical Vignette: A 61-year-old man develops progressive weakness lifting his arms along with stiffness of both legs.

- Deltoid and biceps weakness
- Wasting of biceps
- Biceps reflex reduced
- Triceps reflex relatively preserved
- Lower limbs spastic
- Brisk knee and ankle reflexes
- Bilateral extensor plantar responses
- Sensory impairment beginning around the lateral upper arm/upper trunk
- No cranial nerve abnormality

Q1: CENTRAL OR PERIPHERAL?

Central (Spinal cord myelopathy — UMN signs below lesion + sensory level ± LMN signs at level).

Q2: NEURAXIS LEVEL

Cervical spinal cord around C5 segment

Q3: STRUCTURES & TRACTS INVOLVED

At the level of the lesion there are LMN signs from anterior horn/root involvement.

Q4: DEFINITIVE DIAGNOSIS & SYNDROME

Segmental cervical myelopathy

Teaching Pearl: At the level of the lesion there are LMN signs from anterior horn/root involvement. Below the lesion there are UMN signs from corticospinal tract involvement. So: LMN arms + UMN legs = cervical cord until proven otherwise.

Anatomical Mechanism & Discriminators: At the level of the lesion there are LMN signs from anterior horn/root involvement. Below the lesion there are UMN signs from corticospinal tract involvement. So: LMN arms + UMN legs = cervical cord until proven otherwise. — *Anchor: Cervical spinal cord around C5 segment*

117 Acute Quadriparesis and Diaphragmatic Weakness with Normal Cranial Nerves

44-year-old man • Stage 3: Vertical Cord Level • #3

Clinical Vignette: A 44-year-old man develops rapidly progressive weakness of all four limbs followed by difficulty taking a deep breath.

- Spastic quadriparesis
- Brisk arm and leg reflexes
- Bilateral extensor plantar responses
- Sensory level at approximately the upper neck
- Weak diaphragmatic movement
- Reduced vital capacity
- Face completely normal
- Eye movements normal
- Tongue and palate normal

Q1: CENTRAL OR PERIPHERAL?
Central (Spinal cord myelopathy — UMN signs below lesion + sensory level ± LMN signs at level).

Q2: NEURAXIS LEVEL
High cervical spinal cord, approximately C3–C5

Q3: STRUCTURES & TRACTS INVOLVED
C3–C5 spinal segments contribute to the phrenic nerve.

Q4: DEFINITIVE DIAGNOSIS & SYNDROME
High cervical myelopathy

Teaching Pearl: C3–C5 spinal segments contribute to the phrenic nerve. High cervical lesions can therefore produce quadriparesis plus respiratory failure. Normal cranial nerves strongly favour cervical cord rather than brainstem disease.

Anatomical Mechanism & Discriminators: C3–C5 spinal segments contribute to the phrenic nerve. High cervical lesions can therefore produce quadriparesis plus respiratory failure. Normal cranial nerves strongly favour cervical cord rather than brainstem disease. — *Anchor: High cervical spinal cord, approximately C3–C5*

138 Absent Biceps Reflex with Inverted Brachioradialis Reflex and Hyperactive Lower Limb Jerks

63-year-old man • Stage 3: Vertical Cord Level • #4

Clinical Vignette: A 63-year-old man has progressive hand clumsiness and stiffness while walking.

- Mild weakness of shoulder abduction and elbow flexion
- Biceps reflex reduced
- Brachioradialis reflex produces paradoxical finger flexion
- Triceps reflex brisk
- Finger jerks brisk
- Knee jerks markedly brisk
- Bilateral Babinski signs
- Reduced vibration in feet

Q1: CENTRAL OR PERIPHERAL?
Central (Spinal cord myelopathy — UMN signs below lesion + sensory level ± LMN signs at level).

Q2: NEURAXIS LEVEL
Cervical cord around C5–C6

Q3: STRUCTURES & TRACTS INVOLVED
The lesion damages the C5–C6 reflex arc at its level while corticospinal tract involvement exaggerates reflex activity below it.

Q4: DEFINITIVE DIAGNOSIS & SYNDROME
C5–C6 cervical spondylotic myelopathy with inverted supinator reflex

Teaching Pearl: The lesion damages the C5–C6 reflex arc at its level while corticospinal tract involvement exaggerates reflex activity below it. Thus: reduced reflex at the level + exaggerated reflex below = cord lesion at that level.

Anatomical Mechanism & Discriminators: The lesion damages the C5–C6 reflex arc at its level while corticospinal tract involvement exaggerates reflex activity below it. Thus: reduced reflex at the level + exaggerated reflex below = cord lesion at that level. — *Anchor: Cervical cord around C5–C6*

115 Immediate Flaccid Areflexic Paraplegia and Atonic Bladder Following High-Impact Trauma

30-year-old man • Stage 3: Vertical Cord Level • #5

Clinical Vignette: A 30-year-old man is brought immediately after a road-traffic accident with a T6 spinal fracture.

- Complete paraplegia
- Legs flaccid
- Knee and ankle jerks absent
- Plantar responses absent
- Complete sensory loss below approximately T6
- Abdominal reflexes absent below lesion
- Atonic distended bladder
- Bulbocavernosus reflex absent initially

Q1: CENTRAL OR PERIPHERAL?
Central (Spinal cord myelopathy — UMN signs below lesion + sensory level ± LMN signs at level).

Q2: NEURAXIS LEVEL
Acute severe thoracic spinal cord lesion

Q3: STRUCTURES & TRACTS INVOLVED
A completely transected cord does not immediately produce spastic legs.

Q4: DEFINITIVE DIAGNOSIS & SYNDROME
Spinal shock

Teaching Pearl: A completely transected cord does not immediately produce spastic legs. During spinal shock there is transient loss of reflex activity below the lesion, giving flaccidity and areflexia that can mimic cauda equina disease. The clear truncal sensory level identifies the lesion as spinal cord. Later the legs become spastic and hyperreflexic.

Anatomical Mechanism & Discriminators: A completely transected cord does not immediately produce spastic legs. During spinal shock there is transient loss of reflex activity below the lesion, giving flaccidity and areflexia that can mimic cauda equina disease. The clear truncal sensory level identifies the lesion as spinal cord. Later the legs become spastic and hyperreflexic. — *Anchor: Acute severe thoracic spinal cord lesion*

121 Rapidly Evolving Paraplegia, Sharp Sensory Level at Mid-Thorax, and Bladder Retention

31-year-old woman • Stage 3: Vertical Cord Level • #6

Clinical Vignette: A 31-year-old woman develops back discomfort followed over two days by numbness and rapidly progressive weakness of both legs.

- Bilateral leg weakness
- Initially reduced, subsequently brisk reflexes
- Bilateral extensor plantar responses
- Pinprick sensory level at T8
- Impaired vibration below the level
- Urinary retention
- Upper limbs and cranial nerves normal

Q1: CENTRAL OR PERIPHERAL?
Central (Spinal cord myelopathy — UMN signs below lesion + sensory level ± LMN signs at level).

Q2: NEURAXIS LEVEL
Thoracic spinal cord

Q3: STRUCTURES & TRACTS INVOLVED
The combination of: bilateral motor dysfunction + definite sensory level + autonomic dysfunction is the classic clinical definition of a transverse spinal cord syndrome.

Q4: DEFINITIVE DIAGNOSIS & SYNDROME
Acute transverse myelitis

Teaching Pearl: The combination of: bilateral motor dysfunction + definite sensory level + autonomic dysfunction is the classic clinical definition of a transverse spinal cord syndrome.

Anatomical Mechanism & Discriminators: The combination of: bilateral motor dysfunction + definite sensory level + autonomic dysfunction is the classic clinical definition of a transverse spinal cord syndrome. — *Anchor: Thoracic spinal cord*

122 Severe Spinal Tenderness, Fever, and Rapidly Progressive Paraplegia

55-year-old diabetic man • Stage 3: Vertical Cord Level • #7

Clinical Vignette: A 55-year-old diabetic man has severe thoracic back pain and fever. Two days later he develops difficulty walking and urinary retention.

- Focal severe vertebral tenderness
- Initial thoracic radicular pain
- Progressive spastic paraparesis
- Brisk lower-limb reflexes
- Bilateral Babinski responses
- Sensory level at T7
- Urinary retention
- Upper limbs normal

Q1: CENTRAL OR PERIPHERAL?

Central (Spinal cord myelopathy — UMN signs below lesion + sensory level ± LMN signs at level).

Q2: NEURAXIS LEVEL

Extradural thoracic spinal cord compression

Q3: STRUCTURES & TRACTS INVOLVED

The localisation sequence is valuable: local back pain → root pain → cord compression → sphincter dysfunction This is an extradural compressive myelopathy rather than a peripheral neuropathy or cauda equina lesion.

Q4: DEFINITIVE DIAGNOSIS & SYNDROME

Spinal epidural abscess

Teaching Pearl: The localisation sequence is valuable: local back pain → root pain → cord compression → sphincter dysfunction This is an extradural compressive myelopathy rather than a peripheral neuropathy or cauda equina lesion.

Anatomical Mechanism & Discriminators: The localisation sequence is valuable: local back pain → root pain → cord compression → sphincter dysfunction This is an extradural compressive myelopathy rather than a peripheral neuropathy or cauda equina lesion. — *Anchor: Extradural thoracic spinal cord compression*

123 Electric-Shock Sensation Radiating Down Spine on Neck Flexion

36-year-old woman • Stage 3: Vertical Cord Level • #8

Clinical Vignette: A 36-year-old woman notices an electric-shock sensation running from her neck into both legs whenever she bends her head forward.

- Lhermitte phenomenon on neck flexion
- Reduced vibration in both feet
- Impaired joint-position sense
- Positive Romberg sign
- Brisk knee reflexes
- Bilateral extensor plantar responses
- Mild hand clumsiness
- No cranial nerve signs

Q1: CENTRAL OR PERIPHERAL?

Central (Spinal cord myelopathy — UMN signs below lesion + sensory level ± LMN signs at level).

Q2: NEURAXIS LEVEL

Cervical spinal cord, especially dorsal columns

Q3: STRUCTURES & TRACTS INVOLVED

Lhermitte phenomenon strongly suggests irritation of the cervical dorsal columns.

Q4: DEFINITIVE DIAGNOSIS & SYNDROME

Cervical posterior-column myelopathy

Teaching Pearl: Lhermitte phenomenon strongly suggests irritation of the cervical dorsal columns. When accompanied by proprioceptive loss and pyramidal signs, the lesion is clearly within the cervical cord rather than peripheral nerves.

Anatomical Mechanism & Discriminators: Lhermitte phenomenon strongly suggests irritation of the cervical dorsal columns. When accompanied by proprioceptive loss and pyramidal signs, the lesion is clearly within the cervical cord rather than peripheral nerves. — *Anchor: Cervical spinal cord, especially dorsal columns*

124 Progressive Bilateral Hand Muscle Wasting and Fasciculations with Completely Intact Sensation

27-year-old man • Stage 3: Vertical Cord Level • #9

Clinical Vignette: A 27-year-old man develops progressive weakness and wasting of both hands, with no numbness.

- Marked wasting of intrinsic hand muscles
- Fasciculations in C8–T1 muscles
- Weak finger abduction and grip
- Reduced corresponding upper-limb reflexes
- Pain, temperature, vibration and position sense normal
- Legs initially normal
- No sphincter symptoms

Q1: CENTRAL OR PERIPHERAL?
Central (Spinal cord myelopathy — UMN signs below lesion + sensory level ± LMN signs at level).

Q2: NEURAXIS LEVEL
Anterior horn cells of lower cervical cord, C8–T1

Q3: STRUCTURES & TRACTS INVOLVED
Pure LMN weakness confined to muscles supplied by several different peripheral nerves, but sharing the same spinal segments, points proximal to the individual nerves.

Q4: DEFINITIVE DIAGNOSIS & SYNDROME
Segmental anterior horn cell syndrome

Teaching Pearl: Pure LMN weakness confined to muscles supplied by several different peripheral nerves, but sharing the same spinal segments, points proximal to the individual nerves. Normal sensation favours anterior horn cells rather than roots or plexus.

Anatomical Mechanism & Discriminators: Pure LMN weakness confined to muscles supplied by several different peripheral nerves, but sharing the same spinal segments, points proximal to the individual nerves. Normal sensation favours anterior horn cells rather than roots or plexus. — *Anchor: Anterior horn cells of lower cervical cord, C8–T1*

125 Progressive Leg Weakness and Numbness Aggravated by Walking and Relieved by Rest

67-year-old man • Stage 3: Vertical Cord Level • #10

Clinical Vignette: A 67-year-old man develops gradually progressive difficulty walking. Symptoms fluctuate initially but later become persistent, with urinary urgency.

- Spastic paraparesis
- Brisk knees
- Ankle jerks reduced or absent
- Bilateral extensor plantar responses
- Patchy sensory impairment in legs
- Impaired vibration
- Mild saddle sensory involvement
- Urinary urgency progressing to retention
- Upper limbs normal

Q1: CENTRAL OR PERIPHERAL?
Central / Distal Medullary Cord (Symmetric saddle sensory loss and early autonomic/sphincter dysfunction).

Q2: NEURAXIS LEVEL
Lower thoracic cord/conus region

Q3: STRUCTURES & TRACTS INVOLVED
A spinal dural AV fistula often produces a puzzling mixed UMN and LMN picture because the congestive myelopathy frequently involves the thoracolumbar cord and conus.

Q4: DEFINITIVE DIAGNOSIS & SYNDROME
Spinal dural arteriovenous fistula causing congestive myelopathy

Teaching Pearl: A spinal dural AV fistula often produces a puzzling mixed UMN and LMN picture because the congestive myelopathy frequently involves the thoracolumbar cord and conus. In an older man with progressive unexplained myelopathy and bladder dysfunction, this localisation pattern is particularly important.

Anatomical Mechanism & Discriminators: A spinal dural AV fistula often produces a puzzling mixed UMN and LMN picture because the congestive myelopathy frequently involves the thoracolumbar cord and conus. In an older man with progressive unexplained myelopathy and bladder dysfunction, this localisation pattern is particularly important. — *Anchor: Lower thoracic cord/conus region*

108 Progressive Quadriparesis with Numbness Moving from Hands to Feet and Neck Pain

50-year-old woman • Stage 3: Vertical Cord Level • #11

Clinical Vignette: A 50-year-old woman develops tingling and weakness that began in the hands and then spread to the legs. She has neck pain and a downbeat nystagmus.

- Downbeat nystagmus
- Wasting of the intrinsic hand muscles
- Upper limbs affected before the lower limbs
- "Onion-skin" pattern of facial sensory loss may appear late
- Long-tract signs present
- Occipital headache

Q1: CENTRAL OR PERIPHERAL?

Central (Cerebral cortex — presence of cortical signs such as aphasia, neglect, apraxia, or visual field deficits).

Q2: NEURAXIS LEVEL

Craniocervical junction / foramen magnum

Q3: STRUCTURES & TRACTS INVOLVED

A lesion at the foramen magnum classically produces an "around-the-clock" pattern of progression: upper limbs affected before lower limbs, then the ipsilateral face.

Q4: DEFINITIVE DIAGNOSIS & SYNDROME

Foramen magnum syndrome

Teaching Pearl: A lesion at the foramen magnum classically produces an "around-the-clock" pattern of progression: upper limbs affected before lower limbs, then the ipsilateral face. Downbeat nystagmus and occipital headache are important localising clues to the craniocervical junction.

Anatomical Mechanism & Discriminators: A lesion at the foramen magnum classically produces an "around-the-clock" pattern of progression: upper limbs affected before lower limbs, then the ipsilateral face. Downbeat nystagmus and occipital headache are important localising clues to the craniocervical junction. — *Anchor: Craniocervical junction / foramen magnum*

CURRICULUM CHAPTER 4 • STAGE 4

Stage 4 — Conus Medullaris vs Cauda Equina vs Epiconus

Chapter Learning Objective

Differentiate distal cord (conus) from lumbosacral nerve roots (cauda equina) and spinal canal compression.

Pedagogical Localisation Anchor

Conus -> early symmetric saddle + bladder; Cauda -> asymmetric severe radicular pain + flaccid LMN legs.

Included Clinical Vignettes (7 Cases):
Case 112: Symmetric Saddle Sensory Loss, Early Urinary Retention, and Spared Knee Jerks • Case 111: Severe Asymmetric Radicular Leg Pain, Flaccid Weakness, and Absent Reflexes • Case 113: Bilateral Foot Drop and Lost Ankle Jerks with Preserved Knee Jerks and Sensation • Case 114: Severe Urinary Retention and Saddle Anesthesia with Normal Leg Strength and Reflexes • Case 119: Sharp Radicular Thoracic Pain Followed by Asymmetric Weakness and Contralateral Pain Loss • Case 120: Burning Trunk Pain and Dissociated Sensory Loss with Sacral Sparing • *...and 1 more cases*

112 Symmetric Saddle Sensory Loss, Early Urinary Retention, and Spared Knee Jerks

62-year-old woman • Stage 4: Conus vs Cauda • #1

Clinical Vignette: A 62-year-old woman develops sudden numbness around the perineum and cannot pass urine after an acute L1 vertebral collapse.

- Dense, symmetric saddle anesthesia
- Urinary retention from onset
- Fecal incontinence
- Anal tone markedly reduced
- Bulbocavernosus reflex absent
- Mild symmetric distal leg weakness
- Ankle jerks reduced
- Knee jerks preserved
- Little radicular pain

Q1: CENTRAL OR PERIPHERAL?

Central / Distal Medullary Cord (Symmetric saddle sensory loss and early autonomic/sphincter dysfunction).

Q2: NEURAXIS LEVEL

Conus medullaris

Q3: STRUCTURES & TRACTS INVOLVED

Think conus when sacral dysfunction dominates: symmetric saddle anesthesia early bladder/bowel dysfunction sexual dysfunction relatively symmetric weakness less dramatic radicular pain The conus contains the densely packed sacral cord segments controlling sphincters.

Q4: DEFINITIVE DIAGNOSIS & SYNDROME

Acute conus medullaris syndrome

Teaching Pearl: Think conus when sacral dysfunction dominates: symmetric saddle anesthesia early bladder/bowel dysfunction sexual dysfunction relatively symmetric weakness less dramatic radicular pain The conus contains the densely packed sacral cord segments controlling sphincters.

Anatomical Mechanism & Discriminators: Think conus when sacral dysfunction dominates: symmetric saddle anesthesia early bladder/bowel dysfunction sexual dysfunction relatively symmetric weakness less dramatic radicular pain The conus contains the densely packed sacral cord segments controlling sphincters. — *Anchor: Conus medullaris*

111 Severe Asymmetric Radicular Leg Pain, Flaccid Weakness, and Absent Reflexes

56-year-old man with a large lumbar disc prolapse • Stage 4: Conus vs Cauda • #2

Clinical Vignette: A 56-year-old man with a large lumbar disc prolapse develops severe low-back pain shooting into both legs, followed by progressive right-greater-than-left leg weakness.

- Severe bilateral radicular pain
- Right leg power 2/5, left 4/5
- Distal weakness greater than proximal
- Right knee jerk reduced
- Both ankle jerks absent
- Plantar responses flexor
- Patchy sensory loss involving several dermatomes
- Saddle anesthesia
- Reduced anal tone
- Urinary retention develops later

Q1: CENTRAL OR PERIPHERAL?

Peripheral (Nerve root / Cauda equina level — Lower Motor Neuron signs predominate).

Q2: NEURAXIS LEVEL

Multiple lumbosacral nerve roots within the cauda equina

Q3: STRUCTURES & TRACTS INVOLVED

Cauda equina disease behaves like a multiple-root LMN lesion.

Q4: DEFINITIVE DIAGNOSIS & SYNDROME

Cauda equina syndrome

Teaching Pearl: Cauda equina disease behaves like a multiple-root LMN lesion. The strongest clues are marked asymmetry, radicular pain, areflexia and patchy root sensory loss. Sphincter dysfunction may occur, but is often less abrupt than with a conus lesion.

Anatomical Mechanism & Discriminators: Cauda equina disease behaves like a multiple-root LMN lesion. The strongest clues are marked asymmetry, radicular pain, areflexia and patchy root sensory loss. Sphincter dysfunction may occur, but is often less abrupt than with a conus lesion. — *Anchor: Multiple lumbosacral nerve roots within the cauda equina*

113 Bilateral Foot Drop and Lost Ankle Jerks with Preserved Knee Jerks and Sensation

48-year-old man • Stage 4: Conus vs Cauda • #3

Clinical Vignette: A 48-year-old man develops progressive difficulty walking due to bilateral foot drop. Bladder symptoms appear only later.

- Bilateral foot drop
- Weak ankle dorsiflexion and plantarflexion
- Weak knee flexion
- Knee extension relatively preserved
- Knee jerks preserved
- Ankle jerks absent
- Patchy sensory loss over L5–S2 dermatomes
- One plantar response extensor
- Mild saddle sensory impairment
- Bladder initially preserved

Q1: CENTRAL OR PERIPHERAL?
Central / Distal Medullary Cord (Symmetric saddle sensory loss and early autonomic/sphincter dysfunction).

Q2: NEURAXIS LEVEL
Epiconus — approximately L4–S2 spinal cord segments

Q3: STRUCTURES & TRACTS INVOLVED
The epiconus lies just above the conus and can produce an unusual mixed UMN–LMN picture.

Q4: DEFINITIVE DIAGNOSIS & SYNDROME
Epiconus syndrome

Teaching Pearl: The epiconus lies just above the conus and can produce an unusual mixed UMN–LMN picture. Segmental anterior horn/root involvement causes distal weakness and absent ankle jerks, while nearby corticospinal tract involvement may produce an extensor plantar response. Sphincter disturbance is usually less prominent initially than in true conus syndrome.

Anatomical Mechanism & Discriminators: The epiconus lies just above the conus and can produce an unusual mixed UMN–LMN picture. Segmental anterior horn/root involvement causes distal weakness and absent ankle jerks, while nearby corticospinal tract involvement may produce an extensor plantar response. Sphincter disturbance is usually less prominent initially than in true conus syndrome. — *Anchor: Epiconus — approximately L4–S2 spinal cord segments*

114 Severe Urinary Retention and Saddle Anesthesia with Normal Leg Strength and Reflexes

50-year-old woman • Stage 4: Conus vs Cauda • #4

Clinical Vignette: A 50-year-old woman develops urinary retention and numbness around the anus after surgery involving the lower spinal canal.

- Normal hip, knee and ankle power
- Knee jerks normal
- Ankle jerks preserved
- Plantar responses flexor
- Dense perianal and genital sensory loss
- Anal wink absent
- Bulbocavernosus reflex absent
- Reduced anal tone
- Urinary retention
- Loss of sexual sensation

Q1: CENTRAL OR PERIPHERAL?
Central / Distal Medullary Cord (Symmetric saddle sensory loss and early autonomic/sphincter dysfunction).

Q2: NEURAXIS LEVEL
Sacral S2–S4 cord segments / most distal conus

Q3: STRUCTURES & TRACTS INVOLVED
Normal leg motor examination with profound saddle + sphincter dysfunction points specifically toward the sacral segments.

Q4: DEFINITIVE DIAGNOSIS & SYNDROME
Selective sacral cord syndrome

Teaching Pearl: Normal leg motor examination with profound saddle + sphincter dysfunction points specifically toward the sacral segments. S2–S4 supply perineal sensation and parasympathetic control of bladder, bowel and sexual function.

Anatomical Mechanism & Discriminators: Normal leg motor examination with profound saddle + sphincter dysfunction points specifically toward the sacral segments. S2–S4 supply perineal sensation and parasympathetic control of bladder, bowel and sexual function. — *Anchor: Sacral S2–S4 cord segments / most distal conus*

119 Sharp Radicular Thoracic Pain Followed by Asymmetric Weakness and Contralateral Pain Loss

58-year-old woman • Stage 4: Conus vs Cauda • #5

Clinical Vignette: A 58-year-old woman develops months of band-like pain around the right chest followed by progressive stiffness of the right leg.

- Right thoracic radicular pain
- Right leg spastic weakness
- Right extensor plantar response
- Loss of vibration and position sense in right leg
- Loss of pain and temperature in left leg beginning a few segments below
- Sacral pain-temperature sensation affected relatively early
- Bladder symptoms appear late

Q1: CENTRAL OR PERIPHERAL?

Central (Spinal cord myelopathy — UMN signs below lesion + sensory level ± LMN signs at level).

Q2: NEURAXIS LEVEL

Right-sided extramedullary thoracic cord compression

Q3: STRUCTURES & TRACTS INVOLVED

Extramedullary compression often begins with root pain, followed by asymmetric cord compression.

Q4: DEFINITIVE DIAGNOSIS & SYNDROME

Thoracic spinal meningioma causing incomplete Brown-Séquard syndrome

Teaching Pearl: Extramedullary compression often begins with root pain, followed by asymmetric cord compression. A slowly progressive Brown-Séquard pattern is particularly characteristic of an eccentric extramedullary mass such as a meningioma.

Anatomical Mechanism & Discriminators: Extramedullary compression often begins with root pain, followed by asymmetric cord compression. A slowly progressive Brown-Séquard pattern is particularly characteristic of an eccentric extramedullary mass such as a meningioma. — *Anchor: Right-sided extramedullary thoracic cord compression*

120 Burning Trunk Pain and Dissociated Sensory Loss with Sacral Sparing

42-year-old man • Stage 4: Conus vs Cauda • #6

Clinical Vignette: A 42-year-old man develops slowly progressive bilateral sensory disturbance and weakness without significant radicular pain.

- Bilateral poorly localised burning pain
- Dissociated sensory loss around the trunk
- Pain and temperature impaired more than vibration
- Segmental weakness and wasting at the level of the lesion
- Lower limbs later become spastic
- Perianal sensation relatively preserved despite extensive trunk/leg sensory loss
- Bladder dysfunction occurs late

Q1: CENTRAL OR PERIPHERAL?
Central (Spinal cord myelopathy — UMN signs below lesion + sensory level ± LMN signs at level).

Q2: NEURAXIS LEVEL
Intramedullary spinal cord

Q3: STRUCTURES & TRACTS INVOLVED
Important clues favouring an intramedullary lesion are: poorly localised pain rather than root pain dissociated sensory loss segmental LMN signs sacral sparing relatively late sphincter involvement.

Q4: DEFINITIVE DIAGNOSIS & SYNDROME
Intramedullary cord tumour, e.g. ependymoma

Teaching Pearl: Important clues favouring an intramedullary lesion are: poorly localised pain rather than root pain dissociated sensory loss segmental LMN signs sacral sparing relatively late sphincter involvement

Anatomical Mechanism & Discriminators: Important clues favouring an intramedullary lesion are: poorly localised pain rather than root pain dissociated sensory loss segmental LMN signs sacral sparing relatively late sphincter involvement — *Anchor: Intramedullary spinal cord*

100 Severe Pain Down the Back of Leg to Little Toe with Lost Ankle Jerk

35-year-old man • Stage 4: Conus vs Cauda • #7

Clinical Vignette: A 35-year-old man develops severe low-back pain radiating down the back of the leg to the sole after heavy lifting, and can no longer stand on his toes.

- Weak plantarflexion on the right
- Absent right ankle jerk
- Numbness over the lateral border of the foot and heel
- Straight-leg-raise test positive
- Knee reflex preserved
- Inversion relatively preserved

Q1: CENTRAL OR PERIPHERAL?
Peripheral (Lower motor neuron / peripheral nerve / plexus / root distribution).

Q2: NEURAXIS LEVEL
Right S1 nerve root

Q3: STRUCTURES & TRACTS INVOLVED
S1 supplies the gastrocnemius-soleus complex and the ankle jerk.

Q4: DEFINITIVE DIAGNOSIS & SYNDROME
S1 radiculopathy — lumbosacral disc herniation

Teaching Pearl: S1 supplies the gastrocnemius-soleus complex and the ankle jerk. Loss of the ankle reflex with plantarflexion weakness and pain along the posterior thigh to the sole distinguishes an S1 root lesion from an L5 lesion (dorsiflexion, preserved jerk) and from a sciatic neuropathy.

Anatomical Mechanism & Discriminators: S1 supplies the gastrocnemius-soleus complex and the ankle jerk. Loss of the ankle reflex with plantarflexion weakness and pain along the posterior thigh to the sole distinguishes an S1 root lesion from an L5 lesion (dorsiflexion, preserved jerk) and from a sciatic neuropathy. — *Anchor: Right S1 nerve root*

CURRICULUM CHAPTER 5 • STAGE 5

Stage 5 — Cranial Nerves Individually & Complete Visual Pathways

Chapter Learning Objective

Understand individual cranial nerves and trace the visual pathway from retina to occipital cortex.

Pedagogical Localisation Anchor

Trace nerves along their anatomical exits before tackling complex multi-nerve skull-base syndromes.

Included Clinical Vignettes (17 Cases):
Case 44: Sudden Painless Loss of Vision in One Eye • Case 76: Bitemporal Field Loss with Progressive Visual Impairment and Amenorrhea • Case 83: Ipsilateral Optic Atrophy, Contralateral Papilledema, and Anosmia • Case 101: Incongruous Homonymous Hemianopia with Wernicke Pupillary Phenomenon • Case 102: Homonymous Superior Quadrantic Visual Field Deficit ('Pie in the Sky') • Case 77: Homonymous Inferior Quadrantic Visual Deficit ('Pie in the Floor') • *...and 11 more cases*

44 Sudden Painless Loss of Vision in One Eye

71-year-old man suddenly • Stage 5: Cranial Nerves • #1

Clinical Vignette: A 71-year-old man suddenly develops painless complete loss of vision in the left eye.

- No perception of light in left eye
- Right eye vision normal
- Left direct light reflex absent
- Shining light in left eye produces no constriction of either pupil
- Shining light in right eye constricts both pupils
- Eye movements normal

Q1: CENTRAL OR PERIPHERAL?
Peripheral (Lower motor neuron / peripheral nerve / plexus / root distribution).

Q2: NEURAXIS LEVEL
Left optic nerve or severe left retinal afferent lesion

Q3: STRUCTURES & TRACTS INVOLVED
The pupils can constrict when the normal right eye is illuminated, showing that both efferent CN III pathways work.

Q4: DEFINITIVE DIAGNOSIS & SYNDROME
Left afferent visual pathway lesion

Teaching Pearl: The pupils can constrict when the normal right eye is illuminated, showing that both efferent CN III pathways work. Failure of either pupil to constrict when light enters the left eye identifies a left afferent defect, localising to retina or optic nerve.

Anatomical Mechanism & Discriminators: The pupils can constrict when the normal right eye is illuminated, showing that both efferent CN III pathways work. Failure of either pupil to constrict when light enters the left eye identifies a left afferent defect, localising to retina or optic nerve. — *Anchor: Left optic nerve or severe left retinal afferent lesion*

76 Bitemporal Field Loss with Progressive Visual Impairment and Amenorrhea

42-year-old woman • Stage 5: Cranial Nerves • #2

Clinical Vignette: A 42-year-old woman presents with progressive loss of peripheral vision, bumping into doorways, and secondary amenorrhea.

- Visual acuity 20/30 in both eyes
- Bitemporal hemianopia on confrontation visual field testing
- Bilateral mild optic disc pallor
- Pupillary light reflexes intact
- Normal ocular motility
- Normal limb power and reflexes

Q1: CENTRAL OR PERIPHERAL?
Central (Cerebral cortex — presence of cortical signs such as aphasia, neglect, apraxia, or visual field deficits).

Q2: NEURAXIS LEVEL
Optic chiasm

Q3: STRUCTURES & TRACTS INVOLVED
Lesions compressing the center of the optic chiasm preferentially disrupt the crossing nasal retinal fibers, eliminating the temporal visual field of each eye.

Q4: DEFINITIVE DIAGNOSIS & SYNDROME
Chiasmatic syndrome (e.g., Pituitary macroadenoma)

Teaching Pearl: Lesions compressing the center of the optic chiasm preferentially disrupt the crossing nasal retinal fibers, eliminating the temporal visual field of each eye. * "Pie-in-the-floor" visual field defect

Anatomical Mechanism & Discriminators: Lesions compressing the center of the optic chiasm preferentially disrupt the crossing nasal retinal fibers, eliminating the temporal visual field of each eye. * "Pie-in-the-floor" visual field defect — *Anchor: Optic chiasm*

83 Ipsilateral Optic Atrophy, Contralateral Papilledema, and Anosmia

55-year-old woman • Stage 5: Cranial Nerves • #3

Clinical Vignette: A 55-year-old woman presents with progressive vision loss in her left eye, persistent headaches, and a loss of smell.

- Left optic disc pallor with a dense central scotoma
- Right optic disc swelling (papilledema)
- Anosmia of the left nostril
- Mild frontal disinhibition and apathy
- No focal limb weakness

Q1: CENTRAL OR PERIPHERAL?
Central (Cerebral cortex — presence of cortical signs such as aphasia, neglect, apraxia, or visual field deficits).

Q2: NEURAXIS LEVEL
Left subfrontal / olfactory groove region

Q3: STRUCTURES & TRACTS INVOLVED
A large mass (classically an olfactory groove or sphenoid wing meningioma) causes direct compressive atrophy of the ipsilateral optic nerve and olfactory tract, while raised intracranial pressure produces papilledema in the contralateral eye.

Q4: DEFINITIVE DIAGNOSIS & SYNDROME
Foster Kennedy syndrome

Teaching Pearl: A large mass (classically an olfactory groove or sphenoid wing meningioma) causes direct compressive atrophy of the ipsilateral optic nerve and olfactory tract, while raised intracranial pressure produces papilledema in the contralateral eye. 84 * Hyperorality, hypersexuality, docility, and visual agnosia

Anatomical Mechanism & Discriminators: A large mass (classically an olfactory groove or sphenoid wing meningioma) causes direct compressive atrophy of the ipsilateral optic nerve and olfactory tract, while raised intracranial pressure produces papilledema in the contralateral eye. 84 * Hyperorality, hypersexuality, docility, and visual agnosia — *Anchor: Left subfrontal / olfactory groove region*

101 Incongruous Homonymous Hemianopia with Wernicke Pupillary Phenomenon

47-year-old woman • Stage 5: Cranial Nerves • #4

Clinical Vignette: A 47-year-old woman notices that she keeps missing objects on her left. Her visual field defect is denser in one eye, and a pupillary sign is positive.

- Incongruous left homonymous hemianopia (defect larger in one eye)
- Right relative afferent pupillary defect
- Right optic disc pallor after several months
- Visual acuity largely preserved
- No limb weakness

Q1: CENTRAL OR PERIPHERAL?
Central (Cerebral cortex — presence of cortical signs such as aphasia, neglect, apraxia, or visual field deficits).

Q2: NEURAXIS LEVEL
Right optic tract

Q3: STRUCTURES & TRACTS INVOLVED
Lesions of the optic tract (between the chiasm and the lateral geniculate nucleus) produce a homonymous hemianopia that is usually incongruous, because fibres from the two eyes are not yet aligned.

Q4: DEFINITIVE DIAGNOSIS & SYNDROME
Optic tract lesion — incongruous homonymous hemianopia

Teaching Pearl: Lesions of the optic tract (between the chiasm and the lateral geniculate nucleus) produce a homonymous hemianopia that is usually incongruous, because fibres from the two eyes are not yet aligned. A relative afferent pupillary defect on the side of the disc pallor is characteristic, since the tract carries pupillomotor fibres.

Anatomical Mechanism & Discriminators: Lesions of the optic tract (between the chiasm and the lateral geniculate nucleus) produce a homonymous hemianopia that is usually incongruous, because fibres from the two eyes are not yet aligned. A relative afferent pupillary defect on the side of the disc pallor is characteristic, since the tract carries pupillomotor fibres. — *Anchor: Right optic tract*

102 Homonymous Superior Quadrantic Visual Field Deficit ('Pie in the Sky')

62-year-old woman with a left temporal lobe tumour • Stage 5: Cranial Nerves • #5

Clinical Vignette: A 62-year-old woman with a left temporal lobe tumour reports difficulty seeing objects in the upper right part of her vision.

- Right homonymous superior quadrantanopia
- Central vision preserved
- Pupils normal
- No motor or sensory deficit
- Memory and speech intact

Q1: CENTRAL OR PERIPHERAL?

Central (Cerebral cortex — presence of cortical signs such as aphasia, neglect, apraxia, or visual field deficits).

Q2: NEURAXIS LEVEL

Left temporal lobe — Meyer loop of the optic radiation

Q3: STRUCTURES & TRACTS INVOLVED

Fibres representing the inferior retina (superior visual field) loop forward into the temporal lobe as Meyer loop before reaching the calcarine cortex.

Q4: DEFINITIVE DIAGNOSIS & SYNDROME

Superior quadrantanopia ("pie in the sky")

Teaching Pearl: Fibres representing the inferior retina (superior visual field) loop forward into the temporal lobe as Meyer loop before reaching the calcarine cortex. A temporal lobe lesion therefore produces a contralateral superior quadrantanopia — the opposite of the parietal "pie in the floor."

Anatomical Mechanism & Discriminators: Fibres representing the inferior retina (superior visual field) loop forward into the temporal lobe as Meyer loop before reaching the calcarine cortex. A temporal lobe lesion therefore produces a contralateral superior quadrantanopia — the opposite of the parietal "pie in the floor." — *Anchor: Left temporal lobe — Meyer loop of the optic radiation*

77 Homonymous Inferior Quadrantic Visual Deficit ('Pie in the Floor')

59-year-old man with a stroke • Stage 5: Cranial Nerves • #6

Clinical Vignette: A 59-year-old man with a stroke has difficulty reading and noticing objects in the lower right quadrant of his visual field.

- Right homonymous inferior quadrantanopia
- Visual acuity normal
- Pupillary reflexes normal
- Subtle sensory extinction in the right hand on double simultaneous stimulation
- Power 5/5 bilaterally

Q1: CENTRAL OR PERIPHERAL?

Central (Cerebral cortex — presence of cortical signs such as aphasia, neglect, apraxia, or visual field deficits).

Q2: NEURAXIS LEVEL

Left superior optic radiations (Baum loop) in the left parietal lobe

Q3: STRUCTURES & TRACTS INVOLVED

The superior portion of the optic radiation traverses the parietal lobe carrying fibers from the superior retina (representing the inferior visual field).

Q4: DEFINITIVE DIAGNOSIS & SYNDROME

Homonymous inferior quadrantanopia (Parietal optic radiation lesion)

Teaching Pearl: The superior portion of the optic radiation traverses the parietal lobe carrying fibers from the superior retina (representing the inferior visual field). A parietal lesion causes a contralateral homonymous inferior quadrantanopia ("pie in the floor"), contrasting with temporal Meyer loop lesions ("pie in the sky").

Anatomical Mechanism & Discriminators: The superior portion of the optic radiation traverses the parietal lobe carrying fibers from the superior retina (representing the inferior visual field). A parietal lesion causes a contralateral homonymous inferior quadrantanopia ("pie in the floor"), contrasting with temporal Meyer loop lesions ("pie in the sky"). — *Anchor: Left superior optic radiations (Baum loop) in the left parietal lobe*

134 Dense Homonymous Hemianopia with Sparing of Central Macular Visual Acuity

Adult patient • Stage 5: Cranial Nerves • #7

Clinical Vignette: A 70-year-old man suddenly begins bumping into objects on his left but can still read letters directly in front of him.

- Left homonymous hemianopia
- Small central area of vision preserved
- Visual acuity relatively normal
- Pupillary reflexes normal
- No weakness
- No sensory deficit
- No neglect

Q1: CENTRAL OR PERIPHERAL?

Central (Cerebral cortex — presence of cortical signs such as aphasia, neglect, apraxia, or visual field deficits).

Q2: NEURAXIS LEVEL

Right primary visual cortex — occipital lobe

Q3: STRUCTURES & TRACTS INVOLVED

A highly congruous homonymous hemianopia strongly suggests an occipital lesion.

Q4: DEFINITIVE DIAGNOSIS & SYNDROME

PCA occipital infarction with macular sparing

Teaching Pearl: A highly congruous homonymous hemianopia strongly suggests an occipital lesion. Macular sparing further favours primary visual cortex, traditionally attributed to the macular cortex having relatively robust/overlapping vascular supply.

Anatomical Mechanism & Discriminators: A highly congruous homonymous hemianopia strongly suggests an occipital lesion. Macular sparing further favours primary visual cortex, traditionally attributed to the macular cortex having relatively robust/overlapping vascular supply. — *Anchor: Right primary visual cortex — occipital lobe*

26 Sudden Ptosis and Ophthalmoplegia with Preserved Pupillary Light Reflex

56-year-old diabetic man • Stage 5: Cranial Nerves • #8

Clinical Vignette: A 56-year-old diabetic man develops sudden double vision and drooping of the left eyelid. He has no headache.

- Left ptosis
- Left eye positioned down and out
- Impaired adduction, elevation and depression of left eye
- Left pupil equal in size to right
- Pupillary light reflex preserved
- No limb weakness
- No sensory deficit

Q1: CENTRAL OR PERIPHERAL?

Peripheral (Lower motor neuron / peripheral nerve / plexus / root distribution).

Q2: NEURAXIS LEVEL

Left oculomotor nerve, likely microvascular fascicular/peripheral involvement

Q3: STRUCTURES & TRACTS INVOLVED

Somatic motor fibres supplying extraocular muscles are affected, while superficial parasympathetic fibres controlling the pupil are spared.

Q4: DEFINITIVE DIAGNOSIS & SYNDROME

Pupil-sparing third nerve palsy

Teaching Pearl: Somatic motor fibres supplying extraocular muscles are affected, while superficial parasympathetic fibres controlling the pupil are spared. In an older diabetic patient, this pattern strongly suggests a microvascular third-nerve palsy.

Anatomical Mechanism & Discriminators: Somatic motor fibres supplying extraocular muscles are affected, while superficial parasympathetic fibres controlling the pupil are spared. In an older diabetic patient, this pattern strongly suggests a microvascular third-nerve palsy. — *Anchor: Left oculomotor nerve, likely microvascular fascicular/peripheral involvement*

27 Vertical Diplopia Worse on Downward Gaze and Walking Downstairs

43-year-old woman • Stage 5: Cranial Nerves • #9

Clinical Vignette: A 43-year-old woman complains of vertical diplopia, particularly while reading or walking downstairs. She tilts her head to compensate.

- Right hypertropia
- Diplopia worsens on looking down and left
- Diplopia worsens with head tilt to the right
- Compensatory head tilt to the left
- Pupils normal
- No ptosis

Q1: CENTRAL OR PERIPHERAL?
Peripheral (Lower motor neuron / peripheral nerve / plexus / root distribution).

Q2: NEURAXIS LEVEL
Right trochlear nerve / superior oblique

Q3: STRUCTURES & TRACTS INVOLVED
The superior oblique depresses the adducted eye and intorts it.

Q4: DEFINITIVE DIAGNOSIS & SYNDROME
Right CN IV palsy

Teaching Pearl: The superior oblique depresses the adducted eye and intorts it. A trochlear palsy therefore causes vertical diplopia that is especially troublesome while reading or descending stairs. The Bielschowsky head-tilt pattern helps confirm the side.

Anatomical Mechanism & Discriminators: The superior oblique depresses the adducted eye and intorts it. A trochlear palsy therefore causes vertical diplopia that is especially troublesome while reading or descending stairs. The Bielschowsky head-tilt pattern helps confirm the side. — *Anchor: Right trochlear nerve / superior oblique*

28 Horizontal Diplopia Worse on Looking Toward the Affected Side

65-year-old hypertensive man • Stage 5: Cranial Nerves • #10

Clinical Vignette: A 65-year-old hypertensive man develops sudden horizontal diplopia that is worse when looking to the left.

- Left eye cannot abduct fully
- Right eye movements normal
- Diplopia maximal on left gaze
- Pupils equal and reactive
- No ptosis
- Facial movements normal

Q1: CENTRAL OR PERIPHERAL?
Peripheral (Lower motor neuron / peripheral nerve / plexus / root distribution).

Q2: NEURAXIS LEVEL
Left abducens nerve

Q3: STRUCTURES & TRACTS INVOLVED
The lateral rectus is supplied by CN VI.

Q4: DEFINITIVE DIAGNOSIS & SYNDROME
Left CN VI palsy

Teaching Pearl: The lateral rectus is supplied by CN VI. Failure of one eye to abduct produces horizontal diplopia, maximal when looking toward the affected side. An isolated lesion suggests the peripheral abducens nerve rather than the abducens nucleus.

Anatomical Mechanism & Discriminators: The lateral rectus is supplied by CN VI. Failure of one eye to abduct produces horizontal diplopia, maximal when looking toward the affected side. An isolated lesion suggests the peripheral abducens nerve rather than the abducens nucleus. — *Anchor: Left abducens nerve*

45 Severe Paroxysmal Facial Pain with Depressed Corneal Sensation

54-year-old woman • Stage 5: Cranial Nerves • #11

Clinical Vignette: A 54-year-old woman has months of progressive numbness over the right side of her face and difficulty chewing.

- Reduced sensation over right forehead, cheek and jaw
- Right corneal reflex absent when right cornea is touched
- Corneal reflex present bilaterally when left cornea is touched
- Right masseter weak
- Jaw deviates to the right on opening
- Facial movements otherwise normal

Q1: CENTRAL OR PERIPHERAL?
Peripheral (Lower motor neuron / peripheral nerve / plexus / root distribution).

Q2: NEURAXIS LEVEL
Right trigeminal nerve — CN V

Q3: STRUCTURES & TRACTS INVOLVED
CN V carries facial sensation and provides the afferent limb of the corneal reflex.

Q4: DEFINITIVE DIAGNOSIS & SYNDROME
Right trigeminal neuropathy

Teaching Pearl: CN V carries facial sensation and provides the afferent limb of the corneal reflex. Weak muscles of mastication and jaw deviation toward the weak side indicate motor trigeminal involvement as well.

Anatomical Mechanism & Discriminators: CN V carries facial sensation and provides the afferent limb of the corneal reflex. Weak muscles of mastication and jaw deviation toward the weak side indicate motor trigeminal involvement as well. — *Anchor: Right trigeminal nerve — CN V*

13 Lower Facial Droop with Forehead Movement Preserved

59-year-old diabetic man • Stage 5: Cranial Nerves • #12

Clinical Vignette: A 59-year-old diabetic man notices sudden drooping of the right side of his mouth and mild clumsiness of the right hand.

- Right nasolabial fold flattened
- Mouth deviates left when smiling
- Forehead wrinkles symmetrically
- Both eyes close tightly
- Mild right pronator drift
- Brisk right arm reflexes

Q1: CENTRAL OR PERIPHERAL?
Mixed Central & Peripheral (Simultaneous Upper & Lower Motor Neuron degeneration with sensory sparing).

Q2: NEURAXIS LEVEL
Left corticobulbar pathway

Q3: STRUCTURES & TRACTS INVOLVED
Forehead sparing occurs because the upper facial muscles receive bilateral cortical innervation.

Q4: DEFINITIVE DIAGNOSIS & SYNDROME
right UMN facial palsy

Teaching Pearl: Forehead sparing occurs because the upper facial muscles receive bilateral cortical innervation. Lower facial weakness with limb UMN signs indicates a supranuclear lesion.

Anatomical Mechanism & Discriminators: Forehead sparing occurs because the upper facial muscles receive bilateral cortical innervation. Lower facial weakness with limb UMN signs indicates a supranuclear lesion. — *Anchor: Left corticobulbar pathway*

14 Complete Unilateral Upper and Lower Facial Paralysis

34-year-old woman • Stage 5: Cranial Nerves • #13

Clinical Vignette: A 34-year-old woman wakes with sudden weakness of the entire left side of her face. She reports difficulty drinking water and closing her left eye.

- Left forehead cannot wrinkle
- Left eye cannot close completely
- Left nasolabial fold absent
- Mouth pulled to the right
- No limb weakness
- Normal limb reflexes

Q1: CENTRAL OR PERIPHERAL?
Peripheral (Lower motor neuron / peripheral nerve / plexus / root distribution).

Q2: NEURAXIS LEVEL
Left facial nerve

Q3: STRUCTURES & TRACTS INVOLVED
Weakness of both the upper and lower face on the same side indicates a facial nucleus or peripheral facial nerve lesion.

Q4: DEFINITIVE DIAGNOSIS & SYNDROME
LMN CN VII palsy

Teaching Pearl: Weakness of both the upper and lower face on the same side indicates a facial nucleus or peripheral facial nerve lesion.

Anatomical Mechanism & Discriminators: Weakness of both the upper and lower face on the same side indicates a facial nucleus or peripheral facial nerve lesion. — *Anchor: Left facial nerve*

150 Complete Peripheral Facial Palsy Associated with Hyperacusis and Loss of Anterior Tongue Taste

35-year-old woman • Stage 5: Cranial Nerves • #14

Clinical Vignette: A 35-year-old woman develops complete right facial weakness. Food tastes abnormal and ordinary sounds seem painfully loud in the right ear.

- Right forehead paralysis
- Incomplete right eye closure
- Right lower facial weakness
- Loss of taste over anterior two-thirds of right tongue
- Hyperacusis in right ear
- Reduced lacrimation
- Facial sensation normal
- Hearing acuity otherwise preserved
- No limb signs

Q1: CENTRAL OR PERIPHERAL?
Peripheral (Lower motor neuron / peripheral nerve / plexus / root distribution).

Q2: NEURAXIS LEVEL
Right facial nerve proximal to the nerve to stapedius and chorda tympani, within the facial canal

Q3: STRUCTURES & TRACTS INVOLVED
CN VII localisation can be refined by its branch functions: facial weakness → facial motor fibres hyperacusis → stapedius branch involved taste loss → chorda tympani involved reduced lacrimation → greater petrosal pathway involved A lesion near the stylomastoid foramen would produce facial weakness alone without these proximal features.

Q4: DEFINITIVE DIAGNOSIS & SYNDROME
Proximal LMN facial nerve lesion

Teaching Pearl: CN VII localisation can be refined by its branch functions: facial weakness → facial motor fibres hyperacusis → stapedius branch involved taste loss → chorda tympani involved reduced lacrimation → greater petrosal pathway involved A lesion near the stylomastoid foramen would produce facial weakness alone without these proximal features.

Anatomical Mechanism & Discriminators: CN VII localisation can be refined by its branch functions: facial weakness → facial motor fibres hyperacusis → stapedius branch involved taste loss → chorda tympani involved reduced lacrimation → greater petrosal pathway involved A lesion near the stylomastoid foramen would produce facial weakness alone without these proximal features. — *Anchor: Right facial nerve proximal to the nerve to stapedius and chorda tympani, within the facial canal*

46 Hoarseness, Nasal Regurgitation, and Unilateral Palatal Droop

63-year-old man • Stage 5: Cranial Nerves • #15

Clinical Vignette: A 63-year-old man develops persistent hoarseness, choking on liquids and nasal regurgitation.

- Right side of palate droops
- Uvula deviates to the left
- Gag response reduced on right
- Voice hoarse
- Cough weak
- Tongue normal
- Limb examination normal

Q1: CENTRAL OR PERIPHERAL?
Peripheral (Lower motor neuron / peripheral nerve / plexus / root distribution).

Q2: NEURAXIS LEVEL
Right vagus nerve / nucleus ambiguus pathway

Q3: STRUCTURES & TRACTS INVOLVED
Palatal elevation and laryngeal function depend mainly on the vagus nerve.

Q4: DEFINITIVE DIAGNOSIS & SYNDROME
Right CN X palsy

Teaching Pearl: Palatal elevation and laryngeal function depend mainly on the vagus nerve. With a unilateral vagal lesion, the palate droops on the affected side and the uvula is pulled toward the normal side. Hoarseness reflects recurrent laryngeal involvement.

Anatomical Mechanism & Discriminators: Palatal elevation and laryngeal function depend mainly on the vagus nerve. With a unilateral vagal lesion, the palate droops on the affected side and the uvula is pulled toward the normal side. Hoarseness reflects recurrent laryngeal involvement. — *Anchor: Right vagus nerve / nucleus ambiguus pathway*

47 Inability to Shrug One Shoulder and Weakness of Contralateral Head Rotation

46-year-old woman • Stage 5: Cranial Nerves • #16

Clinical Vignette: A 46-year-old woman notices shoulder drooping and difficulty turning her head after surgery in the posterior triangle of the neck.

- Left shoulder drooped
- Left trapezius wasted
- Weak shoulder shrug on left
- Difficulty abducting left arm above horizontal
- Weak head rotation toward the right
- Sternocleidomastoid bulk reduced on left
- Limb power otherwise normal

Q1: CENTRAL OR PERIPHERAL?
Peripheral (Lower motor neuron / peripheral nerve / plexus / root distribution).

Q2: NEURAXIS LEVEL
Left spinal accessory nerve

Q3: STRUCTURES & TRACTS INVOLVED
CN XI supplies: trapezius → shoulder elevation and scapular rotation sternocleidomastoid → head rotation to the opposite side Posterior-triangle surgery is a classic cause of accessory nerve injury.

Q4: DEFINITIVE DIAGNOSIS & SYNDROME
CN XI palsy

Teaching Pearl: CN XI supplies: trapezius → shoulder elevation and scapular rotation sternocleidomastoid → head rotation to the opposite side Posterior-triangle surgery is a classic cause of accessory nerve injury.

Anatomical Mechanism & Discriminators: CN XI supplies: trapezius → shoulder elevation and scapular rotation sternocleidomastoid → head rotation to the opposite side Posterior-triangle surgery is a classic cause of accessory nerve injury. — *Anchor: Left spinal accessory nerve*

50 Unilateral Tongue Deviation and Atrophy Without Limb Weakness

67-year-old man • Stage 5: Cranial Nerves • #17

Clinical Vignette: A 67-year-old man develops gradually progressive difficulty manipulating food and slurring certain words.

- Right half of tongue wasted
- Fasciculations present on right
- Tongue deviates to right on protrusion
- Palatal movement normal
- Facial movements normal
- Limb strength and reflexes normal

Q1: CENTRAL OR PERIPHERAL?

Peripheral (Lower motor neuron / peripheral nerve / plexus / root distribution).

Q2: NEURAXIS LEVEL

Right hypoglossal nerve

Q3: STRUCTURES & TRACTS INVOLVED

A lower motor neuron hypoglossal lesion causes: ipsilateral tongue wasting fasciculations weakness On protrusion, the tongue deviates toward the weak side because the intact genioglossus pushes it across.

Q4: DEFINITIVE DIAGNOSIS & SYNDROME

Right LMN CN XII palsy

Teaching Pearl: A lower motor neuron hypoglossal lesion causes: ipsilateral tongue wasting fasciculations weakness On protrusion, the tongue deviates toward the weak side because the intact genioglossus pushes it across.

Anatomical Mechanism & Discriminators: A lower motor neuron hypoglossal lesion causes: ipsilateral tongue wasting fasciculations weakness On protrusion, the tongue deviates toward the weak side because the intact genioglossus pushes it across. — *Anchor: Right hypoglossal nerve*

CURRICULUM CHAPTER 6 • STAGE 6

Stage 6 — The Brainstem Crossed Principle: Midbrain, Pons & Medulla

Chapter Learning Objective

Master the master rule: Ipsilateral Cranial Nerve + Contralateral Body Deficit = Brainstem.

Pedagogical Localisation Anchor

Divisible by 12 CNs exit midline (III, IV, VI, XII) with 4 M structures; others exit lateral with 4 S structures.

Included Clinical Vignettes (17 Cases):
Case 2: Unilateral Ptosis with Contralateral Hemiplegia • Case 71: Ptosis and Dilated Pupil with Contralateral Limb Incoordination and Tremor • Case 36: Unilateral Ptosis and Ocular Motor Palsy with Contralateral Limb Tremor • Case 1: Crossed Facial Weakness with Contralateral Hemiparesis • Case 126: Isolated Abducens Palsy Associated with Contralateral Hemiplegia • Case 72: Horizontal Gaze Palsy with Ipsilateral Facial Weakness and Contralateral Hemiplegia • *...and 11 more cases*

2 Unilateral Ptosis with Contralateral Hemiplegia

58-year-old hypertensive man suddenly • Stage 6: Brainstem Crossed • #1

Clinical Vignette: A 58-year-old hypertensive man suddenly develops double vision, drooping of the right eyelid and weakness of the left side.

- Right ptosis
- Right pupil dilated and poorly reactive
- Right eye positioned “down and out”
- Left arm and leg power 2/5
- Brisk left-sided reflexes
- Left extensor plantar response

Q1: CENTRAL OR PERIPHERAL?
Central (Brainstem crossed syndrome — ipsilateral cranial nerve nucleus/fascicle + contralateral long tracts).

Q2: NEURAXIS LEVEL
Right midbrain

Q3: STRUCTURES & TRACTS INVOLVED
Ipsilateral CN III fascicles and the adjacent cerebral peduncle are affected.

Q4: DEFINITIVE DIAGNOSIS & SYNDROME
Weber syndrome

Teaching Pearl: Ipsilateral CN III fascicles and the adjacent cerebral peduncle are affected. This produces third-nerve palsy with contralateral corticospinal weakness.

Anatomical Mechanism & Discriminators: Ipsilateral CN III fascicles and the adjacent cerebral peduncle are affected. This produces third-nerve palsy with contralateral corticospinal weakness. — *Anchor: Right midbrain*

71 Ptosis and Dilated Pupil with Contralateral Limb Incoordination and Tremor

54-year-old woman suddenly • Stage 6: Brainstem Crossed • #2

Clinical Vignette: A 54-year-old woman suddenly develops double vision, drooping of the left eyelid and severe unsteadiness and shaking in her right arm.

- Left ptosis
- Left pupil dilated and unreactive to light
- Left eye positioned "down and out"
- Right upper and lower limb dysmetria and intention tremor
- Right dysdiadochokinesia
- Limb power 5/5 bilaterally
- Plantar responses flexor bilaterally

Q1: CENTRAL OR PERIPHERAL?
Central (Brainstem crossed syndrome — ipsilateral cranial nerve nucleus/fascicle + contralateral long tracts).

Q2: NEURAXIS LEVEL
Left dorsal midbrain tegmentum (CN III fascicles and superior cerebellar peduncle / red nucleus)

Q3: STRUCTURES & TRACTS INVOLVED
Damage to the exiting oculomotor fascicles causes an ipsilateral CN III palsy, while involvement of the superior cerebellar peduncle decussation or red nucleus produces contralateral cerebellar-type ataxia without significant pyramidal weakness.

Q4: DEFINITIVE DIAGNOSIS & SYNDROME
Claude syndrome

Teaching Pearl: Damage to the exiting oculomotor fascicles causes an ipsilateral CN III palsy, while involvement of the superior cerebellar peduncle decussation or red nucleus produces contralateral cerebellar-type ataxia without significant pyramidal weakness. * Horizontal gaze palsy + ipsilateral facial palsy + contralateral hemiparesis 72 .

Anatomical Mechanism & Discriminators: Damage to the exiting oculomotor fascicles causes an ipsilateral CN III palsy, while involvement of the superior cerebellar peduncle decussation or red nucleus produces contralateral cerebellar-type ataxia without significant pyramidal weakness. * Horizontal gaze palsy + ipsilateral facial palsy + contralateral hemiparesis 72 . — *Anchor: Left dorsal midbrain tegmentum (CN III fascicles and superior cerebellar peduncle / red nucleus)*

36 Unilateral Ptosis and Ocular Motor Palsy with Contralateral Limb Tremor

59-year-old man suddenly • Stage 6: Brainstem Crossed • #3

Clinical Vignette: A 59-year-old man suddenly develops drooping of the left eyelid, double vision and severe shaking/incoordination of the right hand.

- Left ptosis
- Left pupil dilated
- Left eye down and out
- Right-sided intention tremor
- Right dysmetria
- Right dysdiadochokinesia
- Limb power relatively preserved

Q1: CENTRAL OR PERIPHERAL?
Central (Brainstem crossed syndrome — ipsilateral cranial nerve nucleus/fascicle + contralateral long tracts).

Q2: NEURAXIS LEVEL
Left midbrain involving CN III fascicles and red nucleus/superior cerebellar pathways

Q3: STRUCTURES & TRACTS INVOLVED
Ipsilateral CN III palsy localises the lesion to the midbrain.

Q4: DEFINITIVE DIAGNOSIS & SYNDROME
Benedikt syndrome

Teaching Pearl: Ipsilateral CN III palsy localises the lesion to the midbrain. Contralateral tremor and cerebellar-type incoordination suggest involvement of the red nucleus and nearby cerebellothalamic fibres.

Anatomical Mechanism & Discriminators: Ipsilateral CN III palsy localises the lesion to the midbrain. Contralateral tremor and cerebellar-type incoordination suggest involvement of the red nucleus and nearby cerebellothalamic fibres. — *Anchor: Left midbrain involving CN III fascicles and red nucleus/superior cerebellar pathways*

1 Crossed Facial Weakness with Contralateral Hemiparesis

62-year-old man • Stage 6: Brainstem Crossed • #4

Clinical Vignette: A 62-year-old man develops sudden vertigo, diplopia and weakness of the right side of the body.

- Left facial weakness involving the whole face
- Failure to close the left eye
- Left abduction weakness
- Right arm and leg power 2/5
- Brisk right-sided reflexes
- Right extensor plantar response
- Sensory examination essentially normal

Q1: CENTRAL OR PERIPHERAL?
Central (Brainstem crossed syndrome — ipsilateral cranial nerve nucleus/fascicle + contralateral long tracts).

Q2: NEURAXIS LEVEL
Left pons

Q3: STRUCTURES & TRACTS INVOLVED
CN VI and VII involvement on the left localises to the caudal pons.

Q4: DEFINITIVE DIAGNOSIS & SYNDROME
Millard-Gubler-type pontine syndrome

Teaching Pearl: CN VI and VII involvement on the left localises to the caudal pons. Adjacent corticospinal tract involvement causes contralateral right hemiparesis.

Anatomical Mechanism & Discriminators: CN VI and VII involvement on the left localises to the caudal pons. Adjacent corticospinal tract involvement causes contralateral right hemiparesis. — *Anchor: Left pons*

126 Isolated Abducens Palsy Associated with Contralateral Hemiplegia

66-year-old hypertensive man suddenly • Stage 6: Brainstem Crossed • #5

Clinical Vignette: A 66-year-old hypertensive man suddenly develops horizontal diplopia and weakness of the left arm and leg.

- Right eye unable to abduct
- No conjugate gaze palsy
- Facial movements normal
- Left arm and leg power 2/5
- Brisk left-sided reflexes
- Left extensor plantar response
- Sensation preserved

Q1: CENTRAL OR PERIPHERAL?
Central (Brainstem crossed syndrome — ipsilateral cranial nerve nucleus/fascicle + contralateral long tracts).

Q2: NEURAXIS LEVEL
Right medial ventral pons — CN VI fascicle plus corticospinal tract

Q3: STRUCTURES & TRACTS INVOLVED
The distinction from a lesion of the abducens nucleus is important.

Q4: DEFINITIVE DIAGNOSIS & SYNDROME
Raymond syndrome

Teaching Pearl: The distinction from a lesion of the abducens nucleus is important. A fascicular CN VI lesion causes an isolated ipsilateral abduction deficit. Nearby corticospinal tract damage causes contralateral hemiplegia.

Anatomical Mechanism & Discriminators: The distinction from a lesion of the abducens nucleus is important. A fascicular CN VI lesion causes an isolated ipsilateral abduction deficit. Nearby corticospinal tract damage causes contralateral hemiplegia. — *Anchor: Right medial ventral pons — CN VI fascicle plus corticospinal tract*

72 Horizontal Gaze Palsy with Ipsilateral Facial Weakness and Contralateral Hemiplegia

66-year-old man • Stage 6: Brainstem Crossed • #6

Clinical Vignette: A 66-year-old man develops acute weakness of the left side of his body, right-sided facial droop and inability to look toward his right side.

- Both eyes unable to look conjugate to the right
- Left gaze preserved
- Right upper and lower facial weakness (inability to close right eye or wrinkle right forehead)
- Left arm and leg power 3/5
- Hyperreflexia on the left side
- Left extensor plantar response
- Normal sensation

Q1: CENTRAL OR PERIPHERAL?
Peripheral (Nerve root / Cauda equina level — Lower Motor Neuron signs predominate).

Q2: NEURAXIS LEVEL
Right caudal pontine tegmentum and base (PPRF / CN VI nucleus, CN VII fascicle, and corticospinal tract)

Q3: STRUCTURES & TRACTS INVOLVED
Involvement of the PPRF or abducens nucleus causes an ipsilateral horizontal conjugate gaze palsy, the looping facial fascicle causes an ipsilateral peripheral-type CN VII palsy, and the corticospinal tract produces contralateral hemiparesis.

Q4: DEFINITIVE DIAGNOSIS & SYNDROME
Foville syndrome (inferior medial pontine syndrome)

Teaching Pearl: Involvement of the PPRF or abducens nucleus causes an ipsilateral horizontal conjugate gaze palsy, the looping facial fascicle causes an ipsilateral peripheral-type CN VII palsy, and the corticospinal tract produces contralateral hemiparesis. * Ipsilateral facial palsy, deafness, ataxia + contralateral pain loss 73 .

Anatomical Mechanism & Discriminators: Involvement of the PPRF or abducens nucleus causes an ipsilateral horizontal conjugate gaze palsy, the looping facial fascicle causes an ipsilateral peripheral-type CN VII palsy, and the corticospinal tract produces contralateral hemiparesis. * Ipsilateral facial palsy, deafness, ataxia + contralateral pain loss 73 . — *Anchor: Right caudal pontine tegmentum and base (PPRF / CN VI nucleus, CN VII fascicle, and corticospinal tract)*

73 Acute Vertigo, Ipsilateral Deafness, Facial Droop, and Contralateral Pain Loss

57-year-old woman • Stage 6: Brainstem Crossed • #7

Clinical Vignette: A 57-year-old woman develops acute severe vertigo, nausea, sudden right ear deafness, right facial numbness, and unsteadiness walking.

- Right sensorineural hearing loss
- Right peripheral facial weakness
- Reduced corneal reflex and pinprick sensation over right face
- Right limb dysmetria and gait ataxia veering to the right
- Reduced pain and temperature over left trunk and extremities
- Normal limb power

Q1: CENTRAL OR PERIPHERAL?
Peripheral (Nerve root / Cauda equina level — Lower Motor Neuron signs predominate).

Q2: NEURAXIS LEVEL
Right lateral caudal pons (AICA territory)

Q3: STRUCTURES & TRACTS INVOLVED
AICA territory ischemia damages the vestibular and cochlear nuclei (vertigo, deafness), spinal trigeminal nucleus (ipsilateral facial sensory loss), facial nucleus/fascicle (ipsilateral LMN facial palsy), middle cerebellar peduncle (ipsilateral ataxia), and spinothalamic tract (contralateral body pain/temperature loss).

Q4: DEFINITIVE DIAGNOSIS & SYNDROME
Lateral pontine syndrome (Marie-Foix syndrome / AICA occlusion)

Teaching Pearl: AICA territory ischemia damages the vestibular and cochlear nuclei (vertigo, deafness), spinal trigeminal nucleus (ipsilateral facial sensory loss), facial nucleus/fascicle (ipsilateral LMN facial palsy), middle cerebellar peduncle (ipsilateral ataxia), and spinothalamic tract (contralateral body pain/temperature loss). * Optic ataxia, ocular apraxia, and simultanagnosia

Anatomical Mechanism & Discriminators: AICA territory ischemia damages the vestibular and cochlear nuclei (vertigo, deafness), spinal trigeminal nucleus (ipsilateral facial sensory loss), facial nucleus/fascicle (ipsilateral LMN facial palsy), middle cerebellar peduncle (ipsilateral ataxia), and spinothalamic tract (contralateral body pain/temperature loss). * Optic ataxia, ocular apraxia, and simultanagnosia *— Anchor: Right lateral caudal pons (AICA territory)*

12 Isolated Adduction Deficit on Lateral Gaze with Abducting Nystagmus

30-year-old woman • Stage 6: Brainstem Crossed • #8

Clinical Vignette: A 30-year-old woman reports intermittent double vision, particularly when looking toward the right.

- On right gaze, left eye does not adduct
- Right eye develops abducting nystagmus
- Convergence preserved
- Pupils normal
- No ptosis

Q1: CENTRAL OR PERIPHERAL?
Central (Cerebral cortex — presence of cortical signs such as aphasia, neglect, apraxia, or visual field deficits).

Q2: NEURAXIS LEVEL
Left medial longitudinal fasciculus

Q3: STRUCTURES & TRACTS INVOLVED
The MLF connects the contralateral abducens nucleus to the ipsilateral medial rectus subnucleus.

Q4: DEFINITIVE DIAGNOSIS & SYNDROME
left INO

Teaching Pearl: The MLF connects the contralateral abducens nucleus to the ipsilateral medial rectus subnucleus. Preserved convergence supports an internuclear rather than peripheral CN III lesion.

Anatomical Mechanism & Discriminators: The MLF connects the contralateral abducens nucleus to the ipsilateral medial rectus subnucleus. Preserved convergence supports an internuclear rather than peripheral CN III lesion. *— Anchor: Left medial longitudinal fasciculus*

11 Horizontal Gaze Palsy to One Side with Adduction Failure to the Other

48-year-old man • Stage 6: Brainstem Crossed • #9

Clinical Vignette: A 48-year-old man develops sudden horizontal diplopia and complains that he cannot move his eyes properly from side to side.

- Neither eye can look to the right
- On looking left, right eye fails to adduct
- Left eye abducts with nystagmus
- Vertical eye movements preserved
- Pupils normal

Q1: CENTRAL OR PERIPHERAL?
Central (Brainstem crossed syndrome — ipsilateral cranial nerve nucleus/fascicle + contralateral long tracts).

Q2: NEURAXIS LEVEL
Right dorsal pons

Q3: STRUCTURES & TRACTS INVOLVED
A right PPRF/abducens nucleus lesion causes loss of rightward gaze.

Q4: DEFINITIVE DIAGNOSIS & SYNDROME
one-and-a-half syndrome

Teaching Pearl: A right PPRF/abducens nucleus lesion causes loss of rightward gaze. Simultaneous right MLF involvement prevents the right eye from adducting during leftward gaze.

Anatomical Mechanism & Discriminators: A right PPRF/abducens nucleus lesion causes loss of rightward gaze. Simultaneous right MLF involvement prevents the right eye from adducting during leftward gaze. — *Anchor: Right dorsal pons*

127 Combined One-and-a-Half Syndrome and Complete Peripheral Facial Palsy

63-year-old man • Stage 6: Brainstem Crossed • #10

Clinical Vignette: A 63-year-old man develops severe difficulty moving his eyes horizontally along with weakness of the entire right side of his face.

- Neither eye can look to the right
- On attempted left gaze, right eye fails to adduct
- Left eye abducts with nystagmus
- Right forehead cannot wrinkle
- Right eye closure weak
- Right lower facial weakness
- Vertical eye movements preserved
- Limb power normal

Q1: CENTRAL OR PERIPHERAL?
Central (Cerebral cortex — presence of cortical signs such as aphasia, neglect, apraxia, or visual field deficits).

Q2: NEURAXIS LEVEL
Right dorsal pontine tegmentum involving PPRF/abducens nucleus, MLF and facial fascicle

Q3: STRUCTURES & TRACTS INVOLVED
One-and-a-half syndrome (1½) + ipsilateral CN VII palsy (7) = 8½.

Q4: DEFINITIVE DIAGNOSIS & SYNDROME
Eight-and-a-half syndrome

Teaching Pearl: One-and-a-half syndrome (1½) + ipsilateral CN VII palsy (7) = 8½. It is an exceptionally useful anatomical signature of a small dorsal pontine lesion.

Anatomical Mechanism & Discriminators: One-and-a-half syndrome (1½) + ipsilateral CN VII palsy (7) = 8½. It is an exceptionally useful anatomical signature of a small dorsal pontine lesion. — *Anchor: Right dorsal pontine tegmentum involving PPRF/abducens nucleus, MLF and facial fascicle*

147 Bilateral Failure of Eye Adduction on Lateral Gaze with Primary Position Exotropia

34-year-old woman with demyelinating disease • Stage 6: Brainstem Crossed • #11

Clinical Vignette: A 34-year-old woman with demyelinating disease develops severe horizontal diplopia.

- Neither eye adducts on attempted horizontal gaze
- Both abducting eyes show nystagmus
- Eyes are exotropic in primary position
- Convergence impaired or variable
- Vertical eye movements preserved
- Pupils normal

Q1: CENTRAL OR PERIPHERAL?
Central (Brainstem crossed syndrome — ipsilateral cranial nerve nucleus/fascicle + contralateral long tracts).

Q2: NEURAXIS LEVEL
Bilateral medial longitudinal fasciculi, often rostral midbrain

Q3: STRUCTURES & TRACTS INVOLVED
A single INO means one MLF.

Q4: DEFINITIVE DIAGNOSIS & SYNDROME
WEBINO syndrome — wall-eyed bilateral internuclear ophthalmoplegia

Teaching Pearl: A single INO means one MLF. Bilateral adduction failure = bilateral MLF lesions. The additional primary-position exotropia produces the characteristic “wall-eyed” appearance.

Anatomical Mechanism & Discriminators: A single INO means one MLF. Bilateral adduction failure = bilateral MLF lesions. The additional primary-position exotropia produces the characteristic “wall-eyed” appearance. — *Anchor: Bilateral medial longitudinal fasciculi, often rostral midbrain*

9 Crossed Facial and Body Pain/Temperature Loss with Dysphagia

55-year-old man suddenly • Stage 6: Brainstem Crossed • #12

Clinical Vignette: A 55-year-old man suddenly develops severe vertigo, vomiting, difficulty swallowing and an unsteady gait.

- Reduced pain and temperature over left face
- Reduced pain and temperature over right arm, trunk and leg
- Left Horner syndrome
- Hoarse voice
- Reduced left palatal elevation
- Limb power preserved
- Severe gait ataxia

Q1: CENTRAL OR PERIPHERAL?
Central (Brainstem crossed syndrome — ipsilateral cranial nerve nucleus/fascicle + contralateral long tracts).

Q2: NEURAXIS LEVEL
Left lateral medulla

Q3: STRUCTURES & TRACTS INVOLVED
Ipsilateral facial pain-temperature loss plus contralateral body pain-temperature loss is a classic lateral medullary pattern.

Q4: DEFINITIVE DIAGNOSIS & SYNDROME
Wallenberg syndrome

Teaching Pearl: Ipsilateral facial pain-temperature loss plus contralateral body pain-temperature loss is a classic lateral medullary pattern. Nucleus ambiguus involvement explains dysphagia and hoarseness.

Anatomical Mechanism & Discriminators: Ipsilateral facial pain-temperature loss plus contralateral body pain-temperature loss is a classic lateral medullary pattern. Nucleus ambiguus involvement explains dysphagia and hoarseness. — *Anchor: Left lateral medulla*

10 Tongue Wasting and Fasciculations with Contralateral Hemiplegia

63-year-old man suddenly • Stage 6: Brainstem Crossed • #13

Clinical Vignette: A 63-year-old man suddenly develops weakness of the right arm and leg and difficulty moving food within his mouth.

- Tongue deviates left on protrusion
- Left half of tongue wasted
- Fasciculations on left side of tongue
- Right arm and leg power 2/5
- Right hyperreflexia
- Right extensor plantar response
- Facial movements preserved

Q1: CENTRAL OR PERIPHERAL?
Central (Brainstem crossed syndrome — ipsilateral cranial nerve nucleus/fascicle + contralateral long tracts).

Q2: NEURAXIS LEVEL
Left medial medulla

Q3: STRUCTURES & TRACTS INVOLVED
Ipsilateral LMN hypoglossal palsy plus contralateral pyramidal weakness localises to the medial medulla.

Q4: DEFINITIVE DIAGNOSIS & SYNDROME
medial medullary syndrome

Teaching Pearl: Ipsilateral LMN hypoglossal palsy plus contralateral pyramidal weakness localises to the medial medulla.

Anatomical Mechanism & Discriminators: Ipsilateral LMN hypoglossal palsy plus contralateral pyramidal weakness localises to the medial medulla. — *Anchor: Left medial medulla*

29 Inability of Both Eyes to Look Conjugate Toward One Side

72-year-old man • Stage 6: Brainstem Crossed • #14

Clinical Vignette: A 72-year-old man develops sudden double vision and weakness of the left side of his body.

- Both eyes cannot look toward the right
- Both eyes can look toward the left
- Vertical gaze preserved
- Left arm and leg weak
- Left-sided hyperreflexia
- Pupils normal

Q1: CENTRAL OR PERIPHERAL?
Mixed Central & Peripheral (Simultaneous Upper & Lower Motor Neuron degeneration with sensory sparing).

Q2: NEURAXIS LEVEL
Right PPRF or abducens nucleus plus nearby corticospinal tract in the pons

Q3: STRUCTURES & TRACTS INVOLVED
A lesion of the abducens nucleus or PPRF causes conjugate gaze palsy toward the side of the lesion.

Q4: DEFINITIVE DIAGNOSIS & SYNDROME
Right pontine gaze palsy with contralateral hemiparesis

Teaching Pearl: A lesion of the abducens nucleus or PPRF causes conjugate gaze palsy toward the side of the lesion. Associated contralateral hemiparesis confirms involvement of adjacent corticospinal fibres within the pons.

Anatomical Mechanism & Discriminators: A lesion of the abducens nucleus or PPRF causes conjugate gaze palsy toward the side of the lesion. Associated contralateral hemiparesis confirms involvement of adjacent corticospinal fibres within the pons. — *Anchor: Right PPRF or abducens nucleus plus nearby corticospinal tract in the pons*

30 Impaired Upward Gaze, Convergence-Retraction Nystagmus and Light-Near Dissociation

24-year-old man • Stage 6: Brainstem Crossed • #15

Clinical Vignette: A 24-year-old man develops progressive headache and double vision. His family notices that he has difficulty looking upward.

- Impaired upward gaze in both eyes
- Convergence-retraction nystagmus on attempted upgaze
- Pupils poorly reactive to light but constrict with accommodation
- Eyelid retraction may be present
- Horizontal eye movements relatively preserved
- No limb weakness

Q1: CENTRAL OR PERIPHERAL?

Central (Brainstem crossed syndrome — ipsilateral cranial nerve nucleus/fascicle + contralateral long tracts).

Q2: NEURAXIS LEVEL

Dorsal rostral midbrain / pretectal region

Q3: STRUCTURES & TRACTS INVOLVED

Vertical gaze pathways lie in the dorsal midbrain.

Q4: DEFINITIVE DIAGNOSIS & SYNDROME

Parinaud syndrome

Teaching Pearl: Vertical gaze pathways lie in the dorsal midbrain. A lesion around the superior colliculus/pretectal region can produce: vertical gaze palsy convergence-retraction nystagmus light-near dissociation eyelid retraction This is a classic dorsal midbrain syndrome.

Anatomical Mechanism & Discriminators: Vertical gaze pathways lie in the dorsal midbrain. A lesion around the superior colliculus/pretectal region can produce: vertical gaze palsy convergence-retraction nystagmus light-near dissociation eyelid retraction This is a classic dorsal midbrain syndrome. — *Anchor: Dorsal rostral midbrain / pretectal region*

38 Vertical Diplopia with Skew Deviation and Ocular Torsion

71-year-old woman • Stage 6: Brainstem Crossed • #16

Clinical Vignette: A 71-year-old woman develops sudden dizziness and vertical diplopia after a small brainstem stroke.

- One eye positioned slightly higher than the other
- Vertical diplopia
- Pattern does not correspond to isolated CN III or IV palsy
- Ocular torsion present
- Associated head tilt
- Mild imbalance
- Pupils normal

Q1: CENTRAL OR PERIPHERAL?

Central (Brainstem crossed syndrome — ipsilateral cranial nerve nucleus/fascicle + contralateral long tracts).

Q2: NEURAXIS LEVEL

Brainstem vestibular pathways

Q3: STRUCTURES & TRACTS INVOLVED

Skew deviation results from disruption of otolith-ocular pathways in the brainstem or cerebellum.

Q4: DEFINITIVE DIAGNOSIS & SYNDROME

Skew deviation

Teaching Pearl: Skew deviation results from disruption of otolith-ocular pathways in the brainstem or cerebellum. Unlike trochlear palsy, the vertical misalignment often does not follow the expected pattern of a single extraocular muscle. It is therefore an important clue to a central lesion.

Anatomical Mechanism & Discriminators: Skew deviation results from disruption of otolith-ocular pathways in the brainstem or cerebellum. Unlike trochlear palsy, the vertical misalignment often does not follow the expected pattern of a single extraocular muscle. It is therefore an important clue to a central lesion. — *Anchor: Brainstem vestibular pathways*

51 Total Quadriplegia and Anarthria with Preserved Vertical Eye Movements and Blinking

60-year-old hypertensive man suddenly collapses and • Stage 6: Brainstem Crossed • #17

Clinical Vignette: A 60-year-old hypertensive man suddenly collapses and becomes completely unable to move or speak. His family notices that he seems awake and can answer questions by blinking.

- Complete paralysis of all four limbs
- Bilateral facial weakness
- Anarthria — unable to speak
- Unable to swallow
- Horizontal eye movements absent
- Vertical eye movements and blinking preserved
- Consciousness intact
- Communicates reliably using eye blinks

Q1: CENTRAL OR PERIPHERAL?

Central (Brainstem crossed syndrome — ipsilateral cranial nerve nucleus/fascicle + contralateral long tracts).

Q2: NEURAXIS LEVEL

Bilateral ventral pons (basis pontis)

Q3: STRUCTURES & TRACTS INVOLVED

Bilateral damage to the ventral pons destroys the corticospinal and corticobulbar tracts, causing quadriplegia, facial paralysis and anarthria.

Q4: DEFINITIVE DIAGNOSIS & SYNDROME

Locked-in syndrome

Teaching Pearl: Bilateral damage to the ventral pons destroys the corticospinal and corticobulbar tracts, causing quadriplegia, facial paralysis and anarthria. The dorsal tegmentum — containing the reticular activating system and the pathways for vertical gaze — is spared. Hence the patient remains fully conscious and can communicate only through blinking and vertical eye movements.

Anatomical Mechanism & Discriminators: Bilateral damage to the ventral pons destroys the corticospinal and corticobulbar tracts, causing quadriplegia, facial paralysis and anarthria. The dorsal tegmentum — containing the reticular activating system and the pathways for vertical gaze — is spared. Hence the patient remains fully conscious and can communicate only through blinking and vertical eye movements. — *Anchor: Bilateral ventral pons (basis pontis)*

CURRICULUM CHAPTER 7 • STAGE 7

Stage 7 — Cerebellum vs Sensory Ataxia & Vestibular Balance

Chapter Learning Objective

Recognize that ataxia is not a localisation; differentiate cerebellar hemisphere, midline vermis, dorsal column, and vestibular causes.

Pedagogical Localisation Anchor

Cerebellar -> dysmetria, intention tremor (Romberg negative); Sensory ataxia -> lost proprioception, Romberg positive, worse in dark.

Included Clinical Vignettes (5 Cases):
Case 3: Unilateral Incoordination and Veering Toward One Side • Case 105: Severe Truncal Ataxia and Titubation with Intact Coordination of Limbs in Bed • Case 109: Acute Severe Prolonged Vertigo and Spontaneous Horizontal Nystagmus Without Deafness • Case 92: Acute Triad of Ataxia, Areflexia, and External Ophthalmoplegia • Case 110: Acute Confusion, Ataxia, and Nystagmus with Ophthalmoparesis in Alcohol Dependency

3 Unilateral Incoordination and Veering Toward One Side

64-year-old woman • Stage 7: Cerebellum vs Ataxia • #1

Clinical Vignette: A 64-year-old woman develops acute unsteadiness while walking and repeatedly veers toward the right.

- Broad-based gait
- Falls toward the right
- Right finger-nose dysmetria
- Right heel-knee-shin impairment
- Right dysdiadochokinesia
- Right intention tremor
- Normal power and sensation

Q1: CENTRAL OR PERIPHERAL?
Central (Cerebral cortex — presence of cortical signs such as aphasia, neglect, apraxia, or visual field deficits).

Q2: NEURAXIS LEVEL
Right cerebellar hemisphere

Q3: STRUCTURES & TRACTS INVOLVED
Cerebellar limb signs are predominantly ipsilateral.

Q4: DEFINITIVE DIAGNOSIS & SYNDROME
Right cerebellar hemisphere

Teaching Pearl: Cerebellar limb signs are predominantly ipsilateral. Right-sided limb incoordination therefore localises to the right cerebellar hemisphere.

Anatomical Mechanism & Discriminators: Cerebellar limb signs are predominantly ipsilateral. Right-sided limb incoordination therefore localises to the right cerebellar hemisphere. — *Anchor: Right cerebellar hemisphere*

105 Severe Truncal Ataxia and Titubation with Intact Coordination of Limbs in Bed

57-year-old alcoholic man • Stage 7: Cerebellum vs Ataxia • #2

Clinical Vignette: A 57-year-old alcoholic man is unable to sit or stand without falling, though his arms move well on finger-nose testing.

- Truncal ataxia with titubation
- Wide-based gait, falls to either side
- Limb ataxia minimal or absent
- No nystagmus on lateral gaze (or only mild)
- No weakness or sensory loss
- Reflexes normal

Q1: CENTRAL OR PERIPHERAL?
Central (Cerebral cortex — presence of cortical signs such as aphasia, neglect, apraxia, or visual field deficits).

Q2: NEURAXIS LEVEL
Cerebellar vermis (midline) / anterior lobe

Q3: STRUCTURES & TRACTS INVOLVED
The vermis coordinates axial and truncal stability, while the hemispheres coordinate ipsilateral limbs.

Q4: DEFINITIVE DIAGNOSIS & SYNDROME
Vermis (midline cerebellar) syndrome

Teaching Pearl: The vermis coordinates axial and truncal stability, while the hemispheres coordinate ipsilateral limbs. Selective truncal ataxia with preserved limb coordination is characteristic of midline vermal disease, classically alcoholic cerebellar degeneration.

Anatomical Mechanism & Discriminators: The vermis coordinates axial and truncal stability, while the hemispheres coordinate ipsilateral limbs. Selective truncal ataxia with preserved limb coordination is characteristic of midline vermal disease, classically alcoholic cerebellar degeneration. — *Anchor: Cerebellar vermis (midline) / anterior lobe*

109 Acute Severe Prolonged Vertigo and Spontaneous Horizontal Nystagmus Without Deafness

45-year-old man • Stage 7: Cerebellum vs Ataxia • #3

Clinical Vignette: A 45-year-old man wakes with severe continuous vertigo, nausea and vomiting. He walks unsteadily but has no weakness or hearing loss.

- Severe spontaneous horizontal-torsional nystagmus
- Nystagmus suppressed by visual fixation
- Head-impulse test positive to the right
- Falls toward the right on standing
- Hearing intact
- No limb weakness and no brainstem signs
- Skew deviation absent

Q1: CENTRAL OR PERIPHERAL?
Peripheral (Lower motor neuron / peripheral nerve / plexus / root distribution).

Q2: NEURAXIS LEVEL
Right vestibular nerve / labyrinth

Q3: STRUCTURES & TRACTS INVOLVED
The combination of an abnormal head-impulse test, unidirectional nystagmus that suppresses with fixation and preserved hearing localises vertigo to the peripheral vestibular nerve rather than the brainstem (where skew deviation and other brainstem signs would appear).

Q4: DEFINITIVE DIAGNOSIS & SYNDROME
Vestibular neuritis

Teaching Pearl: The combination of an abnormal head-impulse test, unidirectional nystagmus that suppresses with fixation and preserved hearing localises vertigo to the peripheral vestibular nerve rather than the brainstem (where skew deviation and other brainstem signs would appear).

Anatomical Mechanism & Discriminators: The combination of an abnormal head-impulse test, unidirectional nystagmus that suppresses with fixation and preserved hearing localises vertigo to the peripheral vestibular nerve rather than the brainstem (where skew deviation and other brainstem signs would appear). — *Anchor: Right vestibular nerve / labyrinth*

92 Acute Triad of Ataxia, Areflexia, and External Ophthalmoplegia

40-year-old man • Stage 7: Cerebellum vs Ataxia • #4

Clinical Vignette: A 40-year-old man develops double vision and unsteadiness ten days after a sore throat.

- Bilateral partial ophthalmoplegia not fitting any single nerve
- Bilateral ptosis
- Global areflexia
- Severe gait and limb ataxia
- Limb power normal
- Sensation normal

Q1: CENTRAL OR PERIPHERAL?
Peripheral (Lower motor neuron / peripheral nerve / plexus / root distribution).

Q2: NEURAXIS LEVEL
Peripheral nerves — GQ1b ganglioside-rich ocular motor and sensory nerves

Q3: STRUCTURES & TRACTS INVOLVED
The triad of ophthalmoplegia, ataxia and areflexia localises to peripheral nerves rather than the brainstem.

Q4: DEFINITIVE DIAGNOSIS & SYNDROME
Miller Fisher syndrome

Teaching Pearl: The triad of ophthalmoplegia, ataxia and areflexia localises to peripheral nerves rather than the brainstem. Anti-GQ1b antibodies attack ocular motor, sensory and muscle-spindle afferent nerves, producing the distinctive combination without pyramidal signs.

Anatomical Mechanism & Discriminators: The triad of ophthalmoplegia, ataxia and areflexia localises to peripheral nerves rather than the brainstem. Anti-GQ1b antibodies attack ocular motor, sensory and muscle-spindle afferent nerves, producing the distinctive combination without pyramidal signs. — *Anchor: Peripheral nerves — GQ1b ganglioside-rich ocular motor and sensory nerves*

110 Acute Confusion, Ataxia, and Nystagmus with Ophthalmoparesis in Alcohol Dependency

40-year-old malnourished alcoholic • Stage 7: Cerebellum vs Ataxia • #5

Clinical Vignette: A 40-year-old malnourished alcoholic is brought in confused, with an unsteady gait and difficulty looking from side to side.

- Global confusion and inattention
- Horizontal and vertical nystagmus
- Bilateral lateral rectus weakness (CN VI)
- Gait and truncal ataxia
- Memory impairment
- No limb weakness

Q1: CENTRAL OR PERIPHERAL?

Central (Deep subcortical / thalamocortical / internal capsule — dense deficits with absent cortical signs).

Q2: NEURAXIS LEVEL

Periaqueductal grey matter, mammillary bodies and medial dorsal thalamus

Q3: STRUCTURES & TRACTS INVOLVED

The triad of ophthalmoplegia, ataxia and confusion results from thiamine deficiency damaging the periaqueductal grey, vestibular and oculomotor nuclei, mammillary bodies and medial thalamus.

Q4: DEFINITIVE DIAGNOSIS & SYNDROME

Wernicke encephalopathy

Teaching Pearl: The triad of ophthalmoplegia, ataxia and confusion results from thiamine deficiency damaging the periaqueductal grey, vestibular and oculomotor nuclei, mammillary bodies and medial thalamus. It is a medical emergency, and the clinical diagnosis alone mandates immediate thiamine. Worked for 1m 9s Yes. I'd continue from 111, since your file already reaches 110. I've deliberately weighted these toward cord/conus/cauda localisation, with the last few returning to brainstem cranial-nerve and stroke patterns.

Anatomical Mechanism & Discriminators: The triad of ophthalmoplegia, ataxia and confusion results from thiamine deficiency damaging the periaqueductal grey, vestibular and oculomotor nuclei, mammillary bodies and medial thalamus. It is a medical emergency, and the clinical diagnosis alone mandates immediate thiamine. Worked for 1m 9s Yes. I'd continue from 111, since your file already reaches 110. I've deliberately weighted these toward cord/conus/cauda localisation, with the last few returning to brainstem cranial-nerve and stroke patterns. — *Anchor: Periaqueductal grey matter, mammillary bodies and medial dorsal thalamus*

CURRICULUM CHAPTER 8 • STAGE 8

Stage 8 — Cerebral Cortex: Syndromes by Lobe & Language Networks

Chapter Learning Objective

Analyze cortical signs by anatomical lobe (Frontal, Parietal, Temporal, Occipital, Association) and the Aphasia Tree.

Pedagogical Localisation Anchor

Cortical signs (aphasia, apraxia, neglect, visual field defects, agnosias) definitively place lesion in cortex.

Included Clinical Vignettes (27 Cases):
Case 4: Nonfluent Speech with Asymmetric Face and Arm Weakness • Case 23: Nonfluent Hesitant Speech with Preserved Repetition and Right Leg Weakness • Case 104: Conjugate Deviation of Eyes Looking Toward the Side of the Hemispheric Lesion • Case 85: Autonomous Involuntary Goal-Directed Movements of One Hand • Case 135: Sudden Isolated Weakness of Hand and Finger Extension Mimicking Peripheral Nerve Injury • Case 137: Inability to Voluntarily Smile, Swallow or Speak Despite Normal Involuntary Responses • *...and 21 more cases*

4 Nonfluent Speech with Asymmetric Face and Arm Weakness

Adult patient • Stage 8: Cortex by Lobe • #1

Clinical Vignette: A 70-year-old right-handed man suddenly becomes unable to speak properly and develops weakness of the right face and arm.

- Speech nonfluent and effortful
- Comprehension relatively preserved
- Repetition impaired
- Right lower facial weakness
- Right arm power 2/5
- Right leg power 4+/5
- Right-sided hyperreflexia

Q1: CENTRAL OR PERIPHERAL?
Central (Cerebral cortex — presence of cortical signs such as aphasia, neglect, apraxia, or visual field deficits).

Q2: NEURAXIS LEVEL
Left frontal lobe

Q3: STRUCTURES & TRACTS INVOLVED
Nonfluent aphasia indicates dominant inferior frontal involvement.

Q4: DEFINITIVE DIAGNOSIS & SYNDROME
dominant hemisphere, Broca region / superior division MCA

Teaching Pearl: Nonfluent aphasia indicates dominant inferior frontal involvement. Greater face-arm than leg weakness suggests adjacent lateral motor cortex.

Anatomical Mechanism & Discriminators: Nonfluent aphasia indicates dominant inferior frontal involvement. Greater face-arm than leg weakness suggests adjacent lateral motor cortex. — *Anchor: Left frontal lobe*

23 Nonfluent Hesitant Speech with Preserved Repetition and Right Leg Weakness

64-year-old woman • Stage 8: Cortex by Lobe • #2

Clinical Vignette: A 64-year-old woman becomes unusually quiet after an acute stroke. She understands questions but gives very short responses.

- Nonfluent, sparse speech
- Comprehension relatively preserved
- Repetition preserved
- Naming impaired
- Mild right leg weakness greater than arm weakness
- Reduced spontaneous initiation

Q1: CENTRAL OR PERIPHERAL?
Central (Cerebral cortex — presence of cortical signs such as aphasia, neglect, apraxia, or visual field deficits).

Q2: NEURAXIS LEVEL
Dominant medial frontal/anterior watershed region

Q3: STRUCTURES & TRACTS INVOLVED
It resembles Broca aphasia, but repetition is preserved.

Q4: DEFINITIVE DIAGNOSIS & SYNDROME
Transcortical motor aphasia

Teaching Pearl: It resembles Broca aphasia, but repetition is preserved. Associated reduced initiation and leg-predominant weakness can occur with medial frontal or ACA–MCA watershed involvement.

Anatomical Mechanism & Discriminators: It resembles Broca aphasia, but repetition is preserved. Associated reduced initiation and leg-predominant weakness can occur with medial frontal or ACA–MCA watershed involvement. — *Anchor: Dominant medial frontal/anterior watershed region*

104 Conjugate Deviation of Eyes Looking Toward the Side of the Hemispheric Lesion

69-year-old man • Stage 8: Cortex by Lobe • #3

Clinical Vignette: A 69-year-old man has an acute left hemiparesis. On examination his eyes are forcibly turned toward the right.

- Conjugate gaze deviation to the right
- Cannot voluntarily look to the left
- Left facial and limb weakness
- Left hyperreflexia and extensor plantar response
- Oculocephalic reflex restores leftward gaze

Q1: CENTRAL OR PERIPHERAL?
Central (Cerebral cortex — presence of cortical signs such as aphasia, neglect, apraxia, or visual field deficits).

Q2: NEURAXIS LEVEL
Right frontal eye field (premotor area 8)

Q3: STRUCTURES & TRACTS INVOLVED
The frontal eye field drives voluntary conjugate gaze to the opposite side.

Q4: DEFINITIVE DIAGNOSIS & SYNDROME
Frontal eye field lesion with contralateral hemiparesis

Teaching Pearl: The frontal eye field drives voluntary conjugate gaze to the opposite side. Its destruction causes the eyes to deviate toward the side of the lesion ("the patient looks at their lesion"), while the preserved vestibulo-ocular reflex still moves the eyes contralaterally.

Anatomical Mechanism & Discriminators: The frontal eye field drives voluntary conjugate gaze to the opposite side. Its destruction causes the eyes to deviate toward the side of the lesion ("the patient looks at their lesion"), while the preserved vestibulo-ocular reflex still moves the eyes contralaterally. — *Anchor: Right frontal eye field (premotor area 8)*

85 Autonomous Involuntary Goal-Directed Movements of One Hand

69-year-old woman who suffered an anterior cerebral artery stroke • Stage 8: Cortex by Lobe • #4

Clinical Vignette: A 69-year-old woman who suffered an anterior cerebral artery stroke complains that her left hand acts independently, grasping door handles or unbuttoning clothes that her right hand has just fastened.

- Involuntary, purposeful, complex movements of the left hand ("intermanual conflict")
- Left hand exhibits a strong, involuntary grasp reflex
- Limb power 5/5 bilaterally
- Normal primary touch, vibration, and joint-position sense
- Patient regards the left hand as foreign or disobedient

Q1: CENTRAL OR PERIPHERAL?
Central (Cerebral cortex — presence of cortical signs such as aphasia, neglect, apraxia, or visual field deficits).

Q2: NEURAXIS LEVEL
Anterior corpus callosum and/or supplementary motor area (medial frontal cortex)

Q3: STRUCTURES & TRACTS INVOLVED
Disconnection between the dominant and non-dominant motor planning regions (callosal) or loss of supplementary motor area inhibitory control over motor execution leads to autonomous, goal-directed limb movements outside conscious voluntary intent.

Q4: DEFINITIVE DIAGNOSIS & SYNDROME
Alien hand syndrome

Teaching Pearl: Disconnection between the dominant and non-dominant motor planning regions (callosal) or loss of supplementary motor area inhibitory control over motor execution leads to autonomous, goal-directed limb movements outside conscious voluntary intent. 86 * Inability to form "OK" sign with normal sensation

Anatomical Mechanism & Discriminators: Disconnection between the dominant and non-dominant motor planning regions (callosal) or loss of supplementary motor area inhibitory control over motor execution leads to autonomous, goal-directed limb movements outside conscious voluntary intent. 86 * Inability to form "OK" sign with normal sensation — *Anchor: Anterior corpus callosum and/or supplementary motor area (medial frontal cortex)*

135 Sudden Isolated Weakness of Hand and Finger Extension Mimicking Peripheral Nerve Injury

Adult patient • Stage 8: Cortex by Lobe • #5

Clinical Vignette: A 66-year-old man suddenly drops his cup and finds his left fingers weak. He has no numbness or facial symptoms.

- Marked weakness of left finger extension
- Mild weakness of finger abduction
- Wrist extension mildly weak
- Shoulder and elbow power normal
- Sensation completely normal
- Reflexes slightly brisk on left
- Subtle left Babinski response
- No facial weakness

Q1: CENTRAL OR PERIPHERAL?
Central (Cerebral cortex — presence of cortical signs such as aphasia, neglect, apraxia, or visual field deficits).

Q2: NEURAXIS LEVEL
Right precentral gyrus — cortical "hand knob"

Q3: STRUCTURES & TRACTS INVOLVED
A very small cortical motor infarct may mimic radial, ulnar or median neuropathy.

Q4: DEFINITIVE DIAGNOSIS & SYNDROME
Cortical hand-knob infarction

Teaching Pearl: A very small cortical motor infarct may mimic radial, ulnar or median neuropathy. The clues to cortex are: abrupt onset no corresponding sensory nerve territory subtle UMN signs weakness crossing several peripheral nerves

Anatomical Mechanism & Discriminators: A very small cortical motor infarct may mimic radial, ulnar or median neuropathy. The clues to cortex are: abrupt onset no corresponding sensory nerve territory subtle UMN signs weakness crossing several peripheral nerves — *Anchor: Right precentral gyrus — cortical "hand knob"*

137 Inability to Voluntarily Smile, Swallow or Speak Despite Normal Involuntary Responses

70-year-old man with previous bilateral strokes • Stage 8: Cortex by Lobe • #6

Clinical Vignette: A 70-year-old man with previous bilateral strokes becomes unable to speak or voluntarily move his lower face and tongue.

- Cannot voluntarily smile
- Cannot voluntarily protrude tongue
- Severe dysarthria/anarthria
- Severe voluntary swallowing difficulty
- Cannot voluntarily cough on command
- Yet smiles spontaneously at a joke
- Yawning produces normal facial movement
- Jaw jerk brisk
- Limb weakness relatively mild

Q1: CENTRAL OR PERIPHERAL?
Central (Cerebral cortex — presence of cortical signs such as aphasia, neglect, apraxia, or visual field deficits).

Q2: NEURAXIS LEVEL
Bilateral frontal opercular cortex / corticobulbar cortical pathways

Q3: STRUCTURES & TRACTS INVOLVED
The striking feature is automatic-voluntary dissociation.

Q4: DEFINITIVE DIAGNOSIS & SYNDROME
Foix–Chavany–Marie syndrome — anterior opercular syndrome

Teaching Pearl: The striking feature is automatic-voluntary dissociation. Voluntary corticobulbar control is lost, while emotional and automatic facial movements remain preserved.

Anatomical Mechanism & Discriminators: The striking feature is automatic-voluntary dissociation. Voluntary corticobulbar control is lost, while emotional and automatic facial movements remain preserved. — *Anchor: Bilateral frontal opercular cortex / corticobulbar cortical pathways*

132 Severe Bilateral Upper Limb Paralysis with Sparing of Face and Lower Limbs

72-year-old man • Stage 8: Cortex by Lobe • #7

Clinical Vignette: A 72-year-old man has prolonged severe hypotension during cardiac surgery. When he awakens, he cannot lift either arm.

- Severe bilateral arm weakness, especially shoulders and upper arms
- Hands less affected
- Legs nearly normal
- Facial movements preserved
- Sensation relatively preserved
- Reflexes brisk in weak arms
- No sensory level
- No cranial-nerve palsy

Q1: CENTRAL OR PERIPHERAL?
Central (Cerebral cortex — presence of cortical signs such as aphasia, neglect, apraxia, or visual field deficits).

Q2: NEURAXIS LEVEL
Bilateral ACA–MCA watershed motor cortex

Q3: STRUCTURES & TRACTS INVOLVED
The proximal upper-limb motor areas lie near the ACA–MCA border zone.

Q4: DEFINITIVE DIAGNOSIS & SYNDROME
Man-in-the-barrel syndrome

Teaching Pearl: The proximal upper-limb motor areas lie near the ACA–MCA border zone. Global hypoperfusion can damage these regions bilaterally, producing striking brachial diplegia with relative sparing of face and legs.

Anatomical Mechanism & Discriminators: The proximal upper-limb motor areas lie near the ACA–MCA border zone. Global hypoperfusion can damage these regions bilaterally, producing striking brachial diplegia with relative sparing of face and legs. — *Anchor: Bilateral ACA–MCA watershed motor cortex*

8 Inattention to Left Visual Field and Left-Sided Body Space

72-year-old woman • Stage 8: Cortex by Lobe • #8

Clinical Vignette: A 72-year-old woman has an acute stroke. Her family notices that she ignores people and objects placed on her left side.

- Ignores objects on the left
- Eats only from the right half of the plate
- Extinction of left-sided stimulus on bilateral simultaneous stimulation
- Left homonymous hemianopia
- Mild left-sided weakness
- Speech fluency and comprehension preserved

Q1: CENTRAL OR PERIPHERAL?
Central (Cerebral cortex — presence of cortical signs such as aphasia, neglect, apraxia, or visual field deficits).

Q2: NEURAXIS LEVEL
Right parietal cortex

Q3: STRUCTURES & TRACTS INVOLVED
Hemispatial neglect is characteristic of damage to the nondominant, usually right, parietal association cortex.

Q4: DEFINITIVE DIAGNOSIS & SYNDROME
nondominant hemisphere

Teaching Pearl: Hemispatial neglect is characteristic of damage to the nondominant, usually right, parietal association cortex.

Anatomical Mechanism & Discriminators: Hemispatial neglect is characteristic of damage to the nondominant, usually right, parietal association cortex. — *Anchor: Right parietal cortex*

32 Acalculia, Agraphia, Finger Agnosia, and Right-Left Disorientation

57-year-old accountant suddenly • Stage 8: Cortex by Lobe • #9

Clinical Vignette: A 57-year-old accountant suddenly becomes unable to perform simple calculations and repeatedly confuses his right and left sides.

- Acalculia
- Agraphia
- Finger agnosia
- Right-left disorientation
- Speech relatively fluent
- Motor examination normal
- Primary sensation normal

Q1: CENTRAL OR PERIPHERAL?
Central (Cerebral cortex — presence of cortical signs such as aphasia, neglect, apraxia, or visual field deficits).

Q2: NEURAXIS LEVEL
Dominant inferior parietal lobule, especially angular gyrus

Q3: STRUCTURES & TRACTS INVOLVED
The classic tetrad is: acalculia agraphia finger agnosia right-left disorientation This strongly localises to the dominant angular gyrus, usually the left hemisphere.

Q4: DEFINITIVE DIAGNOSIS & SYNDROME
Gerstmann syndrome

Teaching Pearl: The classic tetrad is: acalculia agraphia finger agnosia right-left disorientation This strongly localises to the dominant angular gyrus, usually the left hemisphere.

Anatomical Mechanism & Discriminators: The classic tetrad is: acalculia agraphia finger agnosia right-left disorientation This strongly localises to the dominant angular gyrus, usually the left hemisphere. — *Anchor: Dominant inferior parietal lobule, especially angular gyrus*

33 Inability to Perform Learned Motor Acts Despite Normal Muscle Power

Adult patient • Stage 8: Cortex by Lobe • #10

Clinical Vignette: A 65-year-old man recovers from a mild stroke. He can move his limbs normally but cannot demonstrate how to use a toothbrush when asked.

- Normal power
- Normal coordination
- Normal primary sensation
- Understands the command
- Unable to pantomime brushing teeth
- Unable to demonstrate use of a key on command
- Can sometimes perform the action spontaneously

Q1: CENTRAL OR PERIPHERAL?
Central (Cerebral cortex — presence of cortical signs such as aphasia, neglect, apraxia, or visual field deficits).

Q2: NEURAXIS LEVEL
Dominant parietal-premotor network

Q3: STRUCTURES & TRACTS INVOLVED
The patient understands the task and has sufficient strength, coordination and sensation, yet cannot perform a learned motor act on command.

Q4: DEFINITIVE DIAGNOSIS & SYNDROME
Ideomotor apraxia

Teaching Pearl: The patient understands the task and has sufficient strength, coordination and sensation, yet cannot perform a learned motor act on command. This indicates dysfunction of the dominant parietal-premotor praxis network.

Anatomical Mechanism & Discriminators: The patient understands the task and has sufficient strength, coordination and sensation, yet cannot perform a learned motor act on command. This indicates dysfunction of the dominant parietal-premotor praxis network. — *Anchor: Dominant parietal-premotor network*

42 Inability to Identify Objects by Touch Despite Intact Primary Sensation

Adult patient • Stage 8: Cortex by Lobe • #11

Clinical Vignette: A 62-year-old woman recovers from mild right-sided weakness after a stroke. She now complains that she cannot identify objects placed in her right hand unless she looks at them.

- Primary touch sensation preserved
- Pain and temperature preserved
- Joint-position sense reasonably preserved
- Unable to identify a coin or key placed in right hand
- Graphesthesia impaired in right palm
- Two-point discrimination impaired
- Power near normal

Q1: CENTRAL OR PERIPHERAL?
Central (Cerebral cortex — presence of cortical signs such as aphasia, neglect, apraxia, or visual field deficits).

Q2: NEURAXIS LEVEL
Left parietal sensory association cortex

Q3: STRUCTURES & TRACTS INVOLVED
The patient can feel the object but cannot interpret the sensory information.

Q4: DEFINITIVE DIAGNOSIS & SYNDROME
Cortical sensory dysfunction / astereognosis

Teaching Pearl: The patient can feel the object but cannot interpret the sensory information. Preserved primary sensation with impaired stereognosis and graphesthesia indicates a lesion of the contralateral parietal association cortex.

Anatomical Mechanism & Discriminators: The patient can feel the object but cannot interpret the sensory information. Preserved primary sensation with impaired stereognosis and graphesthesia indicates a lesion of the contralateral parietal association cortex. — *Anchor: Left parietal sensory association cortex*

5 Fluent but Meaningless Speech with Impaired Comprehension

66-year-old woman • Stage 8: Cortex by Lobe • #12

Clinical Vignette: A 66-year-old woman is brought because she suddenly started speaking continuously but her family cannot understand what she is saying.

- Fluent speech with normal rhythm
- Frequent paraphasic errors
- Poor comprehension
- Unable to follow simple commands
- Repetition severely impaired
- Right superior quadrantanopia
- Limb power normal

Q1: CENTRAL OR PERIPHERAL?
Central (Cerebral cortex — presence of cortical signs such as aphasia, neglect, apraxia, or visual field deficits).

Q2: NEURAXIS LEVEL
Left posterior temporal lobe

Q3: STRUCTURES & TRACTS INVOLVED
Fluent speech with severely impaired comprehension localises to the dominant posterior superior temporal region.

Q4: DEFINITIVE DIAGNOSIS & SYNDROME
Wernicke aphasia

Teaching Pearl: Fluent speech with severely impaired comprehension localises to the dominant posterior superior temporal region. Superior quadrantanopia may occur from nearby Meyer-loop involvement.

Anatomical Mechanism & Discriminators: Fluent speech with severely impaired comprehension localises to the dominant posterior superior temporal region. Superior quadrantanopia may occur from nearby Meyer-loop involvement. — *Anchor: Left posterior temporal lobe*

24 Fluent Speech with Severely Impaired Comprehension but Intact Repetition

70-year-old man suddenly begins speaking fluently but gives inappropriate answers and seems unable to understand what • Stage 8: Cortex by Lobe • #13

Clinical Vignette: A 70-year-old man suddenly begins speaking fluently but gives inappropriate answers and seems unable to understand what is being said.

- Fluent speech
- Poor comprehension
- Naming impaired
- Repetition surprisingly preserved
- Frequent semantic paraphasias
- No major motor deficit

Q1: CENTRAL OR PERIPHERAL?
Central (Cerebral cortex — presence of cortical signs such as aphasia, neglect, apraxia, or visual field deficits).

Q2: NEURAXIS LEVEL
Dominant posterior temporal-parietal association cortex, outside the classic perisylvian repetition pathway

Q3: STRUCTURES & TRACTS INVOLVED
It resembles Wernicke aphasia, except repetition remains intact.

Q4: DEFINITIVE DIAGNOSIS & SYNDROME
Transcortical sensory aphasia

Teaching Pearl: It resembles Wernicke aphasia, except repetition remains intact. That single finding helps separate transcortical sensory aphasia from Wernicke aphasia.

Anatomical Mechanism & Discriminators: It resembles Wernicke aphasia, except repetition remains intact. That single finding helps separate transcortical sensory aphasia from Wernicke aphasia. — *Anchor: Dominant posterior temporal-parietal association cortex, outside the classic perisylvian repetition pathway*

79 Fluent Speech and Reading with Inability to Comprehend Spoken Words

Adult patient • Stage 8: Cortex by Lobe • #14

Clinical Vignette: A 61-year-old man suddenly stops understanding what people say to him. He can read newspapers, write letters, and speak fluently, and his hearing tests are normal.

- Fluent, grammatically correct spontaneous speech
- Normal reading comprehension and written expression
- Inability to comprehend or repeat spoken words
- Preserved ability to recognize non-verbal sounds (e.g., telephone ring, car horn)
- Normal pure-tone audiogram

Q1: CENTRAL OR PERIPHERAL?
Central (Cerebral cortex — presence of cortical signs such as aphasia, neglect, apraxia, or visual field deficits).

Q2: NEURAXIS LEVEL
Bilateral posterior superior temporal gyri (auditory association cortex) or dominant subcortical auditory radiation disconnection

Q3: STRUCTURES & TRACTS INVOLVED
Primary auditory input cannot access the dominant language area (Wernicke area), leaving spoken language comprehension and repetition isolated, while internal language generation and reading pathways remain intact.

Q4: DEFINITIVE DIAGNOSIS & SYNDROME
Pure word deafness (auditory verbal agnosia)

Teaching Pearl: Primary auditory input cannot access the dominant language area (Wernicke area), leaving spoken language comprehension and repetition isolated, while internal language generation and reading pathways remain intact. 80 * Loss of gag, hoarseness, and shoulder droop

Anatomical Mechanism & Discriminators: Primary auditory input cannot access the dominant language area (Wernicke area), leaving spoken language comprehension and repetition isolated, while internal language generation and reading pathways remain intact. 80 * Loss of gag, hoarseness, and shoulder droop — *Anchor: Bilateral posterior superior temporal gyri (auditory association cortex) or dominant subcortical auditory radiation disconnection*

107 Dense Inability to Form New Declarative Memories Following Herpes Encephalitis

Adult patient • Stage 8: Cortex by Lobe • #15

Clinical Vignette: A 44-year-old man recovered from herpes simplex encephalitis but can no longer remember events from minutes ago, while his older memories remain intact.

- Profound anterograde amnesia
- Inability to retain new information over minutes
- Confabulation may occur
- Immediate digit span preserved
- Remote memory relatively preserved
- Language and motor function intact

Q1: CENTRAL OR PERIPHERAL?
Central (Cerebral cortex — presence of cortical signs such as aphasia, neglect, apraxia, or visual field deficits).

Q2: NEURAXIS LEVEL
Bilateral medial temporal lobes (hippocampi)

Q3: STRUCTURES & TRACTS INVOLVED
The hippocampus converts short-term to long-term memory.

Q4: DEFINITIVE DIAGNOSIS & SYNDROME
Bilateral hippocampal amnesia

Teaching Pearl: The hippocampus converts short-term to long-term memory. Bilateral hippocampal destruction produces severe anterograde amnesia with preserved immediate recall, language and motor skills — the hallmark of a medial temporal lesion.

Anatomical Mechanism & Discriminators: The hippocampus converts short-term to long-term memory. Bilateral hippocampal destruction produces severe anterograde amnesia with preserved immediate recall, language and motor skills — the hallmark of a medial temporal lesion. — *Anchor: Bilateral medial temporal lobes (hippocampi)*

84 Hyperorality, Docility, Loss of Fear, and Visual Agnosia Following Encephalitis

Adult patient • Stage 8: Cortex by Lobe • #16

Clinical Vignette: A 36-year-old man recovering from herpes simplex encephalitis begins placing inappropriate objects in his mouth, showing indiscriminate sexual advances, and losing all fear and anger responses.

- Inability to recognize familiar objects and faces visually (psychic blindness / visual agnosia)
- Tendency to examine all objects orally (hyperorality)
- Marked placidity with loss of emotional fear and aggression
- Compulsive over-attention to visual stimuli (hypermetamorphosis)
- Severe anterograde amnesia

Q1: CENTRAL OR PERIPHERAL?
Central (Cerebral cortex — presence of cortical signs such as aphasia, neglect, apraxia, or visual field deficits).

Q2: NEURAXIS LEVEL
Bilateral anterior temporal lobes (including amygdaloid nuclei and hippocampus)

Q3: STRUCTURES & TRACTS INVOLVED
Bilateral temporal and amygdaloid destruction disconnects visual sensory processing from limbic emotional regulation, causing loss of fear conditioning, abnormal appetite/oral behaviors, and altered sexual drive.

Q4: DEFINITIVE DIAGNOSIS & SYNDROME
Klüver-Bucy syndrome

Teaching Pearl: Bilateral temporal and amygdaloid destruction disconnects visual sensory processing from limbic emotional regulation, causing loss of fear conditioning, abnormal appetite/oral behaviors, and altered sexual drive. 85 * Involuntary, purposeful movements of one hand

Anatomical Mechanism & Discriminators: Bilateral temporal and amygdaloid destruction disconnects visual sensory processing from limbic emotional regulation, causing loss of fear conditioning, abnormal appetite/oral behaviors, and altered sexual drive. 85 * Involuntary, purposeful movements of one hand — *Anchor: Bilateral anterior temporal lobes (including amygdaloid nuclei and hippocampus)*

43 Bumping into Objects on One Side with Loss of Half of Visual Field

Adult patient • Stage 8: Cortex by Lobe • #17

Clinical Vignette: A 59-year-old man suddenly begins bumping into doors and people on his right side. He denies any loss of vision.

- Visual acuity preserved
- Pupils equal and reactive
- Right visual field absent in both eyes
- Fundus normal
- Eye movements full
- Limb power normal
- No sensory deficit

Q1: CENTRAL OR PERIPHERAL?

Central (Cerebral cortex — presence of cortical signs such as aphasia, neglect, apraxia, or visual field deficits).

Q2: NEURAXIS LEVEL

Left retrochiasmal visual pathway

Q3: STRUCTURES & TRACTS INVOLVED

Loss of the same half of the visual field in both eyes means the lesion is behind the optic chiasm.

Q4: DEFINITIVE DIAGNOSIS & SYNDROME

Right homonymous hemianopia

Teaching Pearl: Loss of the same half of the visual field in both eyes means the lesion is behind the optic chiasm. A highly congruous hemianopia without other major neurological findings particularly suggests the contralateral occipital cortex.

Anatomical Mechanism & Discriminators: Loss of the same half of the visual field in both eyes means the lesion is behind the optic chiasm. A highly congruous hemianopia without other major neurological findings particularly suggests the contralateral occipital cortex. — *Anchor: Left retrochiasmal visual pathway*

52 Complete Loss of Vision with Denial of Deficit and Confabulation

74-year-old woman • Stage 8: Cortex by Lobe • #18

Clinical Vignette: A 74-year-old woman is found at home after a stroke. She repeatedly bumps into furniture yet firmly denies any problem with her vision and describes the room around her in plausible but incorrect detail.

- No response to visual threat in either eye
- Cannot count fingers or perceive hand movements
- Denies being blind
- Confabulates descriptions of objects and people
- Pupillary light reflexes normal
- Fundi normal

Q1: CENTRAL OR PERIPHERAL?

Central (Cerebral cortex — presence of cortical signs such as aphasia, neglect, apraxia, or visual field deficits).

Q2: NEURAXIS LEVEL

Bilateral occipital cortex with visual association areas

Q3: STRUCTURES & TRACTS INVOLVED

Bilateral striate cortex destruction causes cortical blindness.

Q4: DEFINITIVE DIAGNOSIS & SYNDROME

Cortical blindness with anosognosia — Anton syndrome

Teaching Pearl: Bilateral striate cortex destruction causes cortical blindness. Additional involvement of visual association cortex removes awareness of the deficit, so the patient denies blindness and confabulates. Preserved pupillary responses prove the pregeniculate light-reflex pathway is intact, separating cortical blindness from optic nerve disease.

Anatomical Mechanism & Discriminators: Bilateral striate cortex destruction causes cortical blindness. Additional involvement of visual association cortex removes awareness of the deficit, so the patient denies blindness and confabulates. Preserved pupillary responses prove the pregeniculate light-reflex pathway is intact, separating cortical blindness from optic nerve disease. — *Anchor: Bilateral occipital cortex with visual association areas*

75 Cortical Blindness with Denial of Visual Deficit and Vivid Confabulation

74-year-old woman • Stage 8: Cortex by Lobe • #19

Clinical Vignette: A 74-year-old woman sustains bilateral PCA infarctions. Although she bumps into walls and furniture, she insists her vision is perfect and describes detailed scenes that are not present.

- Complete bilateral loss of vision (no light perception)
- Bilateral pupillary light reflexes briskly preserved
- Normal fundoscopy
- Unawareness/denial of visual deficit (anosognosia)
- Visual confabulation
- Extraocular movements full

Q1: CENTRAL OR PERIPHERAL?

Central (Cerebral cortex — presence of cortical signs such as aphasia, neglect, apraxia, or visual field deficits).

Q2: NEURAXIS LEVEL

Bilateral primary visual (striate) cortices (Brodmann area 17)

Q3: STRUCTURES & TRACTS INVOLVED

Destruction of bilateral striate cortices causes cortical blindness, while preserved subcortical pretectal pathways maintain pupillary light reflexes.

Q4: DEFINITIVE DIAGNOSIS & SYNDROME

Anton syndrome (visual anosognosia)

Teaching Pearl: Destruction of bilateral striate cortices causes cortical blindness, while preserved subcortical pretectal pathways maintain pupillary light reflexes. Interruption of visual sensory inputs to parietal-frontal monitoring areas leads to visual denial and confabulation. * Loss of outer visual fields in both eyes

Anatomical Mechanism & Discriminators: Destruction of bilateral striate cortices causes cortical blindness, while preserved subcortical pretectal pathways maintain pupillary light reflexes. Interruption of visual sensory inputs to parietal-frontal monitoring areas leads to visual denial and confabulation. * Loss of outer visual fields in both eyes — *Anchor: Bilateral primary visual (striate) cortices (Brodmann area 17)*

53 Inability to Perceive More Than One Object Simultaneously with Misreaching

66-year-old woman recovering from a cardiac arrest now • Stage 8: Cortex by Lobe • #20

Clinical Vignette: A 66-year-old woman recovering from a cardiac arrest now complains that the world looks fragmented. She cannot find the cup in front of her and reaches past objects when she tries to grasp them.

- Perceives only one object at a time — cannot describe a whole scene
- Misreaches for objects under visual guidance
- Difficulty voluntarily directing gaze toward a target
- Visual acuity preserved
- Visual fields full to single targets
- Limb power and sensation normal

Q1: CENTRAL OR PERIPHERAL?

Central (Cerebral cortex — presence of cortical signs such as aphasia, neglect, apraxia, or visual field deficits).

Q2: NEURAXIS LEVEL

Bilateral parieto-occipital association cortex — watershed zones

Q3: STRUCTURES & TRACTS INVOLVED

The classic triad is: simultanagnosia — inability to perceive the visual scene as a whole optic ataxia — impaired visually guided reaching ocular apraxia — difficulty initiating voluntary gaze shifts This reflects bilateral damage to the dorsal visual stream ("where" pathway), classically in the parieto-occipital watershed after global hypoperfusion.

Q4: DEFINITIVE DIAGNOSIS & SYNDROME

Balint syndrome

Teaching Pearl: The classic triad is: simultanagnosia — inability to perceive the visual scene as a whole optic ataxia — impaired visually guided reaching ocular apraxia — difficulty initiating voluntary gaze shifts This reflects bilateral damage to the dorsal visual stream ("where" pathway), classically in the parieto-occipital watershed after global hypoperfusion.

Anatomical Mechanism & Discriminators: The classic triad is: simultanagnosia — inability to perceive the visual scene as a whole optic ataxia — impaired visually guided reaching ocular apraxia — difficulty initiating voluntary gaze shifts This reflects bilateral damage to the dorsal visual stream ("where" pathway), classically in the parieto-occipital watershed after global hypoperfusion. — *Anchor: Bilateral parieto-occipital association cortex — watershed zones*

74 Simultanagnosia, Ocular Motor Apraxia, and Visual Optic Ataxia

68-year-old man recovering from severe hypotension after cardiac arrest • Stage 8: Cortex by Lobe • #21

Clinical Vignette: A 68-year-old man recovering from severe hypotension after cardiac arrest complains that he can only see one thing at a time and repeatedly misses objects he tries to grasp.

- Visual acuity preserved in each eye individually
- Inability to perceive more than one visual stimulus simultaneously (simultanagnosia)
- Inability to direct voluntary saccades to visual targets (ocular motor apraxia)
- Inability to accurately reach for objects under visual guidance despite normal limb strength (optic ataxia)
- Normal motor strength and primary sensation

Q1: CENTRAL OR PERIPHERAL?

Central (Cerebral cortex — presence of cortical signs such as aphasia, neglect, apraxia, or visual field deficits).

Q2: NEURAXIS LEVEL

Bilateral parieto-occipital association cortices (watershed border zones)

Q3: STRUCTURES & TRACTS INVOLVED

Bilateral watershed infarctions between the MCA and PCA territories disrupt the dorsal visual processing stream ("where" pathway), impairing visuospatial integration and visual-motor coordination despite preserved central acuity.

Q4: DEFINITIVE DIAGNOSIS & SYNDROME

Bálint syndrome

Teaching Pearl: Bilateral watershed infarctions between the MCA and PCA territories disrupt the dorsal visual processing stream ("where" pathway), impairing visuospatial integration and visual-motor coordination despite preserved central acuity. * Cortical blindness with denial of visual deficit 75 .

Anatomical Mechanism & Discriminators: Bilateral watershed infarctions between the MCA and PCA territories disrupt the dorsal visual processing stream ("where" pathway), impairing visuospatial integration and visual-motor coordination despite preserved central acuity. * Cortical blindness with denial of visual deficit 75 . — *Anchor: Bilateral parieto-occipital association cortices (watershed border zones)*

54 Inability to Recognize Familiar Faces Despite Normal Visual Acuity

Adult patient • Stage 8: Cortex by Lobe • #22

Clinical Vignette: A 58-year-old man recovers from a right posterior circulation stroke. He can read, name objects and recognise his wife's voice, but fails to recognise her face until she speaks.

- Cannot identify faces of close family members
- Recognises people by voice, gait or clothing
- Can describe age, sex and individual facial features
- Visual acuity normal
- Left homonymous hemianopia present
- Naming and language intact

Q1: CENTRAL OR PERIPHERAL?
Central (Cerebral cortex — presence of cortical signs such as aphasia, neglect, apraxia, or visual field deficits).

Q2: NEURAXIS LEVEL
Right fusiform / inferior occipitotemporal cortex

Q3: STRUCTURES & TRACTS INVOLVED
Face recognition depends on the fusiform face area in the inferior occipitotemporal cortex.

Q4: DEFINITIVE DIAGNOSIS & SYNDROME
Prosopagnosia

Teaching Pearl: Face recognition depends on the fusiform face area in the inferior occipitotemporal cortex. Nondominant (usually right-sided) lesions impair recognition of faces while leaving object naming and language intact.

Anatomical Mechanism & Discriminators: Face recognition depends on the fusiform face area in the inferior occipitotemporal cortex. Nondominant (usually right-sided) lesions impair recognition of faces while leaving object naming and language intact. — *Anchor: Right fusiform / inferior occipitotemporal cortex*

25 Inability to Read Written Words Despite Preserved Ability to Write

58-year-old man suddenly • Stage 8: Cortex by Lobe • #23

Clinical Vignette: A 58-year-old man suddenly complains that printed words no longer make sense. He can still write sentences correctly but cannot read them back.

- Normal speech fluency
- Normal comprehension of spoken language
- Writing preserved
- Unable to read written words
- Right homonymous hemianopia
- Naming of visually presented objects may be impaired
- Limb power preserved

Q1: CENTRAL OR PERIPHERAL?
Central (Cerebral cortex — presence of cortical signs such as aphasia, neglect, apraxia, or visual field deficits).

Q2: NEURAXIS LEVEL
Left occipital cortex with involvement of the splenium of corpus callosum

Q3: STRUCTURES & TRACTS INVOLVED
The left visual cortex cannot receive visual input directly because of the occipital lesion.

Q4: DEFINITIVE DIAGNOSIS & SYNDROME
Alexia without agraphia

Teaching Pearl: The left visual cortex cannot receive visual input directly because of the occipital lesion. Visual information from the right visual cortex also cannot cross through the damaged splenium to reach the dominant language cortex. Hence the patient can write but cannot read.

Anatomical Mechanism & Discriminators: The left visual cortex cannot receive visual input directly because of the occipital lesion. Visual information from the right visual cortex also cannot cross through the damaged splenium to reach the dominant language cortex. Hence the patient can write but cannot read. — *Anchor: Left occipital cortex with involvement of the splenium of corpus callosum*

31 Isolated Object-Naming Deficit with Fluent Grammatical Speech

Adult patient • Stage 8: Cortex by Lobe • #24

Clinical Vignette: A 63-year-old right-handed man develops sudden difficulty finding the names of common objects. His family says he speaks normally but frequently pauses because “the word does not come.”

- Speech fluent
- Comprehension preserved
- Repetition preserved
- Marked difficulty naming common objects
- Can describe the use of the object correctly
- No significant weakness
- No visual field defect

Q1: CENTRAL OR PERIPHERAL?
Central (Cerebral cortex — presence of cortical signs such as aphasia, neglect, apraxia, or visual field deficits).

Q2: NEURAXIS LEVEL
Dominant temporoparietal language network

Q3: STRUCTURES & TRACTS INVOLVED
The predominant deficit is word retrieval.

Q4: DEFINITIVE DIAGNOSIS & SYNDROME
Anomic aphasia

Teaching Pearl: The predominant deficit is word retrieval. Fluency, comprehension and repetition are relatively preserved. Anomia can occur with lesions around the dominant angular/inferior temporal language network and is also common during recovery from other aphasias.

Anatomical Mechanism & Discriminators: The predominant deficit is word retrieval. Fluency, comprehension and repetition are relatively preserved. Anomia can occur with lesions around the dominant angular/inferior temporal language network and is also common during recovery from other aphasias. — *Anchor: Dominant temporoparietal language network*

21 Fluent Spontaneous Speech with Poor Repetition and Phonemic Errors

Adult patient • Stage 8: Cortex by Lobe • #25

Clinical Vignette: A 61-year-old right-handed teacher develops sudden difficulty repeating sentences. She speaks fluently and understands conversation well.

- Fluent spontaneous speech
- Comprehension preserved
- Markedly impaired repetition
- Frequent phonemic paraphasias
- Naming mildly impaired
- No significant limb weakness
- Reading aloud also impaired

Q1: CENTRAL OR PERIPHERAL?
Central (Cerebral cortex — presence of cortical signs such as aphasia, neglect, apraxia, or visual field deficits).

Q2: NEURAXIS LEVEL
Dominant hemisphere arcuate fasciculus / supramarginal region

Q3: STRUCTURES & TRACTS INVOLVED
The key dissociation is: fluent speech preserved comprehension disproportionately poor repetition This suggests disruption of the connection between posterior language comprehension areas and frontal speech-production areas, classically the arcuate fasciculus.

Q4: DEFINITIVE DIAGNOSIS & SYNDROME
Conduction aphasia

Teaching Pearl: The key dissociation is: fluent speech preserved comprehension disproportionately poor repetition This suggests disruption of the connection between posterior language comprehension areas and frontal speech-production areas, classically the arcuate fasciculus.

Anatomical Mechanism & Discriminators: The key dissociation is: fluent speech preserved comprehension disproportionately poor repetition This suggests disruption of the connection between posterior language comprehension areas and frontal speech-production areas, classically the arcuate fasciculus. — *Anchor: Dominant hemisphere arcuate fasciculus / supramarginal region*

22 Severe Reduction in Speech and Comprehension with Preserved Repetition

67-year-old man • Stage 8: Cortex by Lobe • #26

Clinical Vignette: A 67-year-old man develops sudden difficulty speaking and understanding language. Family members notice that he can sometimes repeat phrases spoken to him despite not understanding them.

- Speech markedly reduced
- Poor comprehension
- Naming severely impaired
- Repetition relatively preserved
- Unable to follow verbal commands
- Mild right-sided weakness
- Right visual field defect may be present

Q1: CENTRAL OR PERIPHERAL?
Central (Cerebral cortex — presence of cortical signs such as aphasia, neglect, apraxia, or visual field deficits).

Q2: NEURAXIS LEVEL
Dominant hemisphere language association cortex surrounding, but relatively sparing, the perisylvian repetition pathway

Q3: STRUCTURES & TRACTS INVOLVED
This resembles global aphasia, except repetition is preserved.

Q4: DEFINITIVE DIAGNOSIS & SYNDROME
Transcortical mixed aphasia

Teaching Pearl: This resembles global aphasia, except repetition is preserved. The perisylvian language cortex is functionally isolated from surrounding association cortex, while the repetition circuitry remains relatively intact.

Anatomical Mechanism & Discriminators: This resembles global aphasia, except repetition is preserved. The perisylvian language cortex is functionally isolated from surrounding association cortex, while the repetition circuitry remains relatively intact. — *Anchor: Dominant hemisphere language association cortex surrounding, but relatively sparing, the perisylvian repetition pathway*

103 Severe Headache, Focal Seizures, and Bilateral Leg Weakness in Postpartum Female

34-year-old woman on the combined oral contraceptive pill • Stage 8: Cortex by Lobe • #27

Clinical Vignette: A 34-year-old woman on the combined oral contraceptive pill develops a week of worsening headache, two focal seizures, then progressive weakness of both legs.

- Bilateral lower-limb weakness, left slightly worse
- Brisk lower-limb reflexes
- Bilateral extensor plantar responses
- Upper limbs normal
- Papilloedema present
- No sensory level

Q1: CENTRAL OR PERIPHERAL?
Central (Cerebral cortex — presence of cortical signs such as aphasia, neglect, apraxia, or visual field deficits).

Q2: NEURAXIS LEVEL
Superior sagittal sinus / parasagittal cortex

Q3: STRUCTURES & TRACTS INVOLVED
The parasagittal cortex contains the leg motor areas and drains into the superior sagittal sinus.

Q4: DEFINITIVE DIAGNOSIS & SYNDROME
Superior sagittal sinus thrombosis with parasagittal venous infarction

Teaching Pearl: The parasagittal cortex contains the leg motor areas and drains into the superior sagittal sinus. Bilateral leg-predominant weakness with seizures, headache and papilloedema in a prothrombotic setting points to cortical venous sinus thrombosis.

Anatomical Mechanism & Discriminators: The parasagittal cortex contains the leg motor areas and drains into the superior sagittal sinus. Bilateral leg-predominant weakness with seizures, headache and papilloedema in a prothrombotic setting points to cortical venous sinus thrombosis. — *Anchor: Superior sagittal sinus / parasagittal cortex*

CURRICULUM CHAPTER 9 • STAGE 9

Stage 9 — Deep Hemispheric Lesions: Subcortex, Capsule & Basal Ganglia

Chapter Learning Objective

Apply the negative rule: Dense motor/sensory/movement deficits with CONSPICUOUS ABSENCE of cortical signs.

Pedagogical Localisation Anchor

Internal capsule -> pure motor; Thalamus -> pure sensory/pain; Subthalamic -> hemiballismus; Substantia nigra -> parkinsonism.

Included Clinical Vignettes (11 Cases):
Case 128: Dense Contralateral Hemiplegia, Hemisensory Loss, and Hemianopia Without Aphasia or Neglect • Case 129: Mild Hemiparesis Combined with Prominent Out-of-Proportion Ipsilateral Limb Ataxia • Case 131: Prominent Dysarthria and Clumsy Hand Movements Sparing Language Function • Case 106: Severe Intractable Burning Hemibody Pain Triggered by Light Touch After Stroke • Case 136: Restricted Numbness Involving the Unilateral Perioral Region and Digits of the Hand • Case 133: Sudden Stupor Followed by Vertical Gaze Palsy and Dense Anterograde Amnesia • *...and 5 more cases*

128 Dense Contralateral Hemiplegia, Hemisensory Loss, and Hemianopia Without Aphasia or Neglect

69-year-old man suddenly • Stage 9: Deep Subcortex • #1

Clinical Vignette: A 69-year-old man suddenly develops dense weakness and numbness of the left side and begins missing objects on his left.

- Left UMN facial weakness
- Left arm and leg power 2/5
- Left hyperreflexia
- Left extensor plantar response
- Left hemisensory loss
- Left homonymous hemianopia
- No aphasia
- No neglect
- No gaze deviation

Q1: CENTRAL OR PERIPHERAL?

Central (Deep subcortical / thalamocortical / internal capsule — dense deficits with absent cortical signs).

Q2: NEURAXIS LEVEL

Right posterior limb of internal capsule/retrolenticular region

Q3: STRUCTURES & TRACTS INVOLVED

The classic triad is: contralateral hemiplegia contralateral hemisensory loss contralateral homonymous hemianopia The absence of cortical signs helps separate this deep infarction from a large MCA cortical stroke.

Q4: DEFINITIVE DIAGNOSIS & SYNDROME

Anterior choroidal artery syndrome

Teaching Pearl: The classic triad is: contralateral hemiplegia contralateral hemisensory loss contralateral homonymous hemianopia The absence of cortical signs helps separate this deep infarction from a large MCA cortical stroke.

Anatomical Mechanism & Discriminators: The classic triad is: contralateral hemiplegia contralateral hemisensory loss contralateral homonymous hemianopia The absence of cortical signs helps separate this deep infarction from a large MCA cortical stroke. — *Anchor: Right posterior limb of internal capsule/retrolenticular region*

129 Mild Hemiparesis Combined with Prominent Out-of-Proportion Ipsilateral Limb Ataxia

62-year-old hypertensive man • Stage 9: Deep Subcortex • #2

Clinical Vignette: A 62-year-old hypertensive man develops sudden left-sided weakness but repeatedly overshoots targets with the same weak arm.

- Left arm and leg power 4-/5
- Left hyperreflexia
- Left extensor plantar response
- Marked left finger-nose dysmetria out of proportion to weakness
- Left heel-knee-shin impairment
- No sensory deficit
- No visual field defect
- No aphasia or neglect

Q1: CENTRAL OR PERIPHERAL?

Central (Deep subcortical / thalamocortical / internal capsule — dense deficits with absent cortical signs).

Q2: NEURAXIS LEVEL

Right basis pontis / internal capsule involving corticospinal and pontocerebellar pathways

Q3: STRUCTURES & TRACTS INVOLVED

Weakness and cerebellar-type incoordination occur on the same side of the body.

Q4: DEFINITIVE DIAGNOSIS & SYNDROME

Ataxic hemiparesis — lacunar stroke syndrome

Teaching Pearl: Weakness and cerebellar-type incoordination occur on the same side of the body. That combination without cortical signs is a classic lacunar syndrome, most commonly from a small pontine or internal capsular infarct.

Anatomical Mechanism & Discriminators: Weakness and cerebellar-type incoordination occur on the same side of the body. That combination without cortical signs is a classic lacunar syndrome, most commonly from a small pontine or internal capsular infarct. — *Anchor: Right basis pontis / internal capsule involving corticospinal and pontocerebellar pathways*

131 Prominent Dysarthria and Clumsy Hand Movements Sparing Language Function

64-year-old hypertensive man suddenly • Stage 9: Deep Subcortex • #3

Clinical Vignette: A 64-year-old hypertensive man suddenly develops slurred speech and difficulty using his right hand while eating.

- Marked dysarthria
- Mild right central facial weakness
- Right hand slow and clumsy on rapid finger movements
- Mild right pronator drift
- Leg power nearly normal
- No aphasia
- No neglect
- No visual-field defect
- Sensation normal

Q1: CENTRAL OR PERIPHERAL?

Central (Deep subcortical / thalamocortical / internal capsule — dense deficits with absent cortical signs).

Q2: NEURAXIS LEVEL

Left basis pontis or genu/posterior limb of internal capsule

Q3: STRUCTURES & TRACTS INVOLVED

Dysarthria with disproportionate hand clumsiness, without cortical signs, suggests a small deep infarct affecting corticobulbar and corticospinal fibres.

Q4: DEFINITIVE DIAGNOSIS & SYNDROME

Dysarthria–clumsy hand lacunar syndrome

Teaching Pearl: Dysarthria with disproportionate hand clumsiness, without cortical signs, suggests a small deep infarct affecting corticobulbar and corticospinal fibres. The absence of aphasia is particularly important: slurred speech is not necessarily a cortical language problem.

Anatomical Mechanism & Discriminators: Dysarthria with disproportionate hand clumsiness, without cortical signs, suggests a small deep infarct affecting corticobulbar and corticospinal fibres. The absence of aphasia is particularly important: slurred speech is not necessarily a cortical language problem. — *Anchor: Left basis pontis or genu/posterior limb of internal capsule*

106 Severe Intractable Burning Hemibody Pain Triggered by Light Touch After Stroke

Adult patient • Stage 9: Deep Subcortex • #4

Clinical Vignette: A 66-year-old man had a right thalamic lacunar stroke with left hemianaesthesia. Weeks later he develops agonising burning pain in the previously numb left arm.

- Deep, burning, electric pain in the left arm and face
- Pain triggered by light touch (allodynia)
- Mild left hemisensory loss to pinprick
- Power preserved
- Reflexes symmetric

Q1: CENTRAL OR PERIPHERAL?

Central (Deep subcortical / thalamocortical / internal capsule — dense deficits with absent cortical signs).

Q2: NEURAXIS LEVEL

Right thalamus — ventral posterolateral nucleus

Q3: STRUCTURES & TRACTS INVOLVED

Injury to the ventral posterior thalamic sensory relay can generate spontaneous, paroxysmal central pain in the corresponding body half, often after the initial sensory loss improves.

Q4: DEFINITIVE DIAGNOSIS & SYNDROME

Thalamic pain syndrome (Dejerine-Roussy)

Teaching Pearl: Injury to the ventral posterior thalamic sensory relay can generate spontaneous, paroxysmal central pain in the corresponding body half, often after the initial sensory loss improves. It is a classic central post-stroke pain syndrome.

Anatomical Mechanism & Discriminators: Injury to the ventral posterior thalamic sensory relay can generate spontaneous, paroxysmal central pain in the corresponding body half, often after the initial sensory loss improves. It is a classic central post-stroke pain syndrome. — *Anchor: Right thalamus — ventral posterolateral nucleus*

136 Restricted Numbness Involving the Unilateral Perioral Region and Digits of the Hand

61-year-old hypertensive woman suddenly • Stage 9: Deep Subcortex • #5

Clinical Vignette: A 61-year-old hypertensive woman suddenly notices tingling around the left side of her mouth and in the fingers of her left hand.

- Reduced sensation around left lips
- Reduced pinprick over left thumb and fingers
- Arm, trunk and leg sensation otherwise normal
- Power normal
- Reflexes normal
- No cortical signs

Q1: CENTRAL OR PERIPHERAL?

Central (Brainstem crossed syndrome — ipsilateral cranial nerve nucleus/fascicle + contralateral long tracts).

Q2: NEURAXIS LEVEL

Right thalamus, classically VPM/VPL junction; occasionally pons

Q3: STRUCTURES & TRACTS INVOLVED

The neighbouring sensory representations of the mouth and hand can be affected by a tiny strategically placed lacunar infarct.

Q4: DEFINITIVE DIAGNOSIS & SYNDROME

Cheiro-oral sensory syndrome

Teaching Pearl: The neighbouring sensory representations of the mouth and hand can be affected by a tiny strategically placed lacunar infarct. A small apparently odd sensory pattern can therefore be highly localising.

Anatomical Mechanism & Discriminators: The neighbouring sensory representations of the mouth and hand can be affected by a tiny strategically placed lacunar infarct. A small apparently odd sensory pattern can therefore be highly localising. — *Anchor: Right thalamus, classically VPM/VPL junction; occasionally pons*

133 Sudden Stupor Followed by Vertical Gaze Palsy and Dense Anterograde Amnesia

68-year-old man • Stage 9: Deep Subcortex • #6

Clinical Vignette: A 68-year-old man is found suddenly unresponsive. He improves over several hours but remains very sleepy and confused.

- Fluctuating consciousness
- Marked hypersomnolence
- Vertical gaze palsy
- Pupillary abnormalities may occur
- Severe recent-memory impairment
- Disorientation
- Mild bilateral pyramidal signs
- No major hemispheric cortical deficit

Q1: CENTRAL OR PERIPHERAL?

Central (Brainstem crossed syndrome — ipsilateral cranial nerve nucleus/fascicle + contralateral long tracts).

Q2: NEURAXIS LEVEL

Bilateral paramedian thalami ± rostral midbrain

Q3: STRUCTURES & TRACTS INVOLVED

A single arterial variant may supply both paramedian thalami.

Q4: DEFINITIVE DIAGNOSIS & SYNDROME

Artery of Percheron infarction

Teaching Pearl: A single arterial variant may supply both paramedian thalami. The classic combination is: altered consciousness + memory impairment + vertical gaze abnormality. It is an important bilateral deep-brain stroke pattern.

Anatomical Mechanism & Discriminators: A single arterial variant may supply both paramedian thalami. The classic combination is: altered consciousness + memory impairment + vertical gaze abnormality. It is an important bilateral deep-brain stroke pattern. — *Anchor: Bilateral paramedian thalami ± rostral midbrain*

55 Sudden Involuntary Violent Flinging Movements of One Side of the Body

68-year-old diabetic woman suddenly • Stage 9: Deep Subcortex • #7

Clinical Vignette: A 68-year-old diabetic woman suddenly develops violent, uncontrollable flinging movements of her left arm and leg.

- Large-amplitude rotatory and throwing movements of the left limbs
- Movements continuous while awake, markedly reduced in sleep
- Mild weakness of the left limbs between movements
- No sensory loss
- No aphasia or neglect
- Reflexes symmetric

Q1: CENTRAL OR PERIPHERAL?

Central (Deep subcortical / thalamocortical / internal capsule — dense deficits with absent cortical signs).

Q2: NEURAXIS LEVEL

Right subthalamic nucleus

Q3: STRUCTURES & TRACTS INVOLVED

The subthalamic nucleus normally excites the inhibitory output nucleus of the basal ganglia (GPi).

Q4: DEFINITIVE DIAGNOSIS & SYNDROME

Contralateral hemiballismus

Teaching Pearl: The subthalamic nucleus normally excites the inhibitory output nucleus of the basal ganglia (GPi). A lesion here disinhibits the motor thalamus and cortex, producing violent contralateral ballistic movements. The classic setting is a small deep stroke in an elderly hypertensive or diabetic patient; nonketotic hyperglycaemia is another recognised cause.

Anatomical Mechanism & Discriminators: The subthalamic nucleus normally excites the inhibitory output nucleus of the basal ganglia (GPi). A lesion here disinhibits the motor thalamus and cortex, producing violent contralateral ballistic movements. The classic setting is a small deep stroke in an elderly hypertensive or diabetic patient; nonketotic hyperglycaemia is another recognised cause. — *Anchor: Right subthalamic nucleus*

78 Sudden Uncontrolled High-Amplitude Violent Flinging Movements of One Limb

70-year-old diabetic man • Stage 9: Deep Subcortex • #8

Clinical Vignette: A 70-year-old diabetic man develops sudden, uncontrolled, continuous flinging movements of his right arm and leg.

- Unilateral, high-amplitude, irregular, violent flinging movements of right upper and lower limbs
- Movements disappear during sleep
- Limb power normal when momentarily held
- Tone slightly decreased in right limbs
- Sensation and cranial nerves intact

Q1: CENTRAL OR PERIPHERAL?

Central (Deep subcortical / thalamocortical / internal capsule — dense deficits with absent cortical signs).

Q2: NEURAXIS LEVEL

Contralateral (left) subthalamic nucleus of Luys

Q3: STRUCTURES & TRACTS INVOLVED

The subthalamic nucleus normally excites the internal segment of the globus pallidus (GPi), which inhibits thalamocortical motor drive.

Q4: DEFINITIVE DIAGNOSIS & SYNDROME

Hemiballismus

Teaching Pearl: The subthalamic nucleus normally excites the internal segment of the globus pallidus (GPi), which inhibits thalamocortical motor drive. Lesions of the STN reduce this inhibition, resulting in hyperkinetic, flinging contralateral involuntary movements.

Anatomical Mechanism & Discriminators: The subthalamic nucleus normally excites the internal segment of the globus pallidus (GPi), which inhibits thalamocortical motor drive. Lesions of the STN reduce this inhibition, resulting in hyperkinetic, flinging contralateral involuntary movements. — *Anchor: Contralateral (left) subthalamic nucleus of Luys*

57 Asymmetric Rest Tremor, Cogwheel Rigidity, and Bradykinesia

63-year-old man • Stage 9: Deep Subcortex • #9

Clinical Vignette: A 63-year-old man reports two years of increasing slowness, a softer voice and shrinking handwriting. His wife notices his right hand shakes when he is sitting still.

- Asymmetric 4–6 Hz pill-rolling rest tremor, right hand
- Cogwheel rigidity, right greater than left
- Bradykinesia on finger taps and rapid alternating movements
- Masked facies and hypophonia
- Micrographia
- Stooped posture with reduced right arm swing
- Later: impaired postural reflexes on the pull test
- Cognition, eye movements and sensation initially normal

Q1: CENTRAL OR PERIPHERAL?

Central (Cerebral cortex — presence of cortical signs such as aphasia, neglect, apraxia, or visual field deficits).

Q2: NEURAXIS LEVEL

Substantia nigra pars compacta — nigrostriatal dopaminergic pathway

Q3: STRUCTURES & TRACTS INVOLVED

Degeneration of nigral dopaminergic neurons produces the classic tetrad: rest tremor rigidity bradykinesia postural instability Asymmetric onset, preserved early cognition and a good levodopa response support idiopathic Parkinson disease over atypical parkinsonism.

Q4: DEFINITIVE DIAGNOSIS & SYNDROME

Parkinson disease

Teaching Pearl: Degeneration of nigral dopaminergic neurons produces the classic tetrad: rest tremor rigidity bradykinesia postural instability Asymmetric onset, preserved early cognition and a good levodopa response support idiopathic Parkinson disease over atypical parkinsonism.

Anatomical Mechanism & Discriminators: Degeneration of nigral dopaminergic neurons produces the classic tetrad: rest tremor rigidity bradykinesia postural instability Asymmetric onset, preserved early cognition and a good levodopa response support idiopathic Parkinson disease over atypical parkinsonism. — *Anchor: Substantia nigra pars compacta — nigrostriatal dopaminergic pathway*

56 Progressive Chorea, Executive Decline, and Oculomotor Saccadic Impairment

42-year-old man • Stage 9: Deep Subcortex • #10

Clinical Vignette: A 42-year-old man is brought by his family because of increasing fidgetiness, irritability and forgetfulness over two years. His father developed a similar illness in his forties and died in a psychiatric facility.

- Generalised choreiform movements
- Motor impersistence — cannot keep the tongue protruded ("darting tongue")
- Impaired initiation of saccades; slow, broken smooth pursuit
- Choreic, lurching interruptions of gait
- Brisk tendon reflexes
- Executive dysfunction and irritability without aphasia

Q1: CENTRAL OR PERIPHERAL?

Peripheral (Nerve root / Cauda equina level — Lower Motor Neuron signs predominate).

Q2: NEURAXIS LEVEL

Caudate nucleus and putamen — striatum

Q3: STRUCTURES & TRACTS INVOLVED

Loss of striatal GABAergic neurons produces chorea, characteristic eye-movement abnormalities, motor impersistence and a subcortical dementia with prominent behavioural change.

Q4: DEFINITIVE DIAGNOSIS & SYNDROME

Huntington disease

Teaching Pearl: Loss of striatal GABAergic neurons produces chorea, characteristic eye-movement abnormalities, motor impersistence and a subcortical dementia with prominent behavioural change. The autosomal dominant family history and mid-adult onset strongly suggest Huntington disease.

Anatomical Mechanism & Discriminators: Loss of striatal GABAergic neurons produces chorea, characteristic eye-movement abnormalities, motor impersistence and a subcortical dementia with prominent behavioural change. The autosomal dominant family history and mid-adult onset strongly suggest Huntington disease. — *Anchor: Caudate nucleus and putamen — striatum*

58 Dysarthria, Dysphagia, Hyperactive Jaw Jerk, and Emotional Incontinence

67-year-old hypertensive man with two previous strokes • Stage 9: Deep Subcortex • #11

Clinical Vignette: A 67-year-old hypertensive man with two previous strokes develops progressive slurred speech and difficulty swallowing. His family is distressed by episodes of sudden laughing or crying that do not match his mood.

- Spastic dysarthria
- Dysphagia to solids and liquids
- Brisk jaw jerk
- Exaggerated gag reflex
- Tongue small and slow-moving but not wasted, no fasciculations
- Emotional lability — involuntary laughing and crying
- Bilateral limb hyperreflexia
- Bilateral extensor plantar responses

Q1: CENTRAL OR PERIPHERAL?

Mixed Central & Peripheral (Simultaneous Upper & Lower Motor Neuron degeneration with sensory sparing).

Q2: NEURAXIS LEVEL

Bilateral corticobulbar tracts

Q3: STRUCTURES & TRACTS INVOLVED

Bilateral upper motor neuron interruption of corticobulbar fibres produces spastic bulbar dysfunction: dysarthria, dysphagia, a brisk jaw jerk and emotional incontinence.

Q4: DEFINITIVE DIAGNOSIS & SYNDROME

Pseudobulbar palsy

Teaching Pearl: Bilateral upper motor neuron interruption of corticobulbar fibres produces spastic bulbar dysfunction: dysarthria, dysphagia, a brisk jaw jerk and emotional incontinence. Contrast with bulbar (lower motor neuron) palsy, where the tongue is wasted and fasciculating and the gag reflex is lost. The usual cause is multiple bilateral lacunar infarcts.

Anatomical Mechanism & Discriminators: Bilateral upper motor neuron interruption of corticobulbar fibres produces spastic bulbar dysfunction: dysarthria, dysphagia, a brisk jaw jerk and emotional incontinence. Contrast with bulbar (lower motor neuron) palsy, where the tongue is wasted and fasciculating and the gag reflex is lost. The usual cause is multiple bilateral lacunar infarcts. — *Anchor: Bilateral corticobulbar tracts*

CURRICULUM CHAPTER 10 • STAGE 10

Stage 10 — Peripheral Mimics, Plexus & Grand Rounds Discriminators

Chapter Learning Objective

Master high-stakes discriminators: Root vs Plexus vs Nerve, and Brainstem vs Skull Base foraminal exits.

Pedagogical Localisation Anchor

Root -> myotome/dermatome; Plexus -> multi-nerve non-root; Nerve -> named territory; Skull base -> multiple CNs without long tract signs.

Included Clinical Vignettes (33 Cases):
Case 86: Inability to Flex the Distal Thumb and Index Finger with Normal Sensation • Case 87: Inability to Extend the Fingers with Preserved Radial Wrist Extension • Case 88: Severe Thenar Wasting Disproportionate to Hypothenar with Medial Forearm Numbness • Case 68: Hand Muscle Wasting with Ipsilateral Ptosis, Miosis, and Severe Shoulder Pain • Case 97: Arm Adducted, Internally Rotated, and Pronated in a Newborn • Case 95: Deltoid Weakness and Patch of Numbness Over Lateral Shoulder After Dislocation • *...and 27 more cases*

86 Inability to Flex the Distal Thumb and Index Finger with Normal Sensation

34-year-old painter • Stage 10: Master Discriminators • #1

Clinical Vignette: A 34-year-old painter notices sudden difficulty picking up small coins and pinching keys with his right hand following deep forearm soreness.

- Inability to flex the distal interphalangeal (DIP) joint of the index finger (flexor digitorum profundus I/II)
- Inability to flex the interphalangeal (IP) joint of the thumb (flexor pollicis longus)
- Inability to make a round "OK" sign (forms a flat pad-to-pad pinch)
- Weakness of pronator quadratus with elbow flexed
- Completely normal cutaneous sensation over the hand and fingers

Q1: CENTRAL OR PERIPHERAL?

Peripheral (Lower motor neuron / peripheral nerve / plexus / root distribution).

Q2: NEURAXIS LEVEL

Anterior interosseous nerve (pure motor branch of median nerve) in the proximal forearm

Q3: STRUCTURES & TRACTS INVOLVED

The AIN branches from the median nerve in the upper forearm to supply the deep flexors (FPL, lateral FDP, and pronator quadratus).

Q4: DEFINITIVE DIAGNOSIS & SYNDROME

Anterior interosseous nerve (AIN) syndrome (Kiloh-Nevin syndrome)

Teaching Pearl: The AIN branches from the median nerve in the upper forearm to supply the deep flexors (FPL, lateral FDP, and pronator quadratus). Because it carries no cutaneous sensory fibers, it causes pure motor weakness without sensory loss. 87 * Finger drop with preserved wrist extension and intact sensation

Anatomical Mechanism & Discriminators: The AIN branches from the median nerve in the upper forearm to supply the deep flexors (FPL, lateral FDP, and pronator quadratus). Because it carries no cutaneous sensory fibers, it causes pure motor weakness without sensory loss. 87 * Finger drop with preserved wrist extension and intact sensation — *Anchor: Anterior interosseous nerve (pure motor branch of median nerve) in the proximal forearm*

87 Inability to Extend the Fingers with Preserved Radial Wrist Extension

48-year-old mechanic • Stage 10: Master Discriminators • #2

Clinical Vignette: A 48-year-old mechanic develops progressive weakness extending the fingers of his right hand without arm pain or numbness.

- Inability to extend the metacarpophalangeal (MCP) joints of the right digits 2–5
- Inability to extend and abduct the thumb
- Wrist extension preserved, but deviates radially during extension
- Normal sensation over the anatomical snuffbox and dorsal hand
- Normal triceps and brachioradialis reflexes

Q1: CENTRAL OR PERIPHERAL?

Peripheral (Lower motor neuron / peripheral nerve / plexus / root distribution).

Q2: NEURAXIS LEVEL

Posterior interosseous nerve (deep motor branch of radial nerve) at the arcade of Frohse / supinator muscle

Q3: STRUCTURES & TRACTS INVOLVED

The superficial sensory radial nerve branches before the radial tunnel, so PIN entrapment produces pure motor weakness of the finger/thumb extensors and extensor carpi ulnaris (ECU).

Q4: DEFINITIVE DIAGNOSIS & SYNDROME

Posterior interosseous nerve (PIN) syndrome

Teaching Pearl: The superficial sensory radial nerve branches before the radial tunnel, so PIN entrapment produces pure motor weakness of the finger/thumb extensors and extensor carpi ulnaris (ECU). The extensor carpi radialis longus (ECRL) is spared (innervated proximal to the PIN), preserving radial wrist extension without sensory loss. 88 * Thenar wasting + medial forearm sensory loss

Anatomical Mechanism & Discriminators: The superficial sensory radial nerve branches before the radial tunnel, so PIN entrapment produces pure motor weakness of the finger/thumb extensors and extensor carpi ulnaris (ECU). The extensor carpi radialis longus (ECRL) is spared (innervated proximal to the PIN), preserving radial wrist extension without sensory loss. 88 * Thenar wasting + medial forearm sensory loss — *Anchor: Posterior interosseous nerve (deep motor branch of radial nerve) at the arcade of Frohse / supinator muscle*

88 Severe Thenar Wasting Disproportionate to Hypothenar with Medial Forearm Numbness

28-year-old woman with a congenital cervical band • Stage 10: Master Discriminators • #3

Clinical Vignette: A 28-year-old woman with a congenital cervical band presents with progressive wasting of her right hand and numbness along the inner border of her forearm.

- Severe wasting and weakness of thenar muscles (abductor pollicis brevis) disproportionately greater than hypothenar muscles
- Weakness of all intrinsic hand muscles (interossei)
- Sensory loss along the medial aspect of the forearm and medial 5th digit
- Forearm flexor muscles (median-innervated) preserved
- Reduced medial antebrachial cutaneous sensory nerve action potential

Q1: CENTRAL OR PERIPHERAL?

Peripheral (Lower motor neuron / peripheral nerve / plexus / root distribution).

Q2: NEURAXIS LEVEL

Lower trunk of the brachial plexus (C8–T1 anterior primary rami)

Q3: STRUCTURES & TRACTS INVOLVED

Compression of the lower trunk of the brachial plexus preferentially damages T1 motor fibers (supplying thenar muscles via the median nerve) and C8 sensory fibers (supplying the medial cutaneous nerve of the forearm and ulnar border of the hand).

Q4: DEFINITIVE DIAGNOSIS & SYNDROME

True neurogenic thoracic outlet syndrome (Gilliatt-Sumner hand)

Teaching Pearl: Compression of the lower trunk of the brachial plexus preferentially damages T1 motor fibers (supplying thenar muscles via the median nerve) and C8 sensory fibers (supplying the medial cutaneous nerve of the forearm and ulnar border of the hand). 89 * Sensory ataxia + spastic paraparesis + absent ankle jerks

Anatomical Mechanism & Discriminators: Compression of the lower trunk of the brachial plexus preferentially damages T1 motor fibers (supplying thenar muscles via the median nerve) and C8 sensory fibers (supplying the medial cutaneous nerve of the forearm and ulnar border of the hand). 89 * Sensory ataxia + spastic paraparesis + absent ankle jerks — *Anchor: Lower trunk of the brachial plexus (C8–T1 anterior primary rami)*

68 Hand Muscle Wasting with Ipsilateral Ptosis, Miosis, and Severe Shoulder Pain

60-year-old smoker • Stage 10: Master Discriminators • #4

Clinical Vignette: A 60-year-old smoker develops months of aching shoulder pain radiating into the inner forearm, with progressive wasting of the hand. He has lost weight and has a new cough.

- Wasting and weakness of all intrinsic hand muscles
- Weak long finger flexors
- Sensory loss over the medial forearm and hand
- Right partial ptosis and miosis, worse in the dark
- Reduced sweating over the right face
- Diminished finger-flexor reflex
- Firm fullness in the right supraclavicular fossa

Q1: CENTRAL OR PERIPHERAL?

Peripheral (Lower motor neuron / peripheral nerve / plexus / root distribution).

Q2: NEURAXIS LEVEL

Lower trunk of the brachial plexus (C8–T1) with the cervical sympathetic chain

Q3: STRUCTURES & TRACTS INVOLVED

An apical lung tumour invades the C8–T1 lower trunk, causing hand-muscle wasting and medial forearm sensory loss, and the adjacent stellate ganglion, producing an ipsilateral Horner syndrome.

Q4: DEFINITIVE DIAGNOSIS & SYNDROME

Pancoast tumour — apical lung carcinoma with lower trunk plexopathy and Horner syndrome

Teaching Pearl: An apical lung tumour invades the C8–T1 lower trunk, causing hand-muscle wasting and medial forearm sensory loss, and the adjacent stellate ganglion, producing an ipsilateral Horner syndrome. Hand wasting plus Horner syndrome plus shoulder pain in a smoker should trigger urgent chest imaging.

Anatomical Mechanism & Discriminators: An apical lung tumour invades the C8–T1 lower trunk, causing hand-muscle wasting and medial forearm sensory loss, and the adjacent stellate ganglion, producing an ipsilateral Horner syndrome. Hand wasting plus Horner syndrome plus shoulder pain in a smoker should trigger urgent chest imaging. — *Anchor: Lower trunk of the brachial plexus (C8–T1) with the cervical sympathetic chain*

97 Arm Adducted, Internally Rotated, and Pronated in a Newborn

Adult patient • Stage 10: Master Discriminators • #5

Clinical Vignette: A neonate has a floppy, internally rotated left arm after a difficult shoulder delivery.

- Left arm internally rotated and adducted
- Left forearm extended and pronated (waiter's tip posture)
- Absent left biceps reflex
- Numbness over the lateral forearm (if testable)
- Weak wrist and finger extension
- Legs normal

Q1: CENTRAL OR PERIPHERAL?

Peripheral (Lower motor neuron / peripheral nerve / plexus / root distribution).

Q2: NEURAXIS LEVEL

Left upper trunk of the brachial plexus (C5–C6)

Q3: STRUCTURES & TRACTS INVOLVED

The C5–C6 upper trunk supplies the deltoid, biceps, brachialis and supinator.

Q4: DEFINITIVE DIAGNOSIS & SYNDROME

Erb-Duchenne palsy — upper trunk brachial plexopathy

Teaching Pearl: The C5–C6 upper trunk supplies the deltoid, biceps, brachialis and supinator. Shoulder dystocia stretches these roots, producing the characteristic internally rotated, pronated arm with an absent biceps reflex.

Anatomical Mechanism & Discriminators: The C5–C6 upper trunk supplies the deltoid, biceps, brachialis and supinator. Shoulder dystocia stretches these roots, producing the characteristic internally rotated, pronated arm with an absent biceps reflex. — *Anchor: Left upper trunk of the brachial plexus (C5–C6)*

95 Deltoid Weakness and Patch of Numbness Over Lateral Shoulder After Dislocation

29-year-old rugby player • Stage 10: Master Discriminators • #6

Clinical Vignette: A 29-year-old rugby player develops weakness raising his arm after a shoulder dislocation. He has a numb patch over the shoulder.

- Wasting of the deltoid
- Weak shoulder abduction beyond 15 degrees
- Loss of sensation over the "regimental badge" area
- Biceps, triceps and forearm muscles normal
- Biceps reflex present

Q1: CENTRAL OR PERIPHERAL?
Peripheral (Lower motor neuron / peripheral nerve / plexus / root distribution).

Q2: NEURAXIS LEVEL
Right axillary nerve

Q3: STRUCTURES & TRACTS INVOLVED
The axillary nerve winds around the surgical neck of the humerus, making it vulnerable in shoulder dislocation or fracture.

Q4: DEFINITIVE DIAGNOSIS & SYNDROME
Axillary nerve palsy

Teaching Pearl: The axillary nerve winds around the surgical neck of the humerus, making it vulnerable in shoulder dislocation or fracture. It supplies the deltoid (abduction) and the skin over the lateral shoulder.

Anatomical Mechanism & Discriminators: The axillary nerve winds around the surgical neck of the humerus, making it vulnerable in shoulder dislocation or fracture. It supplies the deltoid (abduction) and the skin over the lateral shoulder. — *Anchor: Right axillary nerve*

96 Prominent Winging of the Scapula Following Neck Dissection Surgery

41-year-old woman • Stage 10: Master Discriminators • #7

Clinical Vignette: A 41-year-old woman notices that her right shoulder blade protrudes from her back when she pushes against a wall, after a mastectomy.

- Medial winging of the right scapula on wall-push
- Winging worsens with forward flexion of the arm
- Serratus anterior weakness
- Shoulder abduction and external rotation power normal
- No sensory loss

Q1: CENTRAL OR PERIPHERAL?
Peripheral (Lower motor neuron / peripheral nerve / plexus / root distribution).

Q2: NEURAXIS LEVEL
Right long thoracic nerve

Q3: STRUCTURES & TRACTS INVOLVED
The long thoracic nerve runs superficially over the chest wall and supplies serratus anterior, which anchors the scapula to the thorax.

Q4: DEFINITIVE DIAGNOSIS & SYNDROME
Long thoracic nerve palsy (winged scapula)

Teaching Pearl: The long thoracic nerve runs superficially over the chest wall and supplies serratus anterior, which anchors the scapula to the thorax. Damage — classically in radical mastectomy — causes medial scapular winging without sensory loss, because the nerve is purely motor.

Anatomical Mechanism & Discriminators: The long thoracic nerve runs superficially over the chest wall and supplies serratus anterior, which anchors the scapula to the thorax. Damage — classically in radical mastectomy — causes medial scapular winging without sensory loss, because the nerve is purely motor. — *Anchor: Right long thoracic nerve*

98 Burning Pain and Numbness Over the Anterolateral Thigh in an Obese Patient

55-year-old obese man • Stage 10: Master Discriminators • #8

Clinical Vignette: A 55-year-old obese man complains of burning and tingling over the outer right thigh. There is no weakness or back pain.

- Dysaesthesia and numbness over the anterolateral right thigh
- Quadriceps power normal
- Patellar reflex normal
- No back tenderness
- Symptoms reproduced by pressure over the inguinal ligament

Q1: CENTRAL OR PERIPHERAL?
Peripheral (Lower motor neuron / peripheral nerve / plexus / root distribution).

Q2: NEURAXIS LEVEL
Right lateral femoral cutaneous nerve

Q3: STRUCTURES & TRACTS INVOLVED
The purely sensory lateral femoral cutaneous nerve may be compressed beneath the inguinal ligament in obesity, tight belts or pregnancy.

Q4: DEFINITIVE DIAGNOSIS & SYNDROME
Meralgia paraesthetica

Teaching Pearl: The purely sensory lateral femoral cutaneous nerve may be compressed beneath the inguinal ligament in obesity, tight belts or pregnancy. The hallmark is isolated anterolateral thigh numbness or pain without any motor or reflex change.

Anatomical Mechanism & Discriminators: The purely sensory lateral femoral cutaneous nerve may be compressed beneath the inguinal ligament in obesity, tight belts or pregnancy. The hallmark is isolated anterolateral thigh numbness or pain without any motor or reflex change. — *Anchor: Right lateral femoral cutaneous nerve*

99 Severe Burning Thigh Pain with Quadriceps Wasting and Lost Knee Jerk in Diabetes

60-year-old diabetic man • Stage 10: Master Discriminators • #9

Clinical Vignette: A 60-year-old diabetic man develops severe pain in the right thigh followed by wasting, weakness and difficulty climbing stairs over several weeks.

- Wasting of the right quadriceps
- Weak hip flexion and knee extension
- Absent right patellar reflex
- Numbness over the anterior thigh and medial leg
- Adductor weakness may be present
- Distal power preserved

Q1: CENTRAL OR PERIPHERAL?
Peripheral (Lower motor neuron / peripheral nerve / plexus / root distribution).

Q2: NEURAXIS LEVEL
Right femoral nerve / L2–L4 plexus

Q3: STRUCTURES & TRACTS INVOLVED
The femoral nerve supplies the iliopsoas, quadriceps and anterior-thigh skin.

Q4: DEFINITIVE DIAGNOSIS & SYNDROME
Diabetic amyotrophy — lumbosacral radiculoplexus neuropathy

Teaching Pearl: The femoral nerve supplies the iliopsoas, quadriceps and anterior-thigh skin. Painful proximal wasting with an absent knee jerk in a diabetic localises to the femoral nerve or upper lumbar plexus; the absence of back pain and preserved distal power help separate it from an L3/L4 disc.

Anatomical Mechanism & Discriminators: The femoral nerve supplies the iliopsoas, quadriceps and anterior-thigh skin. Painful proximal wasting with an absent knee jerk in a diabetic localises to the femoral nerve or upper lumbar plexus; the absence of back pain and preserved distal power help separate it from an L3/L4 disc. — *Anchor: Right femoral nerve / L2–L4 plexus*

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Combined Upper and Lower Motor Neuron Signs with Normal Sensation

58-year-old man • Stage 10: Master Discriminators • #10

Clinical Vignette: A 58-year-old man notices a year of progressive weakness and wasting of his right hand, visible muscle twitching in his arms, and lately slurred speech.

- Wasting and fasciculations of the tongue and limb muscles
- Weakness spreading from one region to contiguous regions
- Paradoxically brisk reflexes in wasted limbs
- Bilateral extensor plantar responses
- Normal sensation throughout
- Normal eye movements
- No bladder involvement

Q1: CENTRAL OR PERIPHERAL?

Mixed Central & Peripheral (Simultaneous Upper & Lower Motor Neuron degeneration with sensory sparing).

Q2: NEURAXIS LEVEL

Upper and lower motor neurons — corticospinal tracts plus anterior horn cells and bulbar motor nuclei

Q3: STRUCTURES & TRACTS INVOLVED

The hallmark is the coexistence of lower motor neuron signs (wasting, fasciculations) and upper motor neuron signs (spasticity, brisk reflexes, extensor plantars) in the same body regions, with entirely normal sensation.

Q4: DEFINITIVE DIAGNOSIS & SYNDROME

Amyotrophic lateral sclerosis

Teaching Pearl: The hallmark is the coexistence of lower motor neuron signs (wasting, fasciculations) and upper motor neuron signs (spasticity, brisk reflexes, extensor plantars) in the same body regions, with entirely normal sensation. Sensory loss, bladder involvement or eye-movement abnormalities argue strongly against the diagnosis.

Anatomical Mechanism & Discriminators: The hallmark is the coexistence of lower motor neuron signs (wasting, fasciculations) and upper motor neuron signs (spasticity, brisk reflexes, extensor plantars) in the same body regions, with entirely normal sensation. Sensory loss, bladder involvement or eye-movement abnormalities argue strongly against the diagnosis. — *Anchor: Upper and lower motor neurons — corticospinal tracts plus anterior horn cells and bulbar motor nuclei*

62 Fluctuating Fatigable Ptosis and Diplopia Worse in the Evening

28-year-old woman • Stage 10: Master Discriminators • #11

Clinical Vignette: A 28-year-old woman reports drooping eyelids and double vision that worsen as the day goes on. Her speech becomes nasal if she talks for long, and chewing is hardest toward the end of a meal.

- Fatigable ptosis, worse with sustained upgaze
- Variable weakness of eye movements not fitting any single nerve
- Pupils normal
- Nasal speech that worsens with counting aloud
- Fatigable proximal limb weakness
- Reflexes normal and symmetrical
- Sensation normal
- Ptosis improves after applying ice to the eyelid

Q1: CENTRAL OR PERIPHERAL?

Peripheral / Neuromuscular Junction (Postsynaptic / Presynaptic NMJ failure without sensory deficit).

Q2: NEURAXIS LEVEL

Neuromuscular junction — postsynaptic acetylcholine receptors

Q3: STRUCTURES & TRACTS INVOLVED

Fluctuating, fatigable weakness that spares the pupils, reflexes and sensation localises to the neuromuscular junction rather than nerve, muscle or brain.

Q4: DEFINITIVE DIAGNOSIS & SYNDROME

Myasthenia gravis

Teaching Pearl: Fluctuating, fatigable weakness that spares the pupils, reflexes and sensation localises to the neuromuscular junction rather than nerve, muscle or brain. The pupil-sparing, patternless ophthalmoplegia with fatigable ptosis is the classic ocular signature of postsynaptic acetylcholine receptor disease.

Anatomical Mechanism & Discriminators: Fluctuating, fatigable weakness that spares the pupils, reflexes and sensation localises to the neuromuscular junction rather than nerve, muscle or brain. The pupil-sparing, patternless ophthalmoplegia with fatigable ptosis is the classic ocular signature of postsynaptic acetylcholine receptor disease. — *Anchor: Neuromuscular junction — postsynaptic acetylcholine receptors*

63 Proximal Weakness and Diminished Reflexes That Facilitate After Exercise

62-year-old man with a heavy smoking history • Stage 10: Master Discriminators • #12

Clinical Vignette: A 62-year-old man with a heavy smoking history develops dry mouth, erectile dysfunction and leg weakness. Oddly, he feels stronger after walking a short distance than when he first stands up.

- Proximal leg weakness greater than arm weakness
- Strength improves briefly after voluntary contraction
- Tendon reflexes reduced or absent at rest
- Reflexes augment after brief exercise
- Autonomic features — dry mouth, sluggish pupils
- Mild ptosis may be present
- Sensation normal

Q1: CENTRAL OR PERIPHERAL?

Peripheral / Neuromuscular Junction (Postsynaptic / Presynaptic NMJ failure without sensory deficit).

Q2: NEURAXIS LEVEL

Presynaptic neuromuscular junction — P/Q-type voltage-gated calcium channels

Q3: STRUCTURES & TRACTS INVOLVED

Antibodies against presynaptic calcium channels reduce acetylcholine release.

Q4: DEFINITIVE DIAGNOSIS & SYNDROME

Lambert-Eaton myasthenic syndrome

Teaching Pearl: Antibodies against presynaptic calcium channels reduce acetylcholine release. Brief exercise transiently raises presynaptic calcium, producing post-exercise facilitation of strength and reflexes — the opposite of myasthenia. In an older smoker, this is a paraneoplastic syndrome of small-cell lung cancer until proven otherwise.

Anatomical Mechanism & Discriminators: Antibodies against presynaptic calcium channels reduce acetylcholine release. Brief exercise transiently raises presynaptic calcium, producing post-exercise facilitation of strength and reflexes — the opposite of myasthenia. In an older smoker, this is a paraneoplastic syndrome of small-cell lung cancer until proven otherwise. — *Anchor: Presynaptic neuromuscular junction — P/Q-type voltage-gated calcium channels*

64 Rapidly Ascending Flaccid Weakness with Areflexia Following Diarrhoeal Illness

30-year-old man • Stage 10: Master Discriminators • #13

Clinical Vignette: A 30-year-old man develops tingling in his toes followed by weakness that climbs from his legs to his trunk and arms over five days, two weeks after a diarrhoeal illness.

- Symmetric flaccid weakness, legs worse than arms
- Universally absent tendon reflexes
- Flexor plantar responses
- Only mild distal sensory loss despite marked weakness
- Bilateral facial weakness developing on day 4
- Blood pressure and heart rate fluctuating
- Respiratory effort weakening

Q1: CENTRAL OR PERIPHERAL?

Peripheral (Lower motor neuron / peripheral nerve / plexus / root distribution).

Q2: NEURAXIS LEVEL

Acute polyradiculoneuropathy — peripheral nerve roots and nerves

Q3: STRUCTURES & TRACTS INVOLVED

Rapidly ascending, symmetric, flaccid weakness with universal areflexia localises to the peripheral nervous system diffusely, not to cord, brain or muscle.

Q4: DEFINITIVE DIAGNOSIS & SYNDROME

Guillain-Barré syndrome (acute inflammatory demyelinating polyradiculoneuropathy)

Teaching Pearl: Rapidly ascending, symmetric, flaccid weakness with universal areflexia localises to the peripheral nervous system diffusely, not to cord, brain or muscle. Antecedent infection (classically Campylobacter), autonomic instability, facial diplegia and later CSF albuminocytological dissociation support the diagnosis.

Anatomical Mechanism & Discriminators: Rapidly ascending, symmetric, flaccid weakness with universal areflexia localises to the peripheral nervous system diffusely, not to cord, brain or muscle. Antecedent infection (classically Campylobacter), autonomic instability, facial diplegia and later CSF albuminocytological dissociation support the diagnosis. — *Anchor: Acute polyradiculoneuropathy — peripheral nerve roots and nerves*

69 Unilateral Deafness, Absent Corneal Reflex, Facial Weakness, and Limb Ataxia

52-year-old woman • Stage 10: Master Discriminators • #14

Clinical Vignette: A 52-year-old woman notices months of progressive right-sided hearing loss with tinnitus and unsteadiness. More recently the right side of her face has become numb and slightly weak.

- Right sensorineural hearing loss
- Right corneal reflex absent when the right cornea is touched
- Mild right lower motor neuron facial weakness
- Right-sided limb ataxia with gait veering to the right
- Horizontal nystagmus
- Papilloedema if the lesion is large
- No long-tract motor or sensory signs in the limbs

Q1: CENTRAL OR PERIPHERAL?

Central (Cerebral cortex — presence of cortical signs such as aphasia, neglect, apraxia, or visual field deficits).

Q2: NEURAXIS LEVEL

Right cerebellopontine angle

Q3: STRUCTURES & TRACTS INVOLVED

CN VIII is affected earliest, causing unilateral sensorineural deafness and tinnitus.

Q4: DEFINITIVE DIAGNOSIS & SYNDROME

Vestibular schwannoma (acoustic neuroma)

Teaching Pearl: CN VIII is affected earliest, causing unilateral sensorineural deafness and tinnitus. Growth then involves CN V (lost corneal reflex — often before facial weakness), CN VII, and the cerebellar peduncle, producing ipsilateral ataxia. Unilateral tinnitus with hearing loss always warrants imaging of the cerebellopontine angle.

Anatomical Mechanism & Discriminators: CN VIII is affected earliest, causing unilateral sensorineural deafness and tinnitus. Growth then involves CN V (lost corneal reflex — often before facial weakness), CN VII, and the cerebellar peduncle, producing ipsilateral ataxia. Unilateral tinnitus with hearing loss always warrants imaging of the cerebellopontine angle. — *Anchor: Right cerebellopontine angle*

70 Hoarseness, Palatal Droop, and Shoulder Shrug Weakness Without Long-Tract Signs

46-year-old man • Stage 10: Master Discriminators • #15

Clinical Vignette: A 46-year-old man develops progressive hoarseness, choking on liquids and weakness of the right shoulder. Imaging shows a mass at the base of the skull.

- Right side of the palate droops; uvula deviates left
- Gag reflex absent on the right
- Hoarse voice
- Loss of taste over the posterior third of the tongue (if tested)
- Right trapezius weak and wasted — weak shoulder shrug
- Weak head rotation to the left (right sternocleidomastoid)
- Tongue movements normal
- No limb weakness or sensory loss

Q1: CENTRAL OR PERIPHERAL?

Central (Cerebral cortex — presence of cortical signs such as aphasia, neglect, apraxia, or visual field deficits).

Q2: NEURAXIS LEVEL

Right jugular foramen — CN IX, X and XI

Q3: STRUCTURES & TRACTS INVOLVED

The glossopharyngeal, vagus and spinal accessory nerves exit the skull together through the jugular foramen.

Q4: DEFINITIVE DIAGNOSIS & SYNDROME

Jugular foramen syndrome — Vernet syndrome

Teaching Pearl: The glossopharyngeal, vagus and spinal accessory nerves exit the skull together through the jugular foramen. A lesion there — classically a glomus jugulare tumour, fracture or venous thrombosis — produces the combination of: dysphagia with palatal droop and absent gag (IX, X) hoarseness (X) trapezius and sternocleidomastoid weakness (XI) The absence of long-tract signs separates this peripheral skull-base syndrome from a brainstem lesion.

Anatomical Mechanism & Discriminators: The glossopharyngeal, vagus and spinal accessory nerves exit the skull together through the jugular foramen. A lesion there — classically a glomus jugulare tumour, fracture or venous thrombosis — produces the combination of: dysphagia with palatal droop and absent gag (IX, X) hoarseness (X) trapezius and sternocleidomastoid weakness (XI) The absence of long-tract signs separates this peripheral skull-base syndrome from a brainstem lesion. — *Anchor: Right jugular foramen — CN IX, X and XI*

80

Loss of Gag Reflex, Palatal Deviation, Hoarseness, and Weak Shoulder Shrug

53-year-old man • Stage 10: Master Discriminators • #16

Clinical Vignette: A 53-year-old man presents with progressive hoarseness, difficulty swallowing liquids, and weakness when elevating his right shoulder.

- Loss of sensation and taste on posterior third of right tongue (CN IX)
- Absent right gag reflex
- Right palatal droop with uvula deviating to the left (CN X)
- Right vocal cord paralysis
- Wasting and weakness of right trapezius and sternocleidomastoid (CN XI)
- Tongue protrudes midline without wasting (CN XII intact)

Q1: CENTRAL OR PERIPHERAL?

Central (Cerebral cortex — presence of cortical signs such as aphasia, neglect, apraxia, or visual field deficits).

Q2: NEURAXIS LEVEL

Right jugular foramen

Q3: STRUCTURES & TRACTS INVOLVED

Cranial nerves IX, X, and XI exit the skull base together via the jugular foramen.

Q4: DEFINITIVE DIAGNOSIS & SYNDROME

Vernet syndrome (jugular foramen syndrome)

Teaching Pearl: Cranial nerves IX, X, and XI exit the skull base together via the jugular foramen. Lesions here paralyze the pharyngeal sensory supply, palatopharyngeal/laryngeal motor supply, and trapezius/SCM muscles while sparing the hypoglossal nerve (which exits via the hypoglossal canal). 81 * Combined CN IX, X, XI, and XII palsies

Anatomical Mechanism & Discriminators: Cranial nerves IX, X, and XI exit the skull base together via the jugular foramen. Lesions here paralyze the pharyngeal sensory supply, palatopharyngeal/laryngeal motor supply, and trapezius/SCM muscles while sparing the hypoglossal nerve (which exits via the hypoglossal canal). 81 * Combined CN IX, X, XI, and XII palsies — *Anchor: Right jugular foramen*

81 Combined CN IX, X, XI, and XII Deficits Without Horner Syndrome

60-year-old man with skull base metastasis • Stage 10: Master Discriminators • #17

Clinical Vignette: A 60-year-old man with skull base metastasis presents with dysphagia, nasal regurgitation, hoarseness, right shoulder weakness, and slurred speech.

- Absent right gag reflex
- Right palatal droop and vocal cord paralysis
- Wasting of right trapezius and sternocleidomastoid
- Right half of the tongue wasted with fasciculations
- Tongue deviates to the right on protrusion
- No Horner syndrome (cervical sympathetic chain spared)

Q1: CENTRAL OR PERIPHERAL?

Central (Cerebral cortex — presence of cortical signs such as aphasia, neglect, apraxia, or visual field deficits).

Q2: NEURAXIS LEVEL

Right retroparotid / posterior lacerated space

Q3: STRUCTURES & TRACTS INVOLVED

Involvement of the posterior retroparotid space affects CN IX, X, and XI (from the jugular foramen) alongside CN XII (from the nearby anterior condylar canal), producing combined lower cranial nerve palsies without sympathetic chain involvement.

Q4: DEFINITIVE DIAGNOSIS & SYNDROME

Collet-Sicard syndrome

Teaching Pearl: Involvement of the posterior retroparotid space affects CN IX, X, and XI (from the jugular foramen) alongside CN XII (from the nearby anterior condylar canal), producing combined lower cranial nerve palsies without sympathetic chain involvement. 82 * Abducens palsy + retro-orbital pain + otorrhoea

Anatomical Mechanism & Discriminators: Involvement of the posterior retroparotid space affects CN IX, X, and XI (from the jugular foramen) alongside CN XII (from the nearby anterior condylar canal), producing combined lower cranial nerve palsies without sympathetic chain involvement. 82 * Abducens palsy + retro-orbital pain + otorrhoea — *Anchor: Right retroparotid / posterior lacerated space*

82 Persistent Otorrhoea with Severe Facial Pain and Lateral Rectus Palsy

45-year-old man with a poorly treated right middle ear infection • Stage 10: Master Discriminators • #18

Clinical Vignette: A 45-year-old man with a poorly treated right middle ear infection develops persistent severe right retro-orbital pain and horizontal double vision.

- Inability of the right eye to abduct (isolated right CN VI palsy)
- Hyperalgesia and numbness in the right ophthalmic (V_1) and maxillary (V_2) distributions
- Purulent discharge from the right external auditory canal
- Other cranial nerves, motor, and sensory testing normal

Q1: CENTRAL OR PERIPHERAL?
Central (Cerebral cortex — presence of cortical signs such as aphasia, neglect, apraxia, or visual field deficits).

Q2: NEURAXIS LEVEL
Right petrous apex / Dorello canal

Q3: STRUCTURES & TRACTS INVOLVED
Infection extending from the middle ear to the petrous apex inflames the abducens nerve in Dorello canal and the adjacent trigeminal ganglion in Meckel cave, generating the classic triad of otorrhoea, V_1/V_2 facial pain, and CN VI palsy.

Q4: DEFINITIVE DIAGNOSIS & SYNDROME
Gradenigo syndrome (petrous apicitis)

Teaching Pearl: Infection extending from the middle ear to the petrous apex inflames the abducens nerve in Dorello canal and the adjacent trigeminal ganglion in Meckel cave, generating the classic triad of otorrhoea, V_1/V_2 facial pain, and CN VI palsy. 83 * Ipsilateral optic atrophy + contralateral papilledema + anosmia

Anatomical Mechanism & Discriminators: Infection extending from the middle ear to the petrous apex inflames the abducens nerve in Dorello canal and the adjacent trigeminal ganglion in Meckel cave, generating the classic triad of otorrhoea, V_1/V_2 facial pain, and CN VI palsy. 83 * Ipsilateral optic atrophy + contralateral papilledema + anosmia — *Anchor: Right petrous apex / Dorello canal*

91 Acute Severe Headache with Ptosis, Down-and-Out Eye, and Widely Dilated Pupil

58-year-old woman • Stage 10: Master Discriminators • #19

Clinical Vignette: A 58-year-old woman is brought in with the "worst headache of her life." Within hours she develops drooping of the right eyelid and double vision.

- Right complete ptosis
- Right pupil dilated and unreactive to light
- Right eye positioned "down and out"
- Neck stiffness present
- No limb weakness
- Plantar responses flexor

Q1: CENTRAL OR PERIPHERAL?
Peripheral (Lower motor neuron / peripheral nerve / plexus / root distribution).

Q2: NEURAXIS LEVEL
Right oculomotor nerve at the posterior communicating artery

Q3: STRUCTURES & TRACTS INVOLVED
Pupillomotor fibres run superficially on the oculomotor nerve, so an extrinsic compressive lesion such as an aneurysm dilates the pupil before limb and extraocular weakness appear.

Q4: DEFINITIVE DIAGNOSIS & SYNDROME
Compressive (surgical) CN III palsy — posterior communicating artery aneurysm

Teaching Pearl: Pupillomotor fibres run superficially on the oculomotor nerve, so an extrinsic compressive lesion such as an aneurysm dilates the pupil before limb and extraocular weakness appear. A "blown pupil" with a third-nerve palsy is a neurosurgical emergency until an aneurysm is excluded.

Anatomical Mechanism & Discriminators: Pupillomotor fibres run superficially on the oculomotor nerve, so an extrinsic compressive lesion such as an aneurysm dilates the pupil before limb and extraocular weakness appear. A "blown pupil" with a third-nerve palsy is a neurosurgical emergency until an aneurysm is excluded. — *Anchor: Right oculomotor nerve at the posterior communicating artery*

148 Sudden Severe Headache with Complete Oculomotor Palsy and a Widely Dilated Unreactive Pupil

52-year-old woman • Stage 10: Master Discriminators • #20

Clinical Vignette: A 52-year-old woman develops an abrupt severe headache followed by right ptosis and diplopia.

- Complete right ptosis
- Right eye “down and out”
- Right pupil markedly dilated
- Right pupil poorly reactive
- No facial sensory loss
- No limb weakness initially

Q1: CENTRAL OR PERIPHERAL?
Peripheral (Lower motor neuron / peripheral nerve / plexus / root distribution).

Q2: NEURAXIS LEVEL
Right oculomotor nerve with superficial parasympathetic fibres

Q3: STRUCTURES & TRACTS INVOLVED
Parasympathetic pupillary fibres run superficially in CN III and are particularly vulnerable to external compression.

Q4: DEFINITIVE DIAGNOSIS & SYNDROME
Compressive CN III palsy — posterior communicating artery aneurysm until proven otherwise

Teaching Pearl: Parasympathetic pupillary fibres run superficially in CN III and are particularly vulnerable to external compression. Thus: painful III palsy + dilated pupil = compressive lesion until proven otherwise. This contrasts with the pupil-sparing microvascular III palsy already represented earlier in the series.

Anatomical Mechanism & Discriminators: Parasympathetic pupillary fibres run superficially in CN III and are particularly vulnerable to external compression. Thus: painful III palsy + dilated pupil = compressive lesion until proven otherwise. This contrasts with the pupil-sparing microvascular III palsy already represented earlier in the series. — *Anchor: Right oculomotor nerve with superficial parasympathetic fibres*

149 Abducens Nerve Palsy Associated with Ipsilateral Oculosympathetic Paralysis

57-year-old woman • Stage 10: Master Discriminators • #21

Clinical Vignette: A 57-year-old woman develops horizontal diplopia and notices that her right pupil is smaller.

- Right eye fails to abduct
- Right partial ptosis
- Right miosis
- Pupillary asymmetry greater in darkness
- Facial sensation largely preserved
- Other ocular movements normal
- No limb signs

Q1: CENTRAL OR PERIPHERAL?
Central (Cerebral cortex — presence of cortical signs such as aphasia, neglect, apraxia, or visual field deficits).

Q2: NEURAXIS LEVEL
Right cavernous sinus / parasellar internal carotid region

Q3: STRUCTURES & TRACTS INVOLVED
Within the cavernous sinus, CN VI runs very close to the internal carotid artery and its sympathetic plexus.

Q4: DEFINITIVE DIAGNOSIS & SYNDROME
Combined CN VI and postganglionic sympathetic lesion

Teaching Pearl: Within the cavernous sinus, CN VI runs very close to the internal carotid artery and its sympathetic plexus. Therefore the unusual combination: abducens palsy + Horner syndrome is a powerful localisation clue to the cavernous sinus/parasellar carotid region.

Anatomical Mechanism & Discriminators: Within the cavernous sinus, CN VI runs very close to the internal carotid artery and its sympathetic plexus. Therefore the unusual combination: abducens palsy + Horner syndrome is a powerful localisation clue to the cavernous sinus/parasellar carotid region. — *Anchor: Right cavernous sinus / parasellar internal carotid region*

145 Combined Lower Cranial Nerve Palsies (IX–XII) Accompanied by Ipsilateral Horner Syndrome

58-year-old man • Stage 10: Master Discriminators • #22

Clinical Vignette: A 58-year-old man develops progressive hoarseness, dysphagia, shoulder weakness and tongue wasting. His right eyelid also begins to droop.

- Right palatal weakness
- Right vocal-cord paralysis
- Right trapezius and SCM weakness
- Right tongue wasting and fasciculations
- Tongue deviates right
- Right miosis and partial ptosis
- No limb long-tract signs

Q1: CENTRAL OR PERIPHERAL?
Central (Cerebral cortex — presence of cortical signs such as aphasia, neglect, apraxia, or visual field deficits).

Q2: NEURAXIS LEVEL
Right retroparotid/posterior parapharyngeal space involving IX–XII plus sympathetic chain

Q3: STRUCTURES & TRACTS INVOLVED
Compare with Collet-Sicard syndrome: IX + X + XI + XII = Collet-Sicard Add Horner syndrome = Villaret syndrome.

Q4: DEFINITIVE DIAGNOSIS & SYNDROME
Villaret syndrome

Teaching Pearl: Compare with Collet-Sicard syndrome: IX + X + XI + XII = Collet-Sicard Add Horner syndrome = Villaret syndrome. The sympathetic involvement pushes the localisation farther into the retroparotid/parapharyngeal space.

Anatomical Mechanism & Discriminators: Compare with Collet-Sicard syndrome: IX + X + XI + XII = Collet-Sicard Add Horner syndrome = Villaret syndrome. The sympathetic involvement pushes the localisation farther into the retroparotid/parapharyngeal space. — *Anchor: Right retroparotid/posterior parapharyngeal space involving IX–XII plus sympathetic chain*

146 Extensive Unilateral Progressive Cranial Nerve Palsies Sparing Long-Tract Pathways

Adult patient • Stage 10: Master Discriminators • #23

Clinical Vignette: A 62-year-old man with a skull-base tumour gradually develops visual loss, ophthalmoplegia, facial numbness, deafness, dysphagia and tongue weakness, all on the right.

- Multiple right-sided cranial-nerve palsies
- Optic dysfunction
- Ophthalmoplegia
- Facial sensory loss
- Facial weakness
- Hearing loss
- Palatal weakness
- Tongue wasting
- No contralateral weakness
- No sensory tract deficit in limbs
- No cerebellar long-tract signs

Q1: CENTRAL OR PERIPHERAL?
Central (Cerebral cortex — presence of cortical signs such as aphasia, neglect, apraxia, or visual field deficits).

Q2: NEURAXIS LEVEL
Extensive unilateral skull base

Q3: STRUCTURES & TRACTS INVOLVED
Multiple unilateral cranial neuropathies without long-tract signs or raised intracranial pressure strongly suggest a skull-base process rather than a brainstem lesion.

Q4: DEFINITIVE DIAGNOSIS & SYNDROME
Garcin syndrome

Teaching Pearl: Multiple unilateral cranial neuropathies without long-tract signs or raised intracranial pressure strongly suggest a skull-base process rather than a brainstem lesion. This is localisation by what is absent as much as by what is present.

Anatomical Mechanism & Discriminators: Multiple unilateral cranial neuropathies without long-tract signs or raised intracranial pressure strongly suggest a skull-base process rather than a brainstem lesion. This is localisation by what is absent as much as by what is present. — *Anchor: Extensive unilateral skull base*

130 Unilateral Vocal Cord Paralysis and Tongue Weakness Following Airway Instrumentation

57-year-old man • Stage 10: Master Discriminators • #24

Clinical Vignette: A 57-year-old man develops hoarseness and difficulty swallowing after prolonged intubation. His tongue also feels weak.

- Right vocal cord paralysis
- Hoarse voice
- Weak right palatal elevation
- Tongue deviates to the right
- Right tongue weakness, with later wasting
- Shoulder shrug normal
- Facial sensation normal
- No limb signs

Q1: CENTRAL OR PERIPHERAL?

Central (Cerebral cortex — presence of cortical signs such as aphasia, neglect, apraxia, or visual field deficits).

Q2: NEURAXIS LEVEL

Extracranial right CN X and CN XII where they run close together in the upper neck

Q3: STRUCTURES & TRACTS INVOLVED

Combined vagus + hypoglossal palsy without IX or XI involvement points away from the jugular foramen and toward the extracranial course of X and XII.

Q4: DEFINITIVE DIAGNOSIS & SYNDROME

Tapia syndrome

Teaching Pearl: Combined vagus + hypoglossal palsy without IX or XI involvement points away from the jugular foramen and toward the extracranial course of X and XII. It may follow difficult airway manipulation or prolonged intubation.

Anatomical Mechanism & Discriminators: Combined vagus + hypoglossal palsy without IX or XI involvement points away from the jugular foramen and toward the extracranial course of X and XII. It may follow difficult airway manipulation or prolonged intubation. — *Anchor: Extracranial right CN X and CN XII where they run close together in the upper neck*

34 Multiple Cranial Nerve Palsies with Facial Pain and Numbness

52-year-old woman • Stage 10: Master Discriminators • #25

Clinical Vignette: A 52-year-old woman develops progressive headache, diplopia and numbness around the right forehead and cheek.

- Right ptosis
- Impaired right eye adduction, elevation and depression
- Right eye abduction also weak
- Reduced sensation over right forehead
- Reduced sensation over right cheek
- Corneal reflex reduced on right
- Vision and optic disc initially normal

Q1: CENTRAL OR PERIPHERAL?

Central (Cerebral cortex — presence of cortical signs such as aphasia, neglect, apraxia, or visual field deficits).

Q2: NEURAXIS LEVEL

Right cavernous sinus

Q3: STRUCTURES & TRACTS INVOLVED

The cavernous sinus contains or is closely related to: CN III CN IV CN V1 CN V2 CN VI Multiple ocular motor palsies with V1/V2 sensory loss strongly point to the cavernous sinus.

Q4: DEFINITIVE DIAGNOSIS & SYNDROME

Cavernous sinus syndrome

Teaching Pearl: The cavernous sinus contains or is closely related to: CN III CN IV CN V1 CN V2 CN VI Multiple ocular motor palsies with V1/V2 sensory loss strongly point to the cavernous sinus.

Anatomical Mechanism & Discriminators: The cavernous sinus contains or is closely related to: CN III CN IV CN V1 CN V2 CN VI Multiple ocular motor palsies with V1/V2 sensory loss strongly point to the cavernous sinus. — *Anchor: Right cavernous sinus*

35 Complete Ophthalmoplegia Associated with Visual Loss

48-year-old man • Stage 10: Master Discriminators • #26

Clinical Vignette: A 48-year-old man develops severe orbital pain, rapidly progressive visual loss and double vision in the left eye.

- Markedly reduced vision in left eye
- Left relative afferent pupillary defect
- Left ptosis
- Impaired movements in multiple directions
- Reduced sensation over left forehead
- Optic disc may initially appear normal
- No facial weakness

Q1: CENTRAL OR PERIPHERAL?
Central (Cerebral cortex — presence of cortical signs such as aphasia, neglect, apraxia, or visual field deficits).

Q2: NEURAXIS LEVEL
Left orbital apex

Q3: STRUCTURES & TRACTS INVOLVED
The key distinction from cavernous sinus syndrome is optic nerve involvement.

Q4: DEFINITIVE DIAGNOSIS & SYNDROME
Orbital apex syndrome

Teaching Pearl: The key distinction from cavernous sinus syndrome is optic nerve involvement. Orbital apex lesions may involve: CN II CN III CN IV CN VI V1 Therefore ophthalmoplegia is accompanied by visual loss.

Anatomical Mechanism & Discriminators: The key distinction from cavernous sinus syndrome is optic nerve involvement. Orbital apex lesions may involve: CN II CN III CN IV CN VI V1 Therefore ophthalmoplegia is accompanied by visual loss. — *Anchor: Left orbital apex*

37 Complete Facial Palsy Combined with Ipsilateral Horizontal Gaze Palsy

64-year-old man • Stage 10: Master Discriminators • #27

Clinical Vignette: A 64-year-old man develops sudden diplopia and weakness of the entire left side of his face.

- Both eyes unable to gaze to the left
- Left facial weakness involving forehead and lower face
- Left eye closure weak
- Right gaze preserved
- Limb strength preserved
- Sensation normal

Q1: CENTRAL OR PERIPHERAL?
Peripheral (Nerve root / Cauda equina level — Lower Motor Neuron signs predominate).

Q2: NEURAXIS LEVEL
Left dorsal caudal pons, around the facial colliculus

Q3: STRUCTURES & TRACTS INVOLVED
A lesion near the facial colliculus can affect: abducens nucleus → ipsilateral conjugate gaze palsy looping facial nerve fascicles → ipsilateral LMN facial palsy This combination is highly localising to the dorsal pons.

Q4: DEFINITIVE DIAGNOSIS & SYNDROME
Facial colliculus syndrome

Teaching Pearl: A lesion near the facial colliculus can affect: abducens nucleus → ipsilateral conjugate gaze palsy looping facial nerve fascicles → ipsilateral LMN facial palsy This combination is highly localising to the dorsal pons.

Anatomical Mechanism & Discriminators: A lesion near the facial colliculus can affect: abducens nucleus → ipsilateral conjugate gaze palsy looping facial nerve fascicles → ipsilateral LMN facial palsy This combination is highly localising to the dorsal pons. — *Anchor: Left dorsal caudal pons, around the facial colliculus*

39 Ipsilateral Ptosis and Miosis with Contralateral Pain and Temperature Loss

56-year-old man • Stage 10: Master Discriminators • #28

Clinical Vignette: A 56-year-old man develops sudden neck pain followed by unsteadiness and altered sensation on the opposite side of the body.

- Right-sided partial ptosis
- Right pupil smaller than left
- Pupillary asymmetry greater in darkness
- Facial sweating reduced on right
- Reduced pain and temperature over left side of body
- Power preserved
- No third-nerve palsy

Q1: CENTRAL OR PERIPHERAL?

Central (Brainstem crossed syndrome — ipsilateral cranial nerve nucleus/fascicle + contralateral long tracts).

Q2: NEURAXIS LEVEL

Right descending sympathetic pathway in the brainstem

Q3: STRUCTURES & TRACTS INVOLVED

Horner syndrome consists of: miosis mild ptosis sometimes anhidrosis When combined with contralateral body sensory abnormalities, it suggests interruption of descending sympathetic fibres within the brainstem, rather than an isolated peripheral sympathetic lesion.

Q4: DEFINITIVE DIAGNOSIS & SYNDROME

Central Horner syndrome

Teaching Pearl: Horner syndrome consists of: miosis mild ptosis sometimes anhidrosis When combined with contralateral body sensory abnormalities, it suggests interruption of descending sympathetic fibres within the brainstem, rather than an isolated peripheral sympathetic lesion.

Anatomical Mechanism & Discriminators: Horner syndrome consists of: miosis mild ptosis sometimes anhidrosis When combined with contralateral body sensory abnormalities, it suggests interruption of descending sympathetic fibres within the brainstem, rather than an isolated peripheral sympathetic lesion. — *Anchor: Right descending sympathetic pathway in the brainstem*

139 Neck Pain Radiating to the Lateral Forearm and Thumb with Weak Biceps

46-year-old man • Stage 10: Master Discriminators • #29

Clinical Vignette: A 46-year-old man develops neck pain radiating down the lateral arm to the thumb after lifting a heavy object.

- Weak elbow flexion
- Weak wrist extension
- Biceps reflex reduced
- Brachioradialis reflex reduced
- Sensory loss over lateral forearm and thumb
- Triceps strength preserved
- Triceps reflex normal
- Legs normal

Q1: CENTRAL OR PERIPHERAL?

Peripheral (Lower motor neuron / peripheral nerve / plexus / root distribution).

Q2: NEURAXIS LEVEL

C6 nerve root

Q3: STRUCTURES & TRACTS INVOLVED

Think C6 when the combination is: biceps/wrist extension weakness + reduced biceps/brachioradialis reflex + thumb sensory symptoms.

Q4: DEFINITIVE DIAGNOSIS & SYNDROME

C6 radiculopathy

Teaching Pearl: Think C6 when the combination is: biceps/wrist extension weakness + reduced biceps/brachioradialis reflex + thumb sensory symptoms. The distribution crosses more than one peripheral nerve, favouring a root lesion.

Anatomical Mechanism & Discriminators: Think C6 when the combination is: biceps/wrist extension weakness + reduced biceps/brachioradialis reflex + thumb sensory symptoms. The distribution crosses more than one peripheral nerve, favouring a root lesion. — *Anchor: C6 nerve root*

140 Neck Pain Radiating to Middle Finger with Weak Triceps and Lost Triceps Jerk

50-year-old woman • Stage 10: Master Discriminators • #30

Clinical Vignette: A 50-year-old woman develops neck pain shooting down the posterior arm into the middle finger.

- Weak elbow extension
- Weak wrist flexion
- Weak finger extension
- Triceps reflex reduced
- Sensory loss centred on middle finger
- Biceps reflex preserved
- Shoulder abduction normal
- Legs normal

Q1: CENTRAL OR PERIPHERAL?
Peripheral (Lower motor neuron / peripheral nerve / plexus / root distribution).

Q2: NEURAXIS LEVEL
C7 nerve root

Q3: STRUCTURES & TRACTS INVOLVED
The classic C7 pattern combines: triceps weakness finger-extension weakness reduced triceps reflex middle-finger sensory symptoms This is one of the most clinically useful cervical-root patterns.

Q4: DEFINITIVE DIAGNOSIS & SYNDROME
C7 radiculopathy

Teaching Pearl: The classic C7 pattern combines: triceps weakness finger-extension weakness reduced triceps reflex middle-finger sensory symptoms This is one of the most clinically useful cervical-root patterns.

Anatomical Mechanism & Discriminators: The classic C7 pattern combines: triceps weakness finger-extension weakness reduced triceps reflex middle-finger sensory symptoms This is one of the most clinically useful cervical-root patterns. — *Anchor: C7 nerve root*

141 Medial Forearm Pain and Hand Weakness Involving Both Median and Ulnar Intrinsic Muscles

44-year-old man • Stage 10: Master Discriminators • #31

Clinical Vignette: A 44-year-old man develops neck pain radiating along the medial forearm into the fourth and fifth fingers.

- Weak finger flexion
- Weak finger abduction and adduction
- Mild weakness of thumb movements
- Sensory loss along medial forearm and little finger
- Finger-flexor reflex reduced
- Triceps relatively preserved
- Weakness involves both median- and ulnar-innervated C8 muscles

Q1: CENTRAL OR PERIPHERAL?
Peripheral (Lower motor neuron / peripheral nerve / plexus / root distribution).

Q2: NEURAXIS LEVEL
C8 nerve root

Q3: STRUCTURES & TRACTS INVOLVED
The important clue is weakness in several muscles supplied by different peripheral nerves but sharing C8 fibres.

Q4: DEFINITIVE DIAGNOSIS & SYNDROME
C8 radiculopathy

Teaching Pearl: The important clue is weakness in several muscles supplied by different peripheral nerves but sharing C8 fibres. That favours a root rather than isolated ulnar neuropathy.

Anatomical Mechanism & Discriminators: The important clue is weakness in several muscles supplied by different peripheral nerves but sharing C8 fibres. That favours a root rather than isolated ulnar neuropathy. — *Anchor: C8 nerve root*

143 Cape-Like Dissociated Sensory Loss Extending to Facial Analgesia and Bulbar Symptoms

42-year-old man with longstanding hand weakness • Stage 10: Master Discriminators • #32

Clinical Vignette: A 42-year-old man with longstanding hand weakness develops progressive numbness of his face, swallowing difficulty and oscillopsia.

- Wasting of intrinsic hand muscles
- Cape-like pain-temperature loss
- Pain-temperature loss over outer face in an “onion-skin” distribution
- Nystagmus
- Palatal weakness
- Tongue weakness may occur
- Limb vibration relatively preserved
- Legs mildly spastic

Q1: CENTRAL OR PERIPHERAL?

Central (Spinal cord myelopathy — UMN signs below lesion + sensory level ± LMN signs at level).

Q2: NEURAXIS LEVEL

Central cervical cord cavity extending upward into medulla

Q3: STRUCTURES & TRACTS INVOLVED

A cervical syrinx extending into the medulla may involve: spinal trigeminal pathways → facial pain-temperature loss vestibular structures → nystagmus lower cranial-nerve nuclei → bulbar weakness The facial findings indicate that a previously spinal process has extended into the brainstem.

Q4: DEFINITIVE DIAGNOSIS & SYNDROME

Syringobulbia accompanying syringomyelia

Teaching Pearl: A cervical syrinx extending into the medulla may involve: spinal trigeminal pathways → facial pain-temperature loss vestibular structures → nystagmus lower cranial-nerve nuclei → bulbar weakness The facial findings indicate that a previously spinal process has extended into the brainstem.

Anatomical Mechanism & Discriminators: A cervical syrinx extending into the medulla may involve: spinal trigeminal pathways → facial pain-temperature loss vestibular structures → nystagmus lower cranial-nerve nuclei → bulbar weakness The facial findings indicate that a previously spinal process has extended into the brainstem. — *Anchor: Central cervical cord cavity extending upward into medulla*

144 Complete Foot Weakness Involving Both Dorsiflexion and Plantarflexion with Hamstring Weakness

40-year-old man • Stage 10: Master Discriminators • #33

Clinical Vignette: A 40-year-old man develops severe pain behind the thigh following hip trauma and subsequently develops foot drop.

- Foot dorsiflexion very weak
- Plantarflexion also weak
- Inversion and eversion both weak
- Hamstring weakness
- Knee extension preserved
- Ankle jerk absent
- Sensory loss over most of foot except medial border
- Hip abduction relatively preserved

Q1: CENTRAL OR PERIPHERAL?

Peripheral (Lower motor neuron / peripheral nerve / plexus / root distribution).

Q2: NEURAXIS LEVEL

Sciatic nerve

Q3: STRUCTURES & TRACTS INVOLVED

A common peroneal lesion produces foot drop but preserves plantarflexion.

Q4: DEFINITIVE DIAGNOSIS & SYNDROME

Sciatic neuropathy

Teaching Pearl: A common peroneal lesion produces foot drop but preserves plantarflexion. An L5 lesion usually preserves the ankle jerk. Combined tibial + peroneal dysfunction with hamstring involvement points proximally to the sciatic nerve.

Anatomical Mechanism & Discriminators: A common peroneal lesion produces foot drop but preserves plantarflexion. An L5 lesion usually preserves the ankle jerk. Combined tibial + peroneal dysfunction with hamstring involvement points proximally to the sciatic nerve. — *Anchor: Sciatic nerve*